

Local Time Against Deployed Code

The Exact Cost Law of Concentrated Liquidity Rebalancing, Its Sharp Floor, and Its Inversion

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August 2026

Keywords concentrated liquidity · liquidity rebalancing · automated market maker · DeFi · mint kerf · local time
impulse control · jump processes · Uniswap V3

JEL: C58, C61, D47, G14, G23

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Abstract. Every floor function, trigger line, and threshold in deployed market code is a surface in price space, and automated liquidity management dissipates value where the price meets them. That dissipation obeys one exact law, developed here in three directions and characterised by six theorems. The law is a pathwise identity splitting dissipated value into three terms of a single expansion: a continuous concavity leg, which occupies the slot priced by the loss-versus-rebalancing literature; an event leg, set by the operator's own geometry; and a monitoring layer. Localised in price, the identity is a measure over the code's discontinuity set. The event leg is priced exactly. An isolated re-placement pays a *mint kerf*, the maximum of two log ratios of target share to current share, a potential difference in the position's composition coordinate computed from the venue's mint arithmetic rather than assumed. The kerf implies a floor. On a diffusive era every band-maintaining policy in the stated classes dissipates at a rate near twice volatility squared over half-width squared; this *tangency constant* is sharp in the narrow limit, approached but not attained by two-sided reflection. Corrective swaps replace the constant by the fee scale. The floor is a property of emission, so retention deletes it rather than lowering it. Inverted, the law makes the deployed population a conditional measuring instrument. The naive pooled reading fails, defeated by the operators' own policy filters; the conditioned small-delay reading identifies integrated jump-tail functionals per pool with explicit error under stated era and monitoring hypotheses, replicating across operator strata within pools. Algebraic, synthetic, production, and bytecode checks support the results: the cost residual at machine precision on 78,908 production rebalances and the exact layer holding against unmodified bytecode on four deployments across three chains. Under the stated landing conditions the jump surcharge closes the loop, hardening the floor at the tails the instrument itself measures.

1. Introduction

A concentrated liquidity manager holds a position on an automated market maker, monitors the price, and acts when a coded condition fires. The condition is a line in price space. So is every floor function in the placement arithmetic that sites the position, every threshold in the sizing rule that fills it, and every range bound at which the holding value changes slope. Seen from the price process, deployed market code is a locally finite set of surfaces. The value that automated liquidity management surrenders is localised where the price meets them.

The subject of this paper is that object: a price process accumulating local time against the discontinuity set of deployed code. At each crossing, the payoff jumps by an exact, source-derivable amount. Crossing behaviour is governed by local time and first-passage structure of the price law. The forward law of the title computes expected value flux as amplitudes integrated against occupa-

tion, local time, and firing intensity over the discontinuity set, displayed as (4) below. (4) is the total mass of a dissipation measure the three acts in turn compute, bound, and invert. Every venue-side quantity entering the amplitudes and the discontinuity set is read from contract source code; the price law, its jump measure, and the monitoring process remain separate inputs. That line separates an exact accounting identity from a statistical model of flows.

Three prior papers derived the same machinery's laws one channel at a time: The Geometric Siphon¹ (GS), The Hidden Microstructure² (ME), and Operator & Quantisation Microstructure³ (OQM). Each published law returns here as a specialisation of a single formula, under its own stated hypotheses; the recoveries therefore serve as consistency checks on the frame. Section 2 details the inheritance.

The accounting that results has more legs than the field's organising concept recognises. The liquidity-provider economics literature prices provision as fees against adverse selection, relative to a rebalancing benchmark held in continuous price.⁴ The dissipation identity decomposes into two legs: a concavity leg, embedding that cost in an exact pathwise expansion, and an event leg. The two legs differ in kind. The concavity leg needs a benchmark portfolio that no chain executes and a model of the price feed, and to leading order it is the cost of holding a concave payoff, indifferent to how the position is run. The event leg is read off the chain and attributed to source, crossing by crossing. Width, trigger fraction, cadence, grid spacing, and sizing arithmetic enter its amplitudes and intensities explicitly, so it is the leg an operator's choices move. Its firing statistics invert for the price law. This two-leg exact accounting, with its monitoring layer, is the paper's dissipation ledger. Nothing in it demotes the continuous layer's dollar weight; what changes is what can be accounted exactly, and what an operator can act on. The move from impermanent loss to loss-versus-rebalancing rested on exactness under a stated model; the ledger continues along the same axis, exact under weaker assumptions and attributed to verifiable code. Fees and emissions enter the same ledger as revenue and are not priced here.

The paper's central result is an exact price for the event leg. An isolated re-placement costs the larger of two log share ratios, one per token. The quantity is a potential difference in the position's composition, and the venue's mint arithmetic fixes it, whereas a transaction-cost model would posit it as a market friction (Theorem 2, stated with the frame in Section 5). I call this quantity the *mint kerf*, after the material a saw destroys at every cut, fixed by the blade and not by the hand that guides it. Here the blade is the venue's mint arithmetic, and the operator chooses only how often to cut and how far. GS's vanishing condition, the ratio-preserving locus, returns as the zero-kerf set. The kerf implies a floor. Every policy that keeps a price in a band must, in a diffusive era, dissipate at a rate near twice volatility squared over half-width squared; in the narrow limit that constant is exact, the class infimum is the floor itself. Three form survives three extensions, though not always the constant. Variable width buys nothing below the fixed-width bound at the cap. Corrective swaps replace the constant by the fee scale without escaping the form. Jump crossings harden the emission-based bounds by a surcharge no return convention avoids. The swap bound, through the deeper landing its execution must correct. The floor is an architectural invariant; it exists for architectures that emit the rebalance surplus, and the retention architecture of ME deletes it rather than lowering it.

Inverted, the law makes the population a measuring instrument. A deployed trigger line is a threshold detector for the price process, and its overshoots and actuation delays are observables of the price law, including the jump measure that diffusive treatments discard. The naive form of the instrument fails. Production operators fire through dwell and penetration filters, and filtered overshoots are invariant in the actuation delay. The pooled census read accordingly matches a diffusion benchmark almost exactly. The same filters are the instrument's aperture. They thin the diffusive background super-polynomially in the short-delay regime where the jump signal concentrates, and the conditioned small-delay read carries an excess of two orders of magnitude over the diffusive null, replicated across operator strata within pools and with its inclusion criteria in Section 22. The instrument was found in the population's record, not designed into it.

The general treatment also resolves a discrepancy between two published statements. GS gives the down-branch boundary magnitude of the residual as $Lw \bar{s}/s_b$, whilst OQM's swap-free credit

evaluates at the same corner to $Lw s_a/s_b$. The two are the interior small-displacement limit and the corner value of a single function: GS's is the limit inside, and OQM's stands at the corner. Corollary 3 reconciles them and Proposition 3 supplies the width condition for strict down-branch monotonicity.

OQM verifies the uniformity of the within-tick offset at firings empirically and explicitly does not claim it as a property of the price process. It is a theorem here, but a conditional one, and the condition has content. The condition is the monitoring. A continuously observed price fires on its trigger line, and the firing-state offset is a point mass; observed on a cadence, the price overshoots the line, and only that overshoot makes the offset uniform. The discreteness of monitoring, not the price law alone, is what the value channel rests on (Proposition 4).

The mathematics assembled here is classical. Itô–Tanaka and the occupation-times formula are textbook material,⁵ and the two-barrier exit probability is a first-passage computation.⁶ The paper claims no novelty for either. The contribution is the object itself, the identity that joins the channels, an equidistribution lemma that turns floor arithmetic into computable flux, and the corollary structure that recovers the published laws from one formula.¹

The contribution has five parts, and they carry different kinds of warrant. The frame is the dissipation measure over the discontinuity set of deployed code. The law inside it is the share-potential identity, which prices each event exactly in the venue's arithmetic. The global consequence is the sharp floor, with its tangency constant and the architecture-dependent replacements that corrective swaps and retention supply. The inversion is the conditional identification of integrated jump-tail functionals from the population's firing record. The verification runs from contract source through deployed bytecode, a machine-checked theorem layer, synthetic ground truth, and production history. Sections 24 and 25 say which of the four preceding parts each surface reaches.

One word does four jobs below, and the four are not interchangeable. A pathwise identity is exact under its stated stochastic hypotheses and carries no approximation error at any step size. A mint law is exact in the venue's own arithmetic, so a disagreement with it is a disagreement with deployed code rather than with a model. A distributional formula is exact under its regularity assumptions, the flat-entry and era hypotheses among them. A numerical check reports finite-precision agreement at a stated tolerance. Where the text says *exact* without qualification, the first two senses are meant and the third and fourth are always named.

Three questions organise what follows, one per act: what law governs the value that deployed code dissipates, how little of it any admissible policy can pay, and what the paying population reveals about the price process it watches. The results ladder accordingly. The dissipation identity supplies the frame; the share-potential identity is the exact law inside it, stated with the frame in Section 5; the floors are the law's global consequence; the monitoring and jump layers extend the frame to deployed observation; the instrument inverts the law; and the jump surcharge joins floor and instrument, the one result that needs all three acts and the reason they share a paper. Section 2 places the work. The first act, Sections 3–9, builds the object and proves the law. The second act, Sections 10–17, prices every admissible policy from below. The third act, Sections 18–22, inverts the law on the population, and the jump surcharge of Section 23 then joins the floor and instrument. Verification, positioning, and limits follow in Sections 24–26. Notation is collected in Appendix A, and proofs are in the appendices, ordered by first body reference. A reader who wants the argument before the algebra can take the italic paragraphs that follow the results, the figures, and the worked example of Section 23 as a spine; together they state all results and their consequences in words.

2. Related work

The cost layer priced by the loss-versus-rebalancing literature^{4,8} is adverse selection against a rebalancing benchmark in continuous price; predictable-loss refinements⁹ price the concavity of

¹The name Kac–Rice is used below only as an analogy. Kac–Rice counts level crossings of differentiable processes;⁷ a diffusion crosses any level infinitely often. The dissipation identity formalises the replacement: local time for crossing counts at static boundaries, first-passage renewals at impulse boundaries.

the holding value between rebalances and derive from it an optimal range width, set by pool profitability net of gas and by concentration risk; and equilibrium treatments¹⁰ embed provision in a market model. The frictionlessness of that benchmark's rebalancing leg has been questioned before by replacing the centralised leg with a frictional one.¹¹ The flows studied here arise at the rebalance itself and are set by trigger geometry, placement arithmetic, and execution. Section 5 shows the two layers are distinct terms of one pathwise expansion; Sections 12–14 bound the impulse term alone from below, an infimum over a policy class rather than an optimum for a single objective under one model. Transaction-level studies of provider outcomes on the chain and across the venues read in Section 22 find that few providers avoid losses and that the successful ones close before a range is traversed.¹²

The floor's mathematical apparatus is that of transaction-cost portfolio theory. Magill and Constantinides¹³ introduced proportional costs into the Merton problem, Davis and Norman¹⁴ characterised the resulting no-trade region, Whalley and Wilmott¹⁵ derived the asymptotic band width under small costs, and Øksendal and Sulem¹⁶ treat the impulse-control formulation, whose verification-function method the fixed-width floor adapts. Shreve and Soner¹⁷ gave the viscosity characterisation of the infinite-horizon problem, from which Janeček and Shreve¹⁸ read off the cube-root scaling of the no-trade region's width in a small proportional cost. In the asymptotics that followed the rebalancing rate is read off the local time of the risky fraction at that region's boundaries, so that the liquidity premium factorises into a spread, a turnover, and a universal constant.¹⁹ The floor separates the same way, into a cost scale and a diffusion scale, and differs in where the cost scale comes from. There it is an exogenous proportional spread; here it is the venue's mint arithmetic, so the constant the verification argument returns is absolute rather than a parameter.

What is inherited is the verification method and the local-time reading of turnover at a boundary. What is new is the cost function the method verifies, the corridor against era-uniformised envelope slopes, the constants with their sharpness, the width-uniform bound in the composition coordinate, and the retention deletion, which is not a reduction in a transaction cost but the removal of the leg the floor prices. On the timing side, optimal-exit treatments^{20,21} optimise an individual withdrawal in continuous price. The floor bounds every band-maintaining policy at once, a complementary normative bound. A quasi-variational formulation of concentrated liquidity range management has been written down and solved numerically for a policy;²² the inequality is solved here in closed form for a bound.

The overshoot of a level by a process with jumps is classical territory; the excess over a boundary has a limiting law given by the ladder-height structure of renewal theory.^{23,24} The econometrics of jumps in high-frequency data separates diffusive and jump variation through bipower and truncation methods^{25,26} applied to regularly sampled prices. Sampling at event times rather than on a calendar grid is a developed alternative, from the duration models of Engle and Russell²⁷ to the price-duration schemes whose realised-variance limit theory was established under hitting times of an irregular grid.^{28,29} When applied to jumps, the scheme sharpens the separation, since a jump wider than the barrier ends the interval at once and leaves its size in the overshoot, and a test built on it distinguishes Brownian exceedances from jump exceedances on tick data.³⁰

The instrument of the third act runs on that mechanism and differs in three respects, the first of which decides what can be proved. A sampling scheme's barrier shrinks with the mesh and takes its consistency from that shrinking; a deployed trigger has whatever width its operator chose and is fixed for the era, so identification is necessarily a bracket with a stated error budget, not a consistent test. Second, a re-anchored ladder begins every interval at exactly the barrier width, where the price approaches the line from an arbitrary gap; that costs the renewal structure and buys the actuation delay as a second observable, and the discriminant here is the invariance of the overshoot in that delay. Third, the barriers are recovered from deployed code rather than being chosen, so the scheme cannot be tuned to the data, and the target is a pool-level integrated jump-tail functional across heterogeneous geometries rather than the presence of jumps in a single record. Section 18 carries the mechanics.

The discreteness literature in traditional microstructure^{31,32} recognised that price grids shape observed flows. Public ledgers add a second kind of observability, since the arithmetic that creates the grid and trigger, and the sizing rule, can be read from source and checked against deployed bytecode. The venue-side claim here is accordingly not that discrete flows are observed, which was already the case, but that their amplitudes are derived from the code that creates them.

Within the programme, GS identified the geometric residual of a rebalance, its closed form and machine-checked sign, and the iff vanishing condition that Theorem 2 upgrades to a cost law; ME promoted the residual to a multi-pool dynamical law and supplies the mint identity behind the retention collapse; and OQM decomposed the resulting dust flows into the frequency and value channels recovered in Section 7, together with the fee, notional, and footprint inputs of the swap-mediated policy class. A concurrent preprint by I. P.³³ derives the swap-free credits independently and develops the cross-pool topology of retained residuals. The empirical companion³⁴ (OF) recovers trigger geometry across the operator population read by the third act; its identification argument reappears in Section 5. On the economic side, a technical note on emissions³⁵ prices the concavity leg in the benchmark literature's coordinates and shows that a gauge-staked provider's break-even scales inversely with volatility squared; the ledger assembled here adds the event leg at the same volatility elasticity, which leaves that regime conclusion intact while making the cost side controllable.

ACT I. THE LAW

The first act builds the object and proves the law. A venue's deployed code is read as a locally finite set of surfaces in price space; the price accumulates local time against them, and the value surrendered at impulses obeys an exact pathwise identity whose event prices are governed by one law, the share-potential identity. All later results consume this act.

3. The object

The first act's object is a price process evolving against the discontinuity set of a coded venue; the next six sections assemble it formally. The price coordinate is scalar throughout, so the surfaces of the introduction are price levels here. A venue whose firing condition reads several state variables presents them as surfaces in the larger state space, and nothing below depends on which case holds. Readers whose interest is the floors may pass from the central law of Section 5.3 directly to Section 10; readers whose interest is the measurements may treat the theory as a guide to the formal claims and pass to Sections 18–22.

3.1. Price process

Throughout, $X = (X_t)_{t \geq 0}$ is a continuous semimartingale on a filtered probability space satisfying the usual conditions, with quadratic variation $\langle X \rangle$ strictly increasing on the eras under consideration. In the concentrated liquidity instance X is the tick coordinate, $X_t = \log P_t / \log 1.0001$, so that grids are integer lattices in X . The local time field $\{\ell_t^a\}$ of X is normalised by the occupation-times formula⁵

$$\int_0^t g(X_s) d\langle X \rangle_s = \int_{\mathbb{R}} g(a) \ell_t^a da \quad \text{for bounded measurable } g.$$

Write $\psi_T(a) = \mathbb{E}[\ell_T^a]$ for the mean occupation density over $[0, T]$. Jump price laws are deferred to Section 9.

3.2. Venue presentation

A venue enters the theory through five slots: a policy state space, an accrual rate, a holding value, a firing set, and an impulse map. The definition fixes the bookkeeping, and the concentrated liquidity instance below fills every slot from contract source.

Definition 1 (Venue Presentation). A venue presentation is a tuple $\mathcal{V} = (\mathcal{R}, c, f, F, \Gamma)$, in which $\mathcal{R}$ is a measurable space of policy states; $c : \mathbb{R} \times \mathcal{R} \rightarrow \mathbb{R}$ is an accrual rate, piecewise continuous in the price coordinate with locally finite discontinuity set $\mathcal{D}_c(R)$; $f : \mathbb{R} \times \mathcal{R} \rightarrow \mathbb{R}$ is a holding value, a difference of convex functions in the price coordinate for each R , with second-derivative measure $f_R''(da)$ and curvature set $\mathcal{D}_f(R)$, the locally finite set carrying that measure's atoms, its singular part, and the discontinuities of its density; $F(R) \subset \mathbb{R}$ is a closed firing set whose boundary $\mathcal{D}_{\text{imp}}(R)$ comprises the trigger surfaces; and $\Gamma : \mathbb{R} \times \mathcal{R} \rightarrow \mathcal{R} \times \mathbb{R}$ is an impulse map, assigning to a firing at price a in state R the post-firing state R' and an emitted flux $J(a, R)$. The discontinuity set of the venue at state R is

$$\mathcal{D}(R) = \mathcal{D}_c(R) \cup \mathcal{D}_f(R) \cup \mathcal{D}_{\text{imp}}(R).$$

In every intended instance the data of Definition 1 are computed from contract source by finitely many field operations and floor functions applied to contract constants and the state. The maps are exact and known rather than estimated.

In the concentrated liquidity instance, with sqrt-price coordinate s of the V3 geometry³⁶ and a position of liquidity L on range $[s_a, s_b]$, the holding value inside the range is $f = L \phi(s, s_a, s_b)$ with $\phi = 2s - s^2/s_b - s_a$ as in GS, all in token1 units. Below the range the position is single-sided in token0 with $f = L(1/s_a - 1/s_b) s^2$, above it the position is single-sided in token1 with $f = L(s_b - s_a)$. The function is C^1 at both bounds, with a jump in the second-derivative density and no atom, so that $f_R''(da)$ is absolutely continuous with density $-2L/s_b$ inside the range, $2L(1/s_a - 1/s_b)$ below it, and zero above it. The firing set is the complement of the trigger band, $F(R) = \{a \leq t_- + \phi W\} \cup \{a \geq t_+ - \phi W\}$ for a range with tick bounds $t_- < t_+$, tick width W , and trigger fraction ϕ .³ The impulse map recentres the range on the grid-floored current tick and re-mints through the binding-side minimum; its floors are the placement floor, acting on $\mathcal{D}_{\text{imp}}$, and the sizing floor, acting on J . The accrual rate is identically zero in this corpus; venues with quantised funding or fee switches populate it, as does the discrete-bin instantiation of Appendix B through its per-bin fee credit. The slot-filling is not only source-read but bytecode-asserted: the fork suite of Section 24 replays the exact layer against the unmodified position managers of three production deployments of this instance on two chains.

Remark 1 (Coordinates). The identity of Section 5 is stated in whichever price coordinate the venue's arithmetic lives in, and its terms transform covariantly under smooth monotone reparametrisation, with local time picking up the derivative factor of the occupation-times formula. Explicitly, the local time of $\psi(X)$ at level $\psi(a)$ is $\psi'(a)$ times the local time of X at a for smooth strictly increasing ψ , so the pairing of local time with the second-derivative measure in (2) does not depend on the coordinate. The concentrated liquidity amplitude geometry is native to the sqrt-price coordinate and Section 6 works there; the era model of Section 5 is native to the tick coordinate, a logarithmic function of s . Nothing below mixes the two coordinates within a single display.

3.3. The controlled pair and the flux

Definition 2 (Controlled Evolution). Given a venue presentation and an initial state R_0 with $X_0 \notin F(R_0)$, set $\tau_0 = 0$ and recursively $\tau_{k+1} = \inf\{t > \tau_k : X_t \in F(R_k)\}$, with $(R_{k+1}, J_{k+1}) = \Gamma(X_{\tau_{k+1}}, R_k)$ and the state constant between firings. Throughout, R_k abbreviates R_{τ_k} , the policy state on the k -th inter-firing interval.

Two standing assumptions govern the impulse structure. The first is uniform interior placement, that there is $d_0 > 0$ with $\text{dist}(X_{\tau_k}, F(R_k)) \geq d_0$ after every firing; a continuous path then admits

only finitely many firings on compacts, since each cycle requires a traversal of length d_0 . In the concentrated liquidity instance $d_0 = (\frac{1}{2} - \varphi)W$ up to placement rounding, and the production window has $\varphi = 0.25$. The second is conservative emission, that the emitted flux is the drop in holding value at the impulse,

$$J(a, R) = f(a, R) - f(a, R'), \quad (1)$$

which in the concentrated liquidity isolated case is taken conservation at the rebalance, the content of GS's vanishing-condition proof. Venues that burn or subsidise at impulses break (1) by an explicit source-derivable term that the identity below carries additively.

The value flux of the controlled pair over $[0, T]$ is the sum of the accrual integral $\int_0^T c(X_s, R_s) ds$ and the impulse sum $\Phi_T = \sum_{\tau_k \leq T} J_k$. The offset coordinate completes the setup. For a grid of spacing $\vartheta > 0$, the offset of a price point a is

$$\zeta_\vartheta(a) = \frac{a}{\vartheta} - \left\lfloor \frac{a}{\vartheta} \right\rfloor \in [0, 1);$$

OQM's within-tick offset is ζ_1 , and the placement offset is $\vartheta \zeta_\vartheta$ on the spacing grid. Figure 1 draws the controlled pair in the concentrated liquidity instance.



Figure 1. The controlled pair in the concentrated liquidity instance. The price path accumulates local time against the bounds of the deployed range (solid, tinted), the curvature set of f ; a crossing of a trigger surface (dashed) fires the impulse, which surrenders J and re-places the range.

4. The equidistribution lemma

Both floors of the corpus consume one distributional input, the uniformity of a sampled state modulo a grid. A sampling density suffices.

Lemma 1 (Offset Uniformity). *Let Z be a real random variable whose law has a density q of finite total variation $\text{TV}(q)$, the variation of the density as a function of the price rather than of the signed measure it carries. Then the law of $\zeta_\vartheta(Z)$ on $[0, 1)$ has a density $\bar{p}_\vartheta$ with*

$$\sup_{z \in [0, 1)} |\bar{p}_\vartheta(z) - 1| \leq \vartheta \text{TV}(q),$$

and total-variation distance between laws at most $\vartheta \text{TV}(q)/2$ from the uniform law.

Proof sketch. An oscillation bound on the pushforward density against the total variation of q ; Appendix C. ■

Lemma 2 (Conditional Offset Uniformity). *Let $\mathcal{G}$ be a sub- σ -algebra such that a regular conditional density $q_{\mathcal{G}}$ of Z given $\mathcal{G}$ exists with $\text{TV}(q_{\mathcal{G}}) \leq \nu$ almost surely. Then the bound of Lemma 1 holds conditionally, and for bounded measurable g on $[0, 1)$ and bounded $\mathcal{G}$ -measurable Y ,*

$$|\mathbb{E}[g(\zeta_{\mathcal{G}}(Z))Y] - \left(\int_0^1 g\right)\mathbb{E}[Y]| \leq \frac{1}{2} \vartheta \nu \|g\|_{\infty} \|Y\|_{\infty}.$$

Proof. Apply Lemma 1 under the regular conditional density, pointwise in ω , and integrate. ■

The conditional form is the working tool. It makes the offset approximately independent of every coarse-scale variable, with the error of that approximation bounded by $\vartheta\nu$, which is how the independence assumption of the value channel becomes a quantified conclusion in Section 7.

Corollary 1 (Occupation Sampling). *Let $\hat{X}$ be the quadratic-variation sampling of the path over $[0, T]$, defined by*

$$\mathbb{P}(\hat{X} \in A) = \frac{\mathbb{E} \int_0^T \mathbf{1}_A(X_s) d\langle X \rangle_s}{\mathbb{E}\langle X \rangle_T}.$$

If ψ_T has finite total variation then $\zeta_{\mathcal{G}}(\hat{X})$ is uniform to

$$\sup_z |\bar{p}_{\mathcal{G}}(z) - 1| \leq \frac{\vartheta \text{TV}(\psi_T)}{\mathbb{E}\langle X \rangle_T}.$$

The sampling here is by occupation. Offsets sampled at firings are a different object, taken up at the end of the section.

For Brownian motion in the tick coordinate, $X = \sigma B$ from the origin, ψ_T is even and unimodal with $\text{TV}(\psi_T) = 2\psi_T(0) = 2\sigma\sqrt{2T/\pi}$ and $\mathbb{E}\langle X \rangle_T = \sigma^2 T$, so the deviation from uniformity is at most $1.60 \vartheta/(\sigma\sqrt{T})$, the constant being $2\sqrt{2}/\sqrt{\pi}$. The same constant bounds the drifted era model, since the total variation of a Gaussian mixture does not see the means (Lemma 12), so the corollary needs no driftless assumption. The single knob is grid spacing over the diffusive range of the window. On OQM's production panel the tick series wanders hundreds of ticks per sampling window, so the tick-grid instance is uniform to well under a per cent, matching the observed near-uniformity of the within-tick offset, while the spacing-grid instance carries a constant one to two orders larger. The latter predicts that the placement-wedge fit should sit slightly below its ideal slope in pools where overshoot spread is not large against the spacing, which is consistent with OQM's reported slope of 0.94; a per-pool test of this attenuation is left to future empirical work.

The pathwise analogue replaces ψ_T with the realised field $a \mapsto \ell_T^a$, which is not of finite variation, so the oscillation argument runs cell by cell against its modulus of continuity and yields an error of order $\vartheta^{1/2-\varepsilon}$ with a random constant for Brownian-type laws.^{5,37} The mean form above is the one that expected flux consumes.

The empirical offsets of the corpus are sampled at firings, and a continuous price monitored continuously fires exactly on its trigger line, so the firing-state offset is a point mass and no uniformity of any kind holds. Deployed systems observe the price on a keeper cadence; the firing state overshoots the line with a conditional density whose scale is set by $\sigma\sqrt{\Delta t}$ over one check interval, and Lemma 2 applies to that density. Uniformity at firings is therefore a joint property of the price law and the monitoring process, quantified in Section 9, and it degrades as monitoring approaches continuity, a prediction testable on existing data across operators of known cadence.

5. The forward identity

5.1. The pathwise dissipation identity

The frame is one identity, exact on paths, whose only inputs are the controlled pair and the convexity structure of the holding value.

Theorem 1 (Dissipation Identity). *Let X be a continuous semimartingale, $\mathcal{V}$ a venue presentation with $f(\cdot, R)$ a difference of convex functions for each R , and let the controlled pair satisfy uniform interior placement and conservative emission (1). Then almost surely, for every $T \geq 0$,*

$$\sum_{k: \tau_k \leq T} J_k = \underbrace{f(X_0, R_0) - f(X_T, R_T)}_{\text{boundary}} + \underbrace{\sum_k \int_{\tau_k \wedge T}^{\tau_{k+1} \wedge T} f'_-(X_s, R_k) dX_s}_{\text{martingale}} + \underbrace{\frac{1}{2} \sum_k \int_{\mathbb{R}} \Delta_k \ell_T^a f''_{R_k}(da)}_{\text{local time}}, \quad (2)$$

where $\Delta_k \ell_T^a = \ell_{\tau_{k+1} \wedge T}^a - \ell_{\tau_k \wedge T}^a$ is the local time accrued on the k -th inter-firing interval. The accrual channel, when present, adds $\int_0^T c(X_s, R_s) ds$ to the left side with its occupation representation on the right.

Proof. On each interval $[\tau_k, \tau_{k+1})$ the policy state is frozen and Itô–Tanaka applies to $f(\cdot, R_k)$.⁵ At τ_k the process $f(X_t, R_t)$ jumps by $f(X_{\tau_k}, R_k) - f(X_{\tau_k}, R_{k-1}) = -J_k$ by (1). Telescoping over the finitely many firings on $[0, T]$ gives (2). ■

In words, everything the policy surrenders at its firings is the decline in holding value over the window, plus what the price path contributes between firings, adjusted by the curvature the price accumulates where the holding value bends. The identity is exact and pathwise; every probabilistic statement below evaluates the expectations of its terms. Taking expectations with X a local martingale on the era and enough integrability for the stochastic integral to vanish in mean, for which it suffices that f'_- is bounded on the price interval the stopped era path visits and $\mathbb{E}\langle X \rangle_T < \infty$ (Appendix C),

$$\mathbb{E} \sum_{\tau_k \leq T} J_k = \mathbb{E}[f(X_0, R_0) - f(X_T, R_T)] + \frac{1}{2} \mathbb{E} \sum_k \int \Delta_k \ell_T^a f''_{R_k}(da). \quad (3)$$

Remark 2 (The Two Cost Layers). In the concentrated liquidity instance the in-range part of the local-time term prices the occupation of a concave payoff by a diffusing price, which is the slot occupied by loss-versus-rebalancing;⁴ the definitions differ, that cost being benchmark-relative, so no equivalence is claimed. The impulse sum is the quantisation and execution flow of OQM. That paper kept the two layers orthogonal as a modelling discipline;(2) derives the orthogonality, the layers being distinct terms of one expansion, so neither can double-count the other.

5.2. The intensity representation

The era model makes the impulse term computable; the proposition assembles four ingredients, a renewal structure, a direction law, a firing-state law, and the resulting flux rate. Let $X_t = \mu t + \sigma B_t$ in tick units with era-constant parameters, let the policy be the concentrated liquidity instance with trigger distances w_u, w_d from the placement point, and let monitoring follow Section 9.

Proposition 1 (Intensity Representation). *Under the era model,*

- (i) *inter-firing cycles are independent and identically distributed given the era, and $N_T/T \rightarrow \lambda = 1/\mathbb{E}[\text{cycle length}]$ almost surely;*

(ii) the firing-direction probabilities $\pi^+ = \mathbb{P}(\text{up})$ and $\pi^- = \mathbb{P}(\text{down})$ satisfy, with $\theta = 2\mu/\sigma^2$,

$$\frac{\pi^+}{\pi^-} = \frac{e^{\theta w_d} - 1}{1 - e^{-\theta w_u}},$$

up to a monitoring correction bounded in Proposition 4;

(iii) the state at firing is the trigger line plus an overshoot whose conditional density satisfies the hypothesis of Lemma 2 at grid scale;

(iv) the era flux rate is

$$\lambda (\pi^+ \mathbb{E}[J \mid \text{up}] + \pi^- \mathbb{E}[J \mid \text{down}]),$$

with the conditional amplitudes computed by integrating the source-derived amplitude against the offset and overshoot laws of (iii).

In words, the era model turns the impulse term into a rate: firing frequency times the direction-weighted mean amplitude, with every factor computable from the source and the era parameters. Item (ii) coincides with the directional firing law of OQM. Writing Λ for the compensator of the firing counting process and $\tilde{c}$ for the accrual rate metered in the quadratic-variation clock, the era-form summary is

$$\begin{aligned} \mathbb{E} \left[\Phi_T + \int_0^T c(X_s, R_s) ds \right] = \\ \underbrace{\int_{\mathbb{R}} \tilde{c}(a) \psi_T(a) da}_{\text{occupation}} + \underbrace{\frac{1}{2} \mathbb{E} \sum_k \int \Delta_k \ell_T^a f_{R_k}''(da)}_{\text{local time}} + \underbrace{\mathbb{E} \int_0^T J(X_t, R_{t-}) d\Lambda_t}_{\text{firing intensity}}, \end{aligned} \quad (4)$$

expected flux as amplitude integrated against the three carriers named beneath it, with Proposition 1 evaluating Λ for the era model. Equating the impulse-term evaluations of (3) and (4) is the bridge between occupation of the discontinuity set and firing intensity; OF's identification argument, that firing intensity is occupation density times hazard, is the folded-coordinate form of this bridge.

The right side of (4) localises in price, and the localisation is the paper's organising object.

Definition 3 (Dissipation Measure). For a venue presentation, an era, and a horizon T , the dissipation measure Ξ_T assigns to a Borel set A the expected flux generated at price levels in A , the three carriers of (4) restricted to A in the same order,

$$\Xi_T(A) = \int_A \tilde{c}(a) \psi_T(a) da + \frac{1}{2} \mathbb{E} \sum_k \int_A \Delta_k \ell_T^a f_{R_k}''(da) + \mathbb{E} \int_0^T \mathbf{1}_{\{X_{t-} \in A\}} J(X_t, R_{t-}) d\Lambda_t.$$

The middle carrier is signed. A concentrated liquidity holding value is concave in the square-root price coordinate, so f_R'' is a negative measure there and the local-time term is a credit against the other two rather than a charge beside them. The concavity leg's cost is its negative, which is the sign in which the benchmark literature reports the same quantity. Ξ_T is a signed measure, and its total mass is an expected flux, not a sum of non-negative parts.

The forward law is the statement that expected total flux is $\Xi_T(\mathbb{R})$. Only the impulse carrier sits on $\mathcal{D}(R)$ exactly, on the trigger surfaces where firings occur. The other two sit on the supports of their densities, and $\mathcal{D}_c$ and $\mathcal{D}_f$ record where those densities break rather than where they live.

How much larger the supports are is a property of the venue class. In the V3 geometry the accrual rate vanishes identically and f_R'' is an interval density that is piecewise constant, so the local-time carrier contributes a smooth background whose only structure, the jumps of that density, sits on $\mathcal{D}_f$; in the discrete-bin geometry of Appendix B the second-derivative measure is purely atomic and its carrier sits on $\mathcal{D}_f$ entirely, while the per-bin fee credit puts the accrual carrier on the occupied bin. Dissipation living on the code's discontinuity set is therefore exact for a discrete-bin venue and exact up to a smooth background for a V3 one. The second act bounds the impulse component from below per unit time over policy classes, and the third act's instrument identifies the jump part of that component from firing data.

5.3. The central law

The impulse amplitudes that Ξ_T integrates are not modelling inputs. In the concentrated liquidity instance they obey an exact law in one state variable, the paper's central result, stated here with the frame because everything that follows consumes it. GS's Theorem 1 characterised when the re-placement cost vanishes; the law states what the cost is, always, at any width pair, the floors of the second act are its global consequence, and the jump surcharge prices jump crossings through its potentials.

Work in the sqrt-price coordinate of Section 3.2. A position $(L, [s_a, s_b])$ with the price s in range has withdrawable amounts $x = L(1/s - 1/s_b)$ and $y = L(s - s_a)$ and value $V = xs^2 + y$ in token1 units, the $L\phi$ of Section 3.2. The holdings shares are

$$\omega_0 = \frac{xs^2}{V}, \quad \omega_1 = \frac{y}{V} = 1 - \omega_0, \quad (5)$$

the token0 and token1 value shares of the withdrawn position. Writing (x_u, y_u) for a candidate range's per-liquidity amounts at the same price, its *target shares* are the same formula applied to those, $\omega'_0 = x_us^2/(x_us^2 + y_u)$ and $\omega'_1 = y_u/(x_us^2 + y_u)$. The share ω_0 is OQM's token0 value share, promoted from bookkeeping to state variable. An isolated re-placement withdraws (x, y) at price s and re-mints into an admissible range containing s through the binding-side minimum $L' = \min(x/x_u, y/y_u)$, emitting the non-binding surplus per (1). Writing V' for the value of the re-minted position at the same price, its mint kerf is the log-cost

$$k = -\ln \frac{V'}{V}. \quad (6)$$

The kerf is a log cost, and the flux it surrenders is exactly $J = V - V' = V(1 - e^{-k})$. Since $1 - e^{-k} \leq k$, the log cost dominates the value fraction and the two agree only to first order. Everything below is stated in log units, kerfs, rates, and floors alike, so a value-fraction rate from another source is not directly comparable.

Theorem 2 (Share-Potential Identity). *For an isolated re-placement at price s , from any range containing s to any admissible range containing s , at any width pair, with holdings shares (ω_0, ω_1) and target shares (ω'_0, ω'_1) as in (5),*

$$k = \max\left(\ln \frac{\omega'_0}{\omega_0}, \ln \frac{\omega'_1}{\omega_1}\right). \quad (7)$$

In particular $k \geq 0$ always, with $k = 0$ exactly when the target shares equal the holdings shares.

Proof sketch. If the token0 constraint binds, $L' = x/x_u$ and the retained value fraction is $(x/x_u)(x_us^2 + y_u)/V = \omega_0/\omega'_0$; symmetrically the token1 branch retains ω_1/ω'_1 . The mint takes

the minimum of the two, so the retained fraction is $\min(\omega_0/\omega'_0, \omega_1/\omega'_1)$, and (7) is its negative logarithm. Full algebra in Appendix D. ■

In words, for an isolated re-placement the mint kerf is exactly the price of moving the position's composition; width, centring, and the price level enter only through the shares. The venue-wide ledger of Definition 3 is signed and has three carriers; this is the one that is non-negative and event-level. Its zero set carries the first consequence.

Corollary 2 (Free Locus). *The kerf vanishes exactly on the share-preserving re-placements, at every width. These are GS's ratio-preserving locus, the zero-kerf set; they do not touch the holdings, and therefore do not move ω_0 .*

Corollary 2 settles the free-riding question, whether a policy can translate its centre along the free locus and track the price without paying. Centre motion is not the charged variable. In the isolated arithmetic a policy may slide and widen without paying a kerf, since its composition does not move, and every bound below charges composition motion alone. A deployed policy still pays gas, fees, and any corrective swap, which the swap class of Section 14 prices separately. The second act works in the law's potential form (Corollary 4), and the two-branch closed form of the next section is its boundary case $\delta' = 0$.

6. The amplitude layer

The first of three specialisations narrows the central law to one geometry, the isolated equal-width rebalance that recentres the range on the current price. Throughout, the old range is $[s_a, s_b]$ in sqrt price with half-width h and midpoint $\bar{s}$, the current sqrt price is $s = \bar{s} + \delta$ with $|\delta| \leq h$, and the new range is $[s - h, s + h]$. The amplitude is the value surrendered at the event,

$$\Delta R(\delta) = V_{\text{old}}(s) - V_{\text{new}}(s) \geq 0,$$

in token1 units, which is the geometric residual of GS in the value-surrendered sign convention adopted throughout. It carries the same content as the kerf of (6) in different units, $k = -\ln(1 - \Delta R/V_{\text{old}})$, and the floors of the second act work in k ; nothing in this section is yet a bound. The amplitude is the boundary case $\delta' = 0$ of the central law's potentials (Corollary 4).

The down-branch geometry below is not new in substance. OQM's swap-free credit publishes the token0 form of the lower branch, its sign corollary states the corner ratio, and the concurrent preprint of I.P.³³ derives the same credits independently. This section adds the assembly, the two-branch closed form in the value-surrendered frame of GS, the bridge to GS's mint-minimum development, the reconciliation of the two published boundary factors, and a monotonicity threshold that neither paper states. Proofs are in Appendix E.

Lemma 3 (Binding Side). *In the mint minimum the token1 term binds exactly when $\delta \leq 0$ and the token0 term exactly when $\delta \geq 0$.*

Proposition 2 (Two-Branch Amplitude). *The value surrendered by the equal-width recentred rebalance is*

$$\Delta R(\delta) = \begin{cases} \frac{L \delta (2\bar{s} + h + \delta)}{\bar{s} + h}, & 0 \leq \delta \leq h, \\ \frac{-L \delta (\bar{s} + \delta) (2\bar{s} + h + \delta)}{(\bar{s} + h)(\bar{s} + h + \delta)}, & -h \leq \delta \leq 0, \end{cases} \quad (8)$$

and the two branches satisfy $\Delta R_{\downarrow}(\delta) = |\Delta R_{\uparrow}^{\text{cont}}(\delta)| \cdot s/s'_b$, where $\Delta R_{\uparrow}^{\text{cont}}$ is the upper branch continued to $\delta < 0$ and $s'_b = s + h$ is the new range's upper bound.

Corollary 3 (Corner Values). *At the range corners the mint is empty and the amplitude equals the entire corner position value,*

$$\Delta R(h) = V(s_b) = L w, \quad \Delta R(-h) = V(s_a) = L w \frac{s_a}{s_b}, \quad w = 2h.$$

The boundary asymmetry factor is s_a/s_b , the corner ratio of OQM's sign corollary; the factor $\bar{s}/s_b$ stated in GS is the interior limit $\Delta R_\downarrow(m)/\Delta R_\uparrow(m) \rightarrow \bar{s}/s_b$ as $m = |\delta| \rightarrow 0^+$, consistent with GS's directional identity.

Proposition 3 (Down-Branch Monotonicity). *$\Delta R_\downarrow$ is strictly increasing in $|\delta|$ on $(0, h]$ if and only if $h/\bar{s} \leq (\sqrt{17} - 3)/2$, equivalently $s_b/s_a \leq (3 + \sqrt{17})/2 \approx 3.562$. Above the threshold the down branch attains an interior maximum and decreases to the corner value.*

In words, the value a recentred rebalance surrenders is a two-branch closed form in the displacement, the two published boundary factors are two limits of that one function, and the down branch is monotone exactly on one side of an explicit width threshold, the side the deployed ranges sit on.

Figure 2 plots both branches for a narrow and a wide range. The gap between the exact down branch and the continuation, and the turnover above the threshold, are shown; the markers at the left edge are the two corner values.²

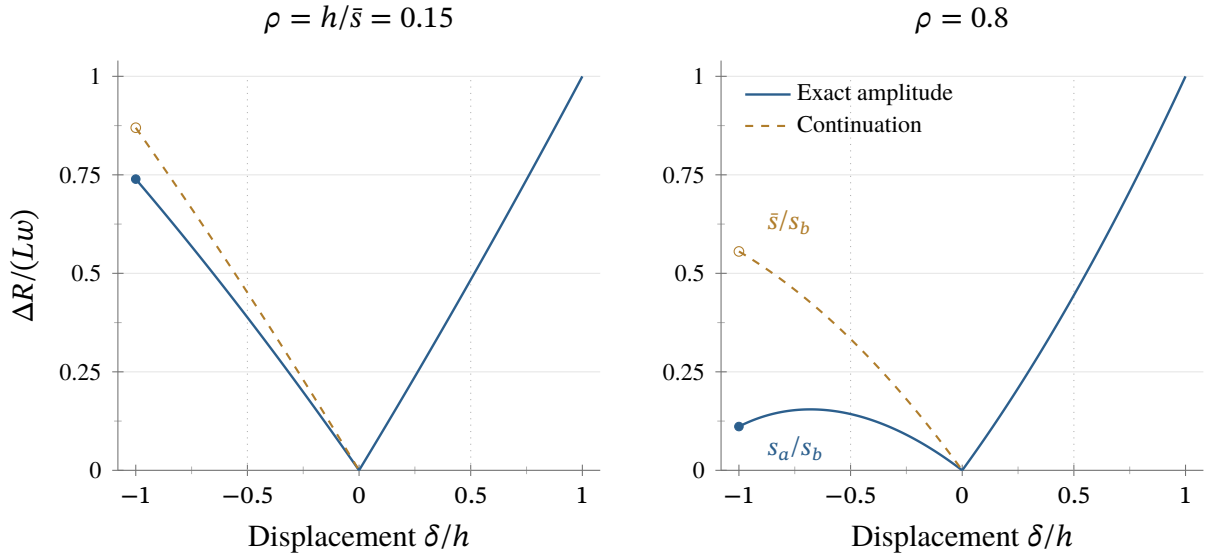


Figure 2. Two-branch amplitude (8) against normalised displacement, for a narrow range (left) and a wide range (right). Solid, the exact amplitude; dashed, the continuation of the up-branch closed form; dotted verticals, the trigger displacement at $\varphi = 1/4$.

The vanishing condition of GS's first theorem enters the present framework as the characterisation of the amplitude's null set, the ratio-preserving locus. The amplitude's sign needs no separate argument, both branches of (8) being manifestly non-negative on their stated domains; it was nonetheless established independently, as the machine-checked non-positivity of the isolated-case residual in GS's companion formalisation.³ Evaluating (8) at the trigger displacement $\delta = (1 - 2\varphi)h$

²As published, GS's monotonicity theorem asserts both branches strictly monotone with no width hypothesis, and Proposition 3 supplies the condition the down branch needs. Production ranges sit one to two orders inside the threshold, so the empirical content of GS is unaffected.

³The formalisation ships in the GS repository (g4nc0r/the-geometric-siphon, directory lean/), machine-checked against a pinned mathlib; the GS paper itself predates it. A citable archive reference is deferred to release.

plugs the amplitude layer into Proposition 1(iv). The same displacement is written xh in the floor act's policy families, so the contract fraction, the policy variable, and the displacement are one quantity, $xh = (1 - 2\varphi)h$.

7. The recovered channels

The second and third specialisations are consistency checks; each recovers a published channel law under the frame's hypotheses, and neither is a claim that the published law holds without them. The second is Proposition 1(ii), OQM's directional firing law. The placement floor enters through the barrier distances. The impulse map centres the new range at $\vartheta[a/\vartheta]$ for spacing ϑ , so the placed midpoint sits below the firing tick by the placement offset $o = \vartheta \zeta_\vartheta(a)$, the barriers become $w_u = \bar{w} + o$ and $w_d = \bar{w} - o$, and the per-pool wedge in the log count ratio is

$$\mathbb{E}[\log(w_d/w_u)] = -\mathbb{E}_o \left[\log \frac{\bar{w} + o}{\bar{w} - o} \right], \quad o \sim U[0, \vartheta], \quad (9)$$

with uniformity of o supplied by Lemma 2 on the spacing grid to the error of Section 4, which is the coarser instance and carries the slope-attenuation prediction stated there. Equation (9) is the quantisation prediction whose per-pool fit (correlation 0.82, slope 0.94) was reported by OQM; the formula itself appears here.

The third concerns the sizing arithmetic, which at a firing computes its target token0 share at the floor tick and executes at the continuous price. OQM's surplus decomposition and tick-race law, taken here as the source-derived amplitude, give the leftover side as $\text{sign}(\zeta + \Delta)$ and the credit as $g|\zeta + \Delta|V$, with ζ the within-tick offset, Δ the corrective swap's tick displacement, and g the local tick sensitivity of the target share, to the fee-and-rounding error order carried there. The forward law adds the distributional layer. The firing-state law of Proposition 1(iii) satisfies the hypothesis of Lemma 2 on the tick grid, Δ is measurable to leading order with respect to the coarse state, and therefore ζ is uniform and decoupled from Δ , both to the error of Lemma 2. The probabilities that the leftover lands opposite the bought token, conditional on direction, follow as

$$q^+ = \mathbb{E}[\min((- \Delta)_+, 1) | \text{up}], \quad q^- = \mathbb{E}[\min((1 + \Delta)_+, 1) | \text{down}], \quad (10)$$

with $z_+ = \max(z, 0)$, each expectation reading off the uniform offset's chance of losing the race. What was an assumption block in OQM, uniformity of ζ and its independence from Δ , is now a lemma plus a stated hypothesis on the firing law, and the empirical checks there (near-uniform ζ , rank correlations within ± 0.04 , q^+ of 0.06 to 0.10 against q^- near 0.5) test the hypothesis rather than stand in for a theorem.

8. The recipe

A new venue enters the forward law through Definition 1. Read the policy state space, the accrual rate, the holding value, the firing set, and the impulse map from contract source; the discontinuity set and amplitudes follow, and (2) and Proposition 1 do the rest. Discrete-bin designs place the grid in the holding value itself, whose second-derivative measure is atomic at the bin prices (Appendix B), and venues with contract-encoded movement modes make the firing set public, which permits exact scoring of blind recovery. What the frame supplies is a fixed interface; what a new venue costs is the slot filling, the event reconstruction, and whatever its own geometry demands. The discrete-bin case is the worked example of that last item, its atomic second-derivative measure being a structure the V3

development never needed.⁴ At the exact layer that claim now has an executed instance; the fork suite of Section 24 asserts the central law’s arithmetic against a second unmodified deployment of the V3 class, so the demonstrated content is portability of the pipeline across deployments. At the venue-class level, Appendix B carries out the first non-V3 instantiation, reading the Trader Joe Liquidity Book through Definition 1 from contract source with a purely atomic second-derivative measure, an instantiation asserted in turn against the venue’s unmodified home deployment (Section 24); the population read of a discrete-bin venue remains open.

9. Jump price laws and monitoring

9.1. Jump price laws

For a semimartingale with jumps, Meyer–Itô replaces Itô–Tanaka in Theorem 1, and the right side of (2) gains the jump-remainder sum

$$\sum_s [f(X_s) - f(X_{s-}) - f'_-(X_{s-}) \Delta X_s]$$

over jump times, the local time being that of the continuous part.³⁸ The sum is pathwise and uncompensated; under the integrability of Section 5 its expectation is evaluated through the compensator of the jump measure. A jump across a boundary accumulates no local time and contributes through this sum. A jump across a trigger surface fires the impulse from the landing point, so the firing compensator splits into a continuous-crossing part, governed by first passage as above, and a jump-crossing part, an integral of the jump measure over straddling jumps (Appendix C). The overshoot past the trigger line at firing is therefore an observable of the jump measure, once the monitoring overshoot of the next subsection is accounted for. Reading integrated jump-tail functionals from population overshoot data, conditionally and per pool rather than recovering the measure itself, is the inverse problem the third act pursues (Sections 18–22).

9.2. Monitoring

Deployed systems evaluate their firing condition at check times (t_j) and fire at the first check with $X_{t_j} \in F(R)$. The proposition below is stated for check times independent of X , deterministic cadence and Poisson checking both admitted; the activity-dependent Cox case, in which the check intensity responds to the observable state, is carried in the third act under the exclusion restriction (F4) of Table 2. Three consequences follow.

Proposition 4 (Monitoring Layer). *Under the era model with check interval Δt and one-check scale $\sigma_1 = \sigma\sqrt{\Delta t}$,*

- (i) *conditional on firing at a surface b , the firing state is $b + O$ with the overshoot O admitting, under the flat-entry hypothesis of Appendix C, the density $\sqrt{2\pi} Q(o/\sigma_1)/\sigma_1$ with Q the standard normal survival function, of total variation $\sqrt{2\pi}/\sigma_1$ and mean $\approx 0.63 \sigma_1$, so the offset-uniformity error at grid ϑ is of order ϑ/σ_1 , degrading to the degenerate trigger offset as $\Delta t \rightarrow 0$;*
- (ii) *the realised firing intensity is, at leading order, that of the continuous problem with each barrier shifted outward by $\beta \sigma_1$ with $\beta \approx 0.5826$, the continuity-correction constant of Broadie, Glasserman, and Kou,³⁹ a modest cadence thinning; where realised firing rates sit well below the raw first-passage prediction, the gap is carried by dead-band and dwell rules, which modify the firing set rather than the monitoring;*

⁴Three further classes fill the slots without new theory. Hook architectures put all five in operator-authored code; repegging gates with hysteresis are firing sets with state-dependent boundaries; and liquidation engines put atoms in the holding value’s second-derivative measure, lighting the local-time channel directly rather than through impulses.

- (iii) the correction to the direction split of Proposition 1(ii) is bounded by the probability of an unobserved excursion to the opposite barrier within one check interval, exponentially small in $(w_u + w_d)/(\sigma\sqrt{\Delta t})$.

In words, discrete monitoring makes every trigger-qualified firing overshoot its line by a known density at the one-check scale, thins the firing rate only mildly, and almost never records a firing on the wrong side. Firings that a dwell or penetration rule has filtered are not drawn from this kernel, which the third act exploits.

Item (i) is the structural fact announced in the introduction. The grids the value channel is measured on come from placement and sizing arithmetic, but the offset on them is random rather than degenerate only because monitoring is discrete, and item (i) quantifies the cadence prediction of Section 4. Item (iii) is why slope-one fits of the directional law survive on production data whose levels are heavily thinned.⁵

ACT II. THE FLOOR

The second act prices every admissible policy from below. The central law makes each re-placement a potential difference in the composition coordinate, so keeping a price in a band has a running cost no policy design evades; three floors state it, and one architecture deletes it. The act consumes the first act's law and geometry, and deployed observation returns in the menu and cadence terms.

10. Policies, classes, and the functional

The second act bounds from below the kerf a policy accumulates per unit time, over a base class and its two independent instrument enrichments, and then treats a fourth architecture that changes the surplus rule rather than the instrument set. The functional is the impulse leg of the dissipation measure (Definition 3) in log units rather than in value units, a distinction Section 10.2 makes exact.

10.1. Setting

Work per era, on the filtered space of Section 3.1, in the sqrt-price coordinate, where the mint geometry and the composition share of Section 5.3 are native. Within an era the sqrt price s_t is a continuous local martingale with $d\langle s \rangle_t = \sigma_s^2 dt$ and σ_s era-constant, confined to the era band $[s_-, s_+]$, a deterministic interval carried as era data and containing every value the stopped era path visits; the integrability convention of Section 5 holds on it. Monitoring is continuous; the cadence layer returns in Proposition 11. A position is a pair $(L, [s_a, s_b])$ with the price in range, with withdrawable amounts, value, and holdings shares (ω_0, ω_1) as in Section 5.3. The narrow-range parameter is $\rho = h/\bar{s}$ for a range of half-width h and midpoint $\bar{s}$; statements invoking the narrow limit $\rho \rightarrow 0$ say so, and the standing hypothesis $h/s_- \leq 1/5$ is assumed throughout the floor sections.

The state coordinate of the second act is the share ω_0 of (5); between firings it evolves continuously with the price and is strictly decreasing in s , and in the narrow limit $\omega_0 \approx (s_b - s)/(s_b - s_a)$.

10.2. Policies and cost

Definition 4 (Policy and Kerf Rate). A policy is a sequence of stopping times $\tau_1 < \tau_2 < \dots$, locally finite, and $\mathcal{F}_{\tau_j}$ -measurable re-placement choices. At τ_j the position is withdrawn and re-minted per the isolated re-placement of Section 5.3, at per-event kerf k_j as in (6), non-negative by Theorem 2.

⁵The same discreteness has been priced on the other leg. Fixed block intervals admit a closed-form per-block benchmark loss, and constant spacing minimises it over inter-block timing patterns;⁴⁰ here discreteness of observation moves the event leg instead, and moves it upward.

The kerf rate of the policy is its accumulated kerf per unit time,

$$r = \limsup_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \sum_{\tau_j \leq T} k_j. \quad (11)$$

The log form compounds, and the numerical harness measures it. The mean identity (3) prices the impulse leg in value units, $\mathbb{E} \sum J_j$ with $J_j = g_j V_{\tau_j-}$ and $g_j = 1 - e^{-k_j}$, so the two accounts agree to first order and satisfy $0 \leq g_j \leq k_j$ exactly. That inequality runs against a lower bound rather than with it, the emitted fraction sitting below the kerf, so a floor on r does not on its own bound the value-flux rate. On events whose kerf is capped, $k_j \leq K$, the comparison is restored with a deflation factor, since $g_j \geq K^{-1}(1 - e^{-K})k_j$ there; a floor $r \geq r_0$ then gives a value-flux rate of at least $K^{-1}(1 - e^{-K})r_0$ times the era's lowest position value. No such cap is free in $\mathcal{A}_0$, a firing at the range boundary surrendering the whole one-sided position, with $g \rightarrow 1$ and $k \rightarrow \infty$. The capped reading is therefore the only value-unit statement claimed here, and every floor below is stated in log units. Nothing below re-prices the concavity leg (Theorem 1); Proposition 12 compares the two.

Definition 5 (Admissible Classes). $\mathcal{A}_1$ and $\mathcal{A}_2$ are two independent enrichments of the base class $\mathcal{A}_0$, each adding one instrument. They are not comparable, and their join, the swap-mediated variable-width class, is treated in Theorem 5. The fourth changes the surplus architecture rather than the instrument set. $\mathcal{A}_0$ (fixed width, isolated) fixes the half-width h , keeps the price in range (a firing occurs at or before every boundary hit), allows re-placement to any equal-width range containing the price, and emits the surplus per the isolated re-placement of Section 5.3. $\mathcal{A}_1$ (corrective swaps) additionally allows an in-pool swap at any impulse, under the execution model of Assumption 1. $\mathcal{A}_2$ (variable width, isolated) allows the width to change at impulses subject to $w \leq W_{\max}$, with emission unchanged. $\mathcal{A}_3$ (retention) replaces emission by retention in a ledger available to later mints, the architecture of ME.²

The band-maintenance requirement of $\mathcal{A}_0$, inherited by every class, is a substantive restriction rather than bookkeeping. A mandate free to let the price leave the range escapes every bound below; the floors price the service of keeping a price in a band, not the holding of a position.

Assumption 1 (Execution Model). A corrective swap that moves the holdings share by $|\omega' - \omega|$ has notional $N_1 = |\omega' - \omega| V$, pays the pool fee η on that notional and a fixed per-event gas cost G_{gas} , and incurs price impact per the constant-liquidity law through the footprint ratio $I = V/(sL_p)$, with L_p the pool's liquidity.

The notional geometry and the impact form are established empirically in OQM;³ they enter here as stated hypotheses, and every $\mathcal{A}_1$ bound is conditional on them.

11. The cost structure

The floor consumes two ingredients, the slope of the cost at zero displacement, which makes the floor non-trivial, and the potential form of the central law. The two-branch amplitude of Proposition 2, reconciling GS's closed form with OQM's swap-free credit, gives the value surrendered by the centred equal-width rebalance at displacement δ from the old midpoint; write $g(\delta)$ for that amplitude as a fraction of position value.

Lemma 4 (Exact Fractional Slopes). $g'(0^+) = 1/h$ exactly, and $g'(0^-) = -(\bar{s}/s_b)(1/h)$ exactly.

The left derivative is negative because the down-branch amplitude falls as δ rises towards zero, the cost itself being non-negative on both sides. A cost first-order in displacement has the local form of a proportional transaction cost, and that local form forces discrete action, since a continuously tracking policy would pay along the price path's total variation, which is infinite for a diffusion. The

novelty against the transaction-cost literature is that Lemma 4 is computed from mint arithmetic, not assumed. The second ingredient is Theorem 2 specialised to the equal-width family, which makes the cost a potential difference in the displacement coordinate. The potentials that follow are not posited. Since $k(\delta \rightarrow 0) = -\ln(1 - g(\delta))$ on each branch, each is the negative logarithm of that branch's retention ratio, fixed up to an additive constant.

Corollary 4 (Placement-Family Potentials). *For equal-width re-placement at price s , from displacement δ (of the price from the old midpoint) to displacement δ' (of the price from the new midpoint), with $P = 2s + h$,*

$$k(\delta \rightarrow \delta') = \max\left(\mathcal{C}^\uparrow(\delta) - \mathcal{C}^\uparrow(\delta'), \mathcal{C}^\downarrow(\delta) - \mathcal{C}^\downarrow(\delta')\right), \quad (12)$$

$$\mathcal{C}^\uparrow(z) = \ln \frac{hP - z^2}{h - z}, \quad \mathcal{C}^\downarrow(z) = \ln \frac{hP - z^2}{(h + z)(s + h - z)}. \quad (13)$$

The potentials depend on the era price s as a parameter, suppressed in the notation. $\mathcal{C}^\uparrow$ is strictly increasing and $\mathcal{C}^\downarrow$ strictly decreasing on $(-h, h)$ under the standing hypothesis, so upward centre translations pay the $\mathcal{C}^\uparrow$ increment, downward ones the $\mathcal{C}^\downarrow$ increment, and costs telescope along same-direction sequences. Every re-placement that moves the displacement is strictly costly, one of the two increments being strictly positive whichever way it moves, so no sequence returning to its starting displacement is free. Setting $\delta' = 0$ recovers the two-branch amplitude on both branches. In the narrow limit the potentials reduce to $-\ln(h - \delta)$ and $-\ln(h + \delta)$ up to constants.

Corollary 4 is the floors' working form. Costs are potential differences, so they telescope along monotone sequences and admit no free round trips, and a lower bound on the dissipation rate becomes a corridor condition on one derivative, a function squeezed between the envelope slopes. The next three sections run that verification argument three times, in the displacement coordinate at fixed width, in the share coordinate at any width, and in the share coordinate again with execution costs in place of mint prices for the swap class.

12. The fixed-width floor

Fix the class $\mathcal{A}_0$ and write $\delta_t = s_t - \bar{s}_t \in [-h, h]$ for the displacement from the current midpoint, so $d\delta = ds$ between impulses. The floor is proved by a verification function in the displacement coordinate. Let $E^\uparrow$ and $E^\downarrow$ denote the era-uniformised envelope slopes, the infimum over the era band of $(\mathcal{C}^\uparrow)'$ and a supremum bound for $(\mathcal{C}^\downarrow)'$, written out in Appendix F, and define

$$c_{\min}(h; s_-, s_+) = \frac{h^2}{2} \inf_{-h < \delta_1 < \delta_2 < h} \frac{E^\uparrow(\delta_2) - E^\downarrow(\delta_1)}{\delta_2 - \delta_1}. \quad (14)$$

Theorem 3 (Fixed-Width Floor). *Every $\mathcal{A}_0$ policy satisfies*

$$r \geq c_{\min}(h; s_-, s_+) \frac{\sigma_s^2}{h^2}, \quad \text{with} \quad c_{\min} \geq 2\left(1 - \frac{3h}{s_-}\right), \quad (15)$$

and $c_{\min} \rightarrow 2$ in the narrow limit.

Proof sketch. A function U on $[-h, h]$ with $U'' \geq 2c/h^2$ as a measure and $U(\delta) - U(\delta') \leq k(\delta \rightarrow \delta')$ for every admissible impulse certifies $r \geq c \sigma_s^2/h^2$, by Itô between impulses, the impulse inequality at them, and telescoping; chattering needs no exclusion since every impulse is priced individually and costs are non-negative. By Corollary 4 the impulse inequality is equivalent to the pointwise corridor $E^\downarrow \leq U' \leq E^\uparrow$, and (14) is the largest c the corridor admits; the extremal U is built as a

supremum of affine functions and has bounded derivative, so the martingale term is genuine. In the narrow limit the corridor is

$$-\frac{1}{h+\delta} \leq U' \leq \frac{1}{h-\delta},$$

and the extremal function is the quadratic $U = 2\delta^2/h^2$, whose derivative $4\delta/h^2$ is tangent to both envelopes at $\pm h/2$, the tangency inequality $4\delta(h-\delta) \leq h^2$ being $(h-2\delta)^2 \geq 0$. The analytic corollary follows from tangent-line convexity with the era corrections bounded by $3/s_-$. Full assembly in Appendix F. ■

Three constants sit in (15) and are not the same object. The first, $c_{\min}$, is the largest constant the corridor admits at the stated width and era band, and carries the class bound. The closed form $2(1-3h/s_-)$ is an analytic under-estimate of it, safe but crude. The narrow-limit value two is sharp as an infimum by Proposition 7, and at finite ρ that sharpness is numerically supported rather than proved.

Proposition 5 (Pathwise Discrete Floor). *Let a policy be monitored at steps $t = 0, 1, \dots, n-1$. Write δ_t for the displacement entering step t and δ'_t for the displacement leaving it, the price having moved between them, and let the impulse between step t and step $t+1$ carry δ'_t to δ_{t+1} at kerf $k_t \geq 0$, with $k_t = 0$ and $\delta_{t+1} = \delta'_t$ when the policy does not fire. Suppose the verification function U is quadratic, so that U'' is constant, and satisfies the impulse inequality $U(\delta'_t) - U(\delta_{t+1}) \leq k_t$ at every t , and suppose the drift pairing is non-negative in aggregate, $\sum_{t < n} U'(\delta_t)(\delta'_t - \delta_t) \geq 0$. Then*

$$\sum_{t < n} k_t \geq \frac{1}{2} U'' \sum_{t < n} (\delta'_t - \delta_t)^2 - [U(\delta_n) - U(\delta_0)], \quad (16)$$

pathwise, with no expectation taken. The extremal $U = 2\delta^2/h^2$ of Theorem 3 gives $\frac{1}{2}U'' = 2/h^2$; the same statement holds in the share coordinate at $U(\omega) = 8(\omega - \frac{1}{2})^2$, where the impulse inequality is exact at every width (Section 13), and in the swap class at the constant of Theorem 5.

In words, under a quadratic certificate and a non-negative drift pairing the floor holds path by path, the same corridor and impulse inequality bounding the accumulated kerf below by the class constant times the realised quadratic variation, with no Itô step and no expectation.

The proposition is not a second cadence charge. Proposition 11 prices what cadence adds to the rate within the continuous frame; (16) is the floor itself, restated for a trajectory that is only ever observed at its monitoring times. What the continuous statement supplies that this one does not is the passage to a rate, since the boundary term is discarded only in the long run and the per-step quadratic increments must be aggregated into σ_s^2 ; what this one supplies is that the mechanism needs none of that machinery. Since deployed operators monitor on a cadence, it is also the closer statement to the object. It is proved in Appendix F and is machine-checked in full (Section 25.2), where the continuous assembly is not.

Proposition 6 (Achievability). *The centred policy that fires at $|\delta| = xh$ and re-replaces at the midpoint has, exactly for the driftless era model,*

$$r(x) = c(x) \frac{\sigma_s^2}{h^2}, \quad c(x) = \frac{-\frac{1}{2} \ln(1 - g_\uparrow(x)) - \frac{1}{2} \ln(1 - g_\downarrow(x))}{x^2}, \quad (17)$$

where $g_\uparrow(x) = \Delta R_\uparrow(xh)/V(\bar{s})$ and $g_\downarrow(x) = \Delta R_\downarrow(-xh)/V(\bar{s})$ are the branch event fractions of (8) and the two halves are the equal up- and down-firing probabilities of the driftless symmetric era. In

the narrow limit $c(x) = -\ln(1-x)/x^2$, minimised at $x^* = 0.7153$ (the root of $x/(1-x) + 2\ln(1-x) = 0$) with $c^* = 2.4554$.

In words, every fixed-width policy pays at least $c_{\min} \sigma_s^2/h^2$, which is two in the narrow limit, and there the best centred policy pays $2.455 \sigma_s^2/h^2$, a fifth above a floor the next proposition shows to be sharp in that limit. Numerically $c_{\min}(\rho) = 1.9522$ at $\rho = 0.05$, 1.9804 at $\rho = 0.02$, and 1.9950 at $\rho = 0.005$, behaving as $2(1 - \rho)$. Figure 3 shows the corridor and both policy families against the floor. The floor scales as $1/h^2$, so concentration makes band maintenance expensive, and the production trigger convention $x = 0.5$ (trigger fraction one quarter³) pays $c(0.5) = 2.77$ in the narrow limit, 13% above c^* .

Proposition 7 (Sharpness of the Floor). *In the narrow limit the corridor floor of Theorem 3 is sharp. For $x \in (0, 1)$ and $\varepsilon > 0$, the two-sided reflection policy that fires at $|\delta| = xh$ and re-replaces with the new displacement $(x - \varepsilon)h$ on the firing side has narrow-limit kerf rate $c_{\text{refl}}^\varepsilon(x) \sigma_s^2/h^2$ with*

$$c_{\text{refl}}^\varepsilon(x) \longrightarrow c_{\text{refl}}(x) = \frac{1}{2x(1-x)} \quad (\varepsilon \rightarrow 0), \quad (18)$$

minimised at the tangency displacement $x = 1/2$ with $c_{\text{refl}} = 2$. Hence the infimum of the narrow-limit rate over $\mathcal{A}_0$ equals the floor $2 \sigma_s^2/h^2$; it is approached, not attained, within locally finite impulse policies.

Proof sketch. Renewal reward on the reflection cycle, using the exact Brownian exit law and the narrow-limit potentials of Corollary 4; each firing costs $\varepsilon/(1-x)$ at leading order on either side, and the mean cycle is $2x\varepsilon h^2/\sigma_s^2$. The lower bound is Theorem 3. Full computation in Appendix F. ■

In words, the floor of two is the exact narrow-limit price of band maintenance; reflecting the displacement at the tangency points $\pm h/2$ by vanishing corrections approaches it, and no policy of the class goes below it. Call this two the *tangency constant*, since it is the value at which a quadratic verification derivative can be tangent to both envelope slopes at once. The extremal structure is singular control. The quadratic verification function satisfies the Itô inequality with equality everywhere, so the only slack in the corridor argument is the impulse inequality, which vanishes for infinitesimal moves exactly at the tangency points; the reflection family closes that slack, its firing rate diverging as $1/\varepsilon$ in the limit, the same boundary phenomenon Remark 3 prices at the corner, where the cost of reflection is instead infinite. The centred renewal family of Proposition 6 cannot approach the infimum because its single shared target makes every far-side excursion pay a large-move cost, per-side targets being the missing instrument; $c^* = 2.4554$ remains the optimum of that family, a fifth above the floor. Away from the narrow limit the sharpness is numerically supported rather than proved; a policy iteration on the displacement grid with no structural restriction converges to the discrete reflection and reproduces the corridor constants at exact kernels, 1.9803 against $c_{\min} = 1.9804$ at $\rho = 0.02$, and no policy among the families and random draws tested on the calibration surface of Section 24 undercuts the floor for its visited band.

Remark 3 (Corner Riding). A policy that pins a range corner to the price by free slides pays nothing per slide, the holding being single-sided at the corner so that re-minting preserves the ratio and lands on the zero-kerf set of Corollary 2, but it does not track for free. The potential $\mathcal{C}^\dagger$ blows up logarithmically at the corner, and the ε -interior version of the slide (fire at $\delta = h - \varepsilon$, re-place to $h - 2\varepsilon$) pays $\ln 2$ per event at cycle length of order $h\varepsilon/\sigma_s^2$, a rate of order $\sigma_s^2/(h\varepsilon)$, diverging as $\varepsilon \rightarrow 0$. Corner riding is the singular-control boundary of the class and is infinitely expensive rather than free; the verification argument prices it through the impulse inequality with no special case.

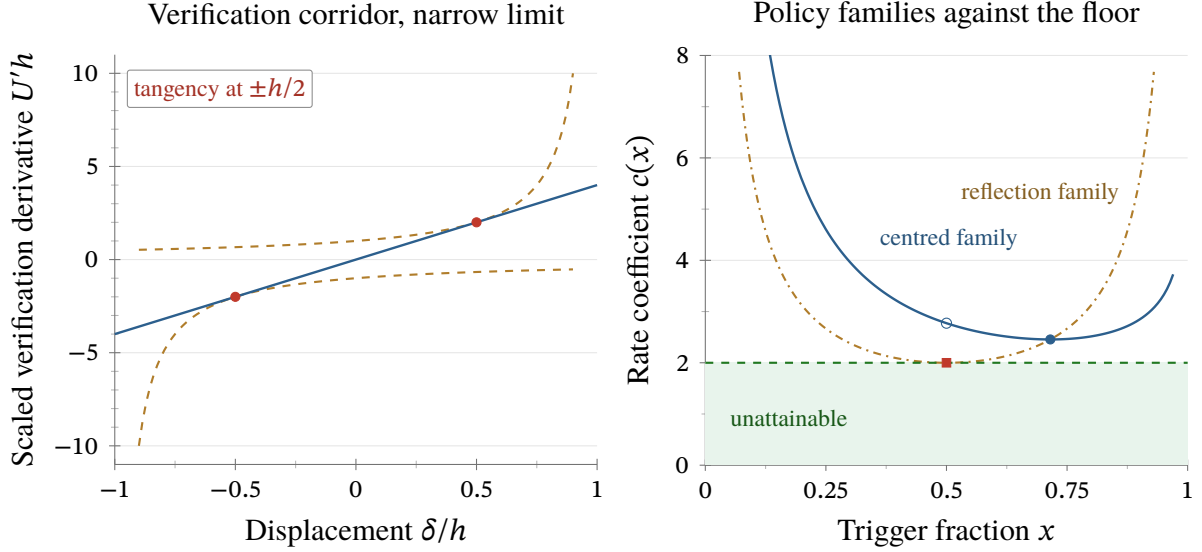


Figure 3. Left, the narrow-limit corridor between the envelope slopes (dashed) and the extremal verification derivative $4\delta/h^2$ (solid), tangent at $\pm h/2$. Right, the centred coefficient $c(x)$ (solid) and the reflection coefficient $c_{\text{refl}}(x)$ (dash-dotted) against the floor at two, with the centred optimum (filled), the production convention (open), and the reflection limit sitting on the floor (square).

13. The width-uniform floor

The second theorem removes the fixed-width hypothesis. Free moves preserve ω_0 at every width by Corollary 2, paid moves are priced by Theorem 2 at every width, and between impulses the share diffuses with the price at a rate the width controls. Write $\rho_W = W_{\max}/s_-$.

Lemma 5 (Clock and Drift Bounds). *Within a range $[a, b]$ of width $w = b - a$ containing the price, $\phi(s, a, b) \leq w$, and*

$$\omega'_0(s) = -\frac{(s-a)^2 + a(b-a)}{b\phi(s, a, b)^2}, \quad |\omega'_0| \geq \frac{a/b}{w}, \quad |\omega''_0| \leq \frac{12b^2}{a^3 w}. \quad (19)$$

Theorem 4 (Width-Uniform Floor). *Every $\mathcal{A}_2$ policy, variable in width and isolated, with width cap $W_{\max}$ and $\rho_W \leq 1/5$, satisfies*

$$r \geq \frac{8\sigma_s^2}{W_{\max}^2}(1 - K_0\rho_W) = \frac{2\sigma_s^2}{h_{\max}^2}(1 - K_0\rho_W), \quad h_{\max} = W_{\max}/2, \quad (20)$$

with K_0 an absolute constant; $K_0 = 10$ suffices on the bounds of Lemma 5 and is not optimised.

Proof sketch. Run the verification argument on the state $\omega = \omega_0 \in (0, 1)$ with $U(\omega) = 8(\omega - \frac{1}{2})^2$. The impulse inequality is the corridor $-1/\omega \leq U' \leq 1/(1 - \omega)$ against the share potentials of Theorem 2, with tangency at $\omega = 1/4$ and $3/4$, the inequalities $(4\omega - 1)^2 \geq 0$ and $(4\omega - 3)^2 \geq 0$; this part is exact and carries no narrow-limit correction at any width. The path of ω is continuous across free impulses, its quadratic variation is at least $(1 - \rho_W)^2\sigma_s^2/W_{\max}^2$ per unit time by Lemma 5 and the cap, and its drift is $O(\rho_W)$ against the clock term by the second-derivative bound. Assembly in Appendix F. ■

In words, variable width buys nothing below the fixed-width floor at the cap; small widths only speed the composition clock and raise the bound, so no lower width bound is needed within the

stated era and cap regime. The two ingredients are of different kinds. The share corridor is exact at every width, being Theorem 2 itself, and the factor $1 - K_0 \rho_w$ is entirely the cost of converting price variation into the share clock. The consistency check is exact, since for fixed width the ω -form reproduces the δ -form constant. The tangency constant is therefore not an artefact of one coordinate; the same argument in the share coordinate, tangent at $\omega = 1/4$ and $3/4$ rather than at $\pm h/2$, returns the same two. The harness's widen-slide family, policies that widen free towards the cap and pay only to slide back, never beats (20), and its best member pays above the best fixed-width-at-cap policy (Section 24).

14. The swap-mediated floor

Production operators do not surrender the mint surplus; they swap first. The class $\mathcal{A}_1$ prices an impulse that moves the holdings share from ω to ω' at

$$k \geq \underbrace{\eta |\omega' - \omega|}_{\text{fee}} + \underbrace{I(\omega' - \omega)^2}_{\text{impact}} + \underbrace{\gamma}_{\text{gas}}, \quad I = \frac{V}{sL_p}, \quad \gamma = \frac{G_{\text{gas}}}{V}, \quad (21)$$

the three legs of Assumption 1 on the notional N_1 . Dust-mediated motion costs at least one log-unit per unit of share by Theorem 2, against the swap leg's fee scale, and mixed moves price leg by leg, so (21) covers every $\mathcal{A}_1$ impulse (Appendix F).

Theorem 5 (Swap-Mediated Floor). *Every $\mathcal{A}_1$ policy at fixed width w satisfies*

$$r \geq A(\eta, \gamma, I) \frac{(1 - \rho_w)^2 \sigma_s^2}{w^2}, \quad \rho_w = \frac{w}{s_-}, \quad A = \min_{d \in (0,1)} \frac{\gamma + \eta d + I d^2}{d(1-d)}. \quad (22)$$

The swap-mediated variable-width class, the join of $\mathcal{A}_1$ and $\mathcal{A}_2$, satisfies the same bound at $w = W_{\max}$. In the two tractable regimes,

$$A = \eta \quad (\gamma = I = 0), \quad A = \eta + 2\sqrt{\gamma\eta} + O(\gamma) \quad (I = 0, \gamma \ll \eta),$$

the second with minimiser $d^ = (\sqrt{\gamma^2 + \gamma\eta} - \gamma)/\eta$.*

Proposition 8 (Return-Point Achievability). *The renewal policy that fires at $|\delta| = xh$ and corrects back by mh on the same side pays*

$$r(x, m) = \frac{\sigma_s^2}{h^2} \cdot \frac{\eta m/2 + \gamma + I m^2/4}{m(2x - m)}, \quad (23)$$

decreasing in the trigger x , so the trigger optimum is the band edge and the optimisation lives in the correction size m . In the fee-only case, $r(1, m) \rightarrow \eta \sigma_s^2/w^2$ as $m \rightarrow 0$; the sandwich of Theorem 5 is tight, with no gap. With gas the optimal correction is $m^ \approx 2\sqrt{\gamma/\eta}$, the Whalley–Wilmott scaling.*

In words, with swaps, and on the execution objective alone, the cheapest way to hold a band is to fire as late as possible and correct as little as possible, and the cost of doing so approaches fee times volatility squared over width squared, an infimum rather than a rate any finite correction attains. Edge-hugging is admissible here because the fee cost is linear rather than logarithmic; the corner blow-up of Remark 3 does not occur. It is a cost-side idealisation, and Proposition 11 and revenue considerations move the practical optimum inward.

Three consequences follow. First, late firing, which the population measurements of OF find implemented as dwell and penetration rules, is cost-optimal in direction, there being no interior trigger optimum from fee, impact, and gas alone. Second, the deployed convention (fire at $x = 0.5$, correct fully to centre) pays $(\eta + 4\gamma) \sigma_s^2/h^2$, between three and four times the floor at production parameters. Table 1 works the numbers at the standing production anchors, price volatility 80 % per square-root year so $\sigma_s/\bar{s} = 0.40$, $\rho = 0.02$, $\eta = 10^{-4}$, gas \$0.03 per event. Third, the floor is a benchmark. For a position of width w , realised fractional dissipation per unit time (fees plus gas plus impact plus residual dust, over mean value) divided by $A(1 - \rho_w)^2 \hat{\sigma}_s^2/w^2$ is a dimensionless score with the floor at one; the score prices the correction-size and width choices, and does not price the trigger, whose late-firing direction is already optimal. The isolated-class optimum $x^* = 0.7153$ of Proposition 6 is not the reference for any swap-mediated operator.

Table 1. The swap-mediated floor at production anchors.

	$V = \$10k$	$V = \$100k$	$V = \$1M$
$\gamma = G_{\text{gas}}/V$	3.0×10^{-6}	3.0×10^{-7}	3.0×10^{-8}
d^*	0.146	0.052	0.017
A	1.41×10^{-4}	1.12×10^{-4}	1.04×10^{-4}
floor $A \sigma_s^2/w^2$	1.41 %/yr	1.12 %/yr	1.04 %/yr
deployed convention	4.5 %/yr	4.05 %/yr	4.0 %/yr

15. The retention collapse

The floors so far assume the surplus is lost. The last theorem-grade statement concerns the architecture that keeps it. In $\mathcal{A}_3$ the state carries standing balances (x_d, y_d) ; an impulse mints from $\hat{x} = x_w + x_d - x_{\text{sw}}$ and $\hat{y} = y_w + y_d - y_{\text{sw}}$ through the same binding-side minimum, and the surplus is credited back to the balances rather than emitted, the ledger-inclusive mint identity established in ME.² Define the system value $V_{\text{sys}} = V + V_d$, position plus ledger, with the ledger valued at the impulse price.

Proposition 9 (Retention Collapse). *In $\mathcal{A}_3$, the impulse being atomic so that withdrawal and re-mint price at one s , V_{sys} is conserved at every swap-free impulse, and at a swap-corrected impulse its loss is the swap’s fee, impact, and gas under Assumption 1. The impulse leg of the dissipation functional on V_{sys} therefore carries no term of order σ_s^2/h^2 or $\eta \sigma_s^2/w^2$; the floors of Theorems 3, 4, and 5 are deleted, not lowered.*

Proof. Token conservation at the impulse is the arithmetic of ME’s jump theorem, whose credited balances sum with the mint’s consumption to the available amounts token by token; valuing both sides at the impulse price gives value conservation, and the swap legs are priced as in (21). No verification function is needed. ■

In words, an architecture that keeps the residuals inside the system carries no emission term in the impulse leg of its system value, only the execution costs of its corrective swaps; the floor is a property of emission, and retention deletes it from that leg rather than lowering it. The concavity leg and the exposure of the retained inventory are untouched. The emission precondition is the one GS established with a stock-manager fork test and ME formalised, and the concurrent topology preprint³³ gives the multi-pool circulation condition for it. Three residual costs survive retention. The ledger is idle inventory with one-sided price exposure, an opportunity cost the functional does not price; forcing its redeployment through swaps pays the $\mathcal{A}_1$ fee scale on the recycled flux; and the concavity leg is untouched by architecture. For the Siphonic design, retention collapses the rebalancing dissipation floor to fees, gas, and execution impact.

16. The menu and the cadence term

The floors take the venue's rounding rule and the operator's monitoring as given. Two propositions price them as additive terms, the first designer-facing, the second operator-facing. Both consume the offset and monitoring machinery of Sections 4 and 9.

Proposition 10 (Quantisation Menu). *Let the impulse map centre ranges on a grid of spacing ϑ and size at the floor tick, and let the firing-state offset be uniform to the error of Lemma 2. Then the rate carries two additive terms conditional on the deployed rule.*

- (i) *Placement. The re-placement lands off-centre by the placement offset $o \in [0, \vartheta)$, so the rate is the renewal family's at $\delta_0 = o$ averaged over the offset law; at the family optimum the penalty is second-order in ϑ , and at a deployed non-optimal configuration it carries the first-order term $\partial r / \partial \delta_0 \cdot \vartheta / 2$, signed by the one-sidedness of the floor.*
- (ii) *Sizing. Floor-tick sizing adds the per-event expected fraction $g \mathbb{E}[\zeta + \Delta]$ of OQM's value channel, a rate term $\lambda g \mathbb{E}[\zeta + \Delta]$; nearest-tick sizing halves the offset contribution and continuous-price sizing deletes it.*

In words, where the range is allowed to land and how its size is rounded are two priced lines of the rate, and each deployed rule carries its own per-event price, additive at the offset and renewal conditions of the proposition. Rounding is thereby a policy instrument.

Proposition 11 (Cadence Term). *For an operator whose trigger sits at displacement xh , firing at the first check after the crossing, with check interval Δt , the firing displacement carries the flat-entry overshoot with mean $\sqrt{2\pi} \sigma_1 / 4 \approx 0.63 \sigma_1$ of Proposition 4(i), and to leading order the rate gains the additive term*

$$\lambda [k'(xh) \mathbb{E}[O] - k(xh) \Theta],$$

the overshoot priced along the envelope slope against the cadence thinning credit Θ of the barrier-shift correction; the first term dominates in the isolated class and the two are comparable in the swap class.

In words, an operator who checks every Δt pays for the displacement the price builds past the line between checks, less a small credit from the crossings the gap lets escape. The population measurements of OF find most production firing gated by dwell and penetration rules rather than cadence, with overshoot roughly invariant in actuation delay, which is a chosen trigger shift rather than a monitoring cost. A deterministic penetration threshold re-parametrises the trigger, which Proposition 8 makes cheaper per event in the swap class, and whose benefit is the avoided round-trip cost of whipsawed firings, both potential increments of Corollary 4. Pricing that trade in full requires the out-of-range leg of the dissipation identity and is left to future work on the churn accounting.

17. The two legs compared

The comparison between the two legs reverses across architectures, which refines the additivity reading of OQM.

Proposition 12 (Leg Comparison). *For a centred narrow range the concavity leg runs at fractional rate approximately $\sigma_s^2 / (2\bar{s}h)$. The isolated-class floor exceeds it by the factor $4/\rho$ at the floor constant, two orders of magnitude at production concentration, while the swap-mediated floor is smaller than it by approximately $\eta/(2\rho)$, roughly $1/400$ at the production anchors.*

In words, in the isolated architecture the event leg is parametrically the leading cost, while for swap-mediated operators the event floor is a small correction to the concavity cost, which is why

the floor is a benchmark rather than a headline expense. The reversal is the practical content of the architecture choice; an isolated venue's operators face the event leg as their dominant cost, a swap venue's operators face it as a controllable correction, and only retention removes it from the ledger altogether (Section 15).

ACT III. THE INSTRUMENT

The third act inverts the law. A deployed trigger line is a threshold detector for the price process, and the population's firing records are read, through the jump and monitoring layers of Section 9, for the diffusive scale and the integrated jump tails the operators themselves face; the naive reading fails, and the conditioned reading is the instrument, and it is conditional throughout, on the geometry census, the era, the monitoring admissions, and the discreteness cut. Notation is the first act's; Appendix A collects it.

18. The instrument

The third act derives a conditional inverse map from the firing records of the deployed population to the jump part of the dissipation measure's impulse component (Definition 3). What it identifies is a family of integrated jump-tail functionals per pool, under the geometry, era, monitoring, and discreteness admissions, and not the jump measure itself.

18.1. Price, detectors, observables

Within an era the price coordinate follows

$$X_t = x_0 + \mu t + \sigma B_t + X_t^J, \quad (24)$$

with B standard Brownian motion and X^J a pure-jump part whose compensator is time-homogeneous per era, $\nu^X(dt, dz) = dt \nu(dz)$, with $\int(|z| \wedge 1) \nu(dz) < \infty$. Hypothesis (F5) below adds first-moment integrability where the tail functionals require it. Eras are the stationarity windows of the population work of OF; every statement in this paper is per era.

An operator holds a trigger line at b (the up-crossing side is treated throughout; the down side is symmetric), monitors at check times (t_j) with intensity χ_t , deterministic cadence Δt , Poisson, and Cox-type checking all admitted, and fires at the first check at which its firing condition holds. The policy filter is the pair $(p_{\text{pen}}, d_{\text{dw}})$, a penetration threshold and a dwell requirement, era-constant per operator; the plain threshold policy is $(0, 0)$. Per firing event the observables are the overshoot $O = X_{\tau_f} - b$ and the actuation delay $t_d = \tau_f - \tau_c$, where τ_c is the most recent entry time of the price into $\{x \geq b\}$ before the firing. This is what the data supply. The line b is recovered per operator from the firing-position discontinuity,³⁴ and τ_c is measured on the pool's public tick stream by walking back from the firing block to the last inside-to-outside transition. The detector is not the analyst's construction; its line, its cadence and its filter are read from deployed code and the record that code leaves.

Crossings split by type, following the firing split $N = N^c + N^J$ of the jump-crossing compensator of Appendix C. The split is at the level of the compensator, and no event in the data carries its type as a label. A continuous crossing has $X_{\tau_c} = b$. A jump crossing has $X_{\tau_c-} = b - Y < b$ and $X_{\tau_c} = b - Y + z$ for a jump of size $z > Y$, and the straddle depth is $O_J = z - Y > 0$, latent at every event and reached only through the overshoot law. The observability remark of Section 9.1, that the overshoot at a jump firing is bounded below by the straddle depth, becomes an exact per-event law below and is then inverted. Figure 4 draws the two crossing types.

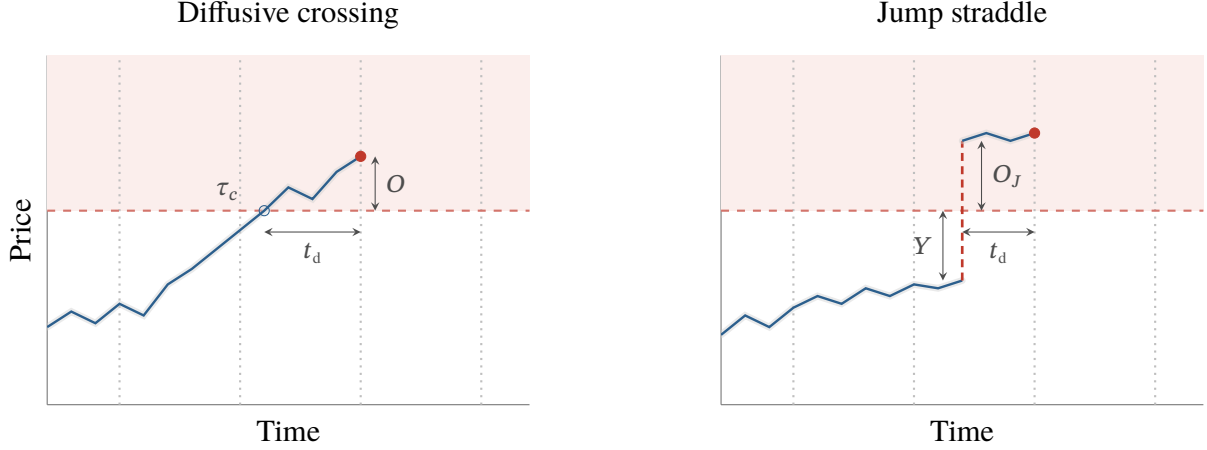


Figure 4. The two crossing types read by the instrument, with check times dotted, the trigger b dashed, and the region beyond it tinted. Left, a diffusive crossing at τ_c (open) fires at the next admitted check (filled) with monitoring overshoot O ; right, a jump from gap Y below the line lands O_J beyond it, and the firing overshoot is bounded below by that straddle depth.

18.2. Hypotheses

Six hypotheses carry the theory; Table 2 states each beside the measurement that certifies it. Certification here means that the hypothesis's designed check passes at its stated power, reported where each check runs; it is measured consistency, not proof. The exclusion restriction (F4) says that checking responds to activity, not to the size of the individual jump; (F5) is the one purely assumed member, a moment hypothesis the tail functionals consume.

Table 2. The instrument's hypotheses and their certifications.

Hypothesis	Certified by
(F1) era homogeneity; the parameters of (24) and the policy filters constant per era	era-half replication, binding per pool (Section 22.6)
(F2) flat entry at the one-check scale near the line; flat gap occupation where the jump tail varies, the exact kernel available where flatness fails	drift and leakage bound, budget row 1; kernel form of Lemma 8
(F3) policy filter era-constant per operator	within-pool agreement design (Section 21.2)
(F4) given a jump crossing and the volatility state, check intensity independent of the straddle depth	exceedance shape across cadence strata (Section 22.6)
(F5) $m_1 = \int z^+ \nu(dz) < \infty$	moment hypothesis, not separately tested
(F6) crossing time measured with bounded staleness	dense-crossing admission, budget row 4

19. The forward map

19.1. The diffusive branch

The monitoring kernel and the delay law follow from the flat-entry hypothesis exactly as in Appendix C, but resolved in the delay variable rather than marginalised over it.

Lemma 6 (Conditional Kernel). *Under the plain threshold policy, conditional on a continuous crossing at τ_c and firing at the next check, at delay $t_d = a$, the overshoot has the folded normal*

density at scale $\sigma\sqrt{a}$,

$$g_{O|a}(o) = \frac{2}{\sigma\sqrt{a}} n(o/\sigma\sqrt{a}), \quad o > 0,$$

and the per-check qualification probability given a is $1/2$, independent of a . Penetration and dwell filters change that probability and enter through the policy filter of Section 19.3.

Lemma 7 (Flat-Entry Delay Law). *Under flat entry, conditional on firing at a check, the crossing-to-check delay has density*

$$g_{\text{act}}(a) = \frac{1}{2\sqrt{\Delta t} \sqrt{\Delta t - a}}, \quad a \in (0, \Delta t),$$

increasing towards $a = \Delta t$, so observed delays are long. The two readings are one fact, crossings clustering early in the check interval because states arbitrarily close to the line cross immediately.

Proposition 13 (Exact Reproduction). *The a -marginal of the joint law of Lemmas 6–7 is exactly the flat-entry overshoot density of Proposition 4(i),*

$$\int_0^{\Delta t} g_{\text{act}}(a) g_{O|a}(o) da = \frac{\sqrt{2\pi}}{\sigma_1} Q(o/\sigma_1).$$

In words, resolving the overshoot by delay and then averaging over the delay law returns exactly the overshoot density the monitoring layer promised, so the delay-resolved instrument and the marginal law are one construction.

The proof (Appendix G) reduces the integral, after normalising to unit scale, by the substitution $a = 1/(1 + s^2)$ to the integrand of Owen's T ,⁴¹ then evaluates it by an inner antiderivative and an interchange rather than by quoting the special function's limit value. That evaluation is machine-checked in the formalisation. The substitution is improper at both endpoints and is not, so the harness's fifteen-digit assertion of the assembled identity now covers that step alone. Three corrections are carried rather than re-derived; drift shifts the kernel mean by $\mu a = o(\sigma\sqrt{a})$ under flat entry, sequential conditioning on earlier failed checks is the barrier-shift correction of Broadie, Glasserman, and Kou³⁹ at leading order, and Poisson or Cox checking replaces Lemma 7 by the first-check law of the χ process after τ_c while Lemma 6 is untouched, since it conditions only on the delay's value.

19.2. The jump branch

Lemma 8 (Straddle Law). *On jump crossings, with $\psi_b(y)$, $y > 0$, the occupation density of the pre-jump gap $Y = b - X_{\tau_c-}$ over the excursion preceding the crossing, normalised to a probability density in y ,*

$$\mathbb{P}(O_J > o) = \frac{\int_0^\infty \psi_b(y) \bar{\nu}(y + o) dy}{\int_0^\infty \psi_b(y) \bar{\nu}(y) dy}, \quad \lambda_J = \int_0^\infty \psi_b(y) \bar{\nu}(y) dy,$$

where $\bar{\nu}(y) = \nu((y, \infty))$ and λ_J is the occupation-averaged jump-crossing intensity, per unit time under that normalisation, which is also the normalisation Proposition 15 compares across operators. If ψ_b is flat at the scale where $\bar{\nu}$ varies and $m_1 = \int_0^\infty \bar{\nu}(z) dz < \infty$, the straddle depth has density

and survival

$$g_J(o) = \frac{\bar{v}(o)}{m_1}, \quad \mathbb{P}(O_J > o) = \frac{\mathcal{J}(o)}{\mathcal{J}(0)}, \quad \mathcal{J}(o) := \int_0^\infty \bar{v}(z) dz.$$

This is the ladder-height structure of renewal theory²⁴ inherited through the occupation measure rather than through renewal cycles. The flat form is the leading-order reading; the exact kernel form is what the multiscale theorem uses, since ψ_b is estimable per cell from the tick stream and the recovered placement, and the tail exponent survives any integrable ψ_b because $\int \psi_b(y) \bar{v}(y+o) dy \sim \bar{v}(o) \int \psi_b$ as $o \rightarrow \infty$ under regular variation.

Lemma 9 (Post-Crossing Composition). *Conditional on a jump crossing with straddle depth o_J and firing at delay a , the overshoot is $O = o_J + \mu a + \sigma\sqrt{a} Z_a$ with Z_a the post-crossing displacement normalised to unit scale, conditioned on the policy filter's qualification event; for the plain threshold policy and $o_J \gg \sigma\sqrt{a}$ the conditioning is negligible and Z_a is standard normal. The straddle component stays at o_J whatever the delay, while the post-crossing motion contributes the drift μa and a monitoring component of order $\sigma\sqrt{a}$.*

19.3. The policy filter

Policy filters decide which crossings reach the record, and they leave two signatures at opposite ends of the delay axis. At long delay a dwell or penetration rule holds the overshoot at its own scale, which is the filter's signature; at small delay the same rule suppresses the diffusive branch far faster than the jump branch, which is the signature the identification uses.

Lemma 10 (Delay Invariance). *Consider an era-constant dwell or penetration policy on an ergodic era model, and assume the excursion above the line, conditioned on survival, converges past a relaxation time to a quasi-stationary law with finite second moment. Then the conditional law of O given $t_d = a$ converges as $a \rightarrow \infty$ to an operator-specific limit that does not depend on a , so $\mathbb{E}[O^2 \mid t_d = a]/(\sigma^2 a) \rightarrow 0$; the pure-lag alternative, firing at the first sparse check with no filter, gives $\mathbb{E}[O^2 \mid t_d = a] \asymp \sigma^2 a$ with no hypothesis beyond the era model.*

The dichotomy is the discriminating measurement between policy and plumbing, and the population data fall on the policy side (Section 22). At population level the invariance appears as clustering, each operator's events concentrated near its own filter scale, so the cross-population scan mixes operator clusters; the per-operator scan needs the penetration form, whose attainment time spreads the delay axis, a point the synthetic harness makes exact.

Lemma 11 (Jump Enrichment). *Under a penetration filter $p_{\text{pen}} > 0$, the continuous branch's probability of firing within delay a of its crossing is $O(\exp(-p_{\text{pen}}^2/2\sigma^2 a))$, while the jump branch's is bounded below by $\mathbb{P}(O_J > p_{\text{pen}}) \mathbb{P}(\text{check within } a)$, linear in the check intensity. Dwell gates both branches equally. Among operators whose filter is penetration-dominated, conditioning on small t_d therefore enriches the jump share beyond the unconditional rate ratio $\lambda_J/(\lambda_J + \lambda_c)$.*

19.4. The instrument response

Proposition 14 (Instrument Response). *Under (F1)–(F6), the joint law of overshoot and actuation delay at a firing of an operator with policy filter $(p_{\text{pen}}, d_{\text{dw}})$, era parameters (μ, σ, ν) , and check intensity χ is the two-branch mixture*

$$\mathbb{P}(O \in do, t_d \in da) = \underbrace{\pi_c \Pi_c(da) K(do \mid a; 0)}_{\text{diffusive branch}} + \underbrace{\pi_J \Pi_J(da) \int_0^\infty g_J(o_J) K(do \mid a; o_J) do_J}_{\text{jump branch}},$$

where π_c, π_J are the check-weighted crossing-type shares, Π_c, Π_J are the crossing-type delay laws (Lemma 7 in the deterministic-cadence flat-entry specialisation; the χ first-arrival law thinned by qualification in general; both supported on $a \geq d_{dw}$ under dwell), and $K(\cdot \mid a; o_J)$ is the law of $o_J + \mu a + \sigma\sqrt{a}Z_a$ conditioned on qualification past $b + p_{pen}$ per Lemmas 6 and 9, with g_J as in Lemma 8.

In words, every crossing-triggered firing of an admitted operator is a draw from a two-branch mixture, a diffusive branch whose overshoot shrinks with the actuation delay and a jump branch whose overshoot does not, weighted by how the checks land on the two crossing types.

Three specialisations recover known statements. Setting $\nu = 0$ with the plain filter and deterministic cadence collapses to Proposition 4(i) exactly (Proposition 13); $a \rightarrow \infty$ under a filter gives the delay-invariant policy plateau (Lemma 10); and $a \rightarrow 0$ suppresses the first branch at rate $\sqrt{a}$, faster under penetration (Lemma 11), leaving g_J convolved with an $O(\sigma\sqrt{a})$ kernel, which is the identification limit.

Check-intensity dependence is carried, not assumed away. If keepers densify checks in volatile windows and jumps cluster with volatility, the jump branch's delay law is stochastically smaller than the continuous branch's, and the crossing-type shares are check-weighted rather than time-averaged. Under (F4) the conditional law of O_J given a jump crossing and a delay value does not depend on χ , so every tail-shape functional read from the small-delay subsample is clean; what χ contaminates is the level, the conversion of shares into absolute rates, which is identified relative to the continuous rate or jointly with a recovered cadence. The harness verifies both halves of this split.

20. Identification

Fix a pool-era with local diffusive scale σ . The target functionals are the integrated tails $\mathcal{J}$ of Lemma 8, whose normalised version is the straddle survival, together with the per-geometry jump-crossing rates. Define, for a multiplier $\kappa > 0$ and delay cut $\bar{a}$, the exceedance functional

$$\mathcal{E}(\kappa, \bar{a}) = \mathbb{P}(O > \kappa\sigma\sqrt{t_d} \mid 0 < t_d \leq \bar{a}),$$

and write $\pi_J(\bar{a})$ for the jump-crossing share of firings with $t_d \leq \bar{a}$.

Theorem 6 (Small-Delay Identification). *Under (F1)–(F6), for every $\kappa \geq 1$ and $\bar{a}$ with $|\mu|\bar{a} \leq \frac{1}{2}\kappa\sigma\sqrt{\bar{a}}$,*

$$|\mathcal{E}(\kappa, \bar{a}) - \pi_J(\bar{a})| \leq \underbrace{\frac{2Q(\kappa/2)}{\text{diffusive leakage}}}_{\text{diffusive leakage}} + \underbrace{\pi_J(\bar{a})\left(1 - \frac{\mathcal{J}(2\kappa\sigma\sqrt{\bar{a}})}{\mathcal{J}(0)}\right)}_{\text{resolution floor}} + \underbrace{\varepsilon_{\text{cens}}}_{\text{staleness}},$$

rows 1, 2, and 4 of the bias budget (Table 3), with $\varepsilon_{\text{cens}}$ the one-sided crossing-staleness error controlled by the dense-crossing admission (F6). In particular $\mathcal{E} \rightarrow \pi_J(0^+)$ as $\bar{a} \rightarrow 0$ with $\kappa \rightarrow \infty$ and $\kappa\sqrt{\bar{a}} \rightarrow 0$, and the excess of $\mathcal{E}$ over the folded normal baseline $2Q(\kappa)$ is a consistent reading of the small-delay jump share.

In words, at small actuation delay the diffusive branch cannot manufacture large overshoots, so the rate of exceedances beyond a few local scales reads the jump share directly, with a three-term error the design controls.

Corollary 5 (Tail Shape). *With $o_{\min} = \kappa\sigma\sqrt{\bar{a}}$, for every $o \geq 2o_{\min}$,*

$$\mathbb{P}(O > o \mid t_d \leq \bar{a}, O > o_{\min}) = \frac{\mathcal{J}(o)}{\mathcal{J}(o_{\min})} \left(1 + O(\sigma\sqrt{\bar{a}} \|g_J\|_{\text{Lip}} \cdot \mathcal{J}(0)/\mathcal{J}(o_{\min})) + O(Q(\kappa/2)/\varepsilon)\right),$$

so the normalised integrated tail is identified on $[o_{\min}, \infty)$, the resolution floor shrinking as $\sqrt{\bar{a}}$; under the exact kernel form of Lemma 8 the identified object is the ψ_b -weighted analogue, with ψ_b estimated rather than assumed.

In words, above a resolution floor that shrinks with the check interval, the shape of the observed overshoot tail is the shape of the integrated jump tail.

Proposition 15 (Multiscale Identification). *Let one pool-era host operators $i = 1, \dots, k$ with certified-distinct geometries, recovered line distances and estimated gap-occupation kernels ψ_i , so that operator i 's jump-crossing rate is $\lambda_j^{(i)} = \int \psi_i(y) \bar{v}(y) dy$. Then (i) each $\lambda_j^{(i)}$ is identified from operator i 's firing record by Theorem 6 and the relative-rate conversion; (ii) if $\bar{v}$ and the ψ_i lie in $L^2(0, \infty)$, the vector $(\lambda_j^{(1)}, \dots, \lambda_j^{(k)})$ identifies the orthogonal projection of $\bar{v}$ onto $\text{span}\{\psi_i\}$ in that space if and only if the Gram matrix $\int \psi_i \psi_j$ is nonsingular, a condition computable from the estimated kernels; (iii) if $\bar{v}$ is regularly varying with index $-\alpha$, $\alpha > 1$, and two kernel scales $s_1 < s_2$ are separated, then $\lambda_j^{(2)}/\lambda_j^{(1)} = (s_2/s_1)^{-\alpha}(1 + O(\varepsilon_{\ker}))$, so $k \geq 2$ separated geometries identify the index α . The amplitude follows only under a stated power-law convention $\bar{v}(y) \simeq C y^{-\alpha}$, regular variation fixing the ratio and not the constant. Precision degrades as $s_2/s_1 \rightarrow 1$, with $\varepsilon_{\ker}$ bounded by the certified kernel deviation.*

In words, one operator identifies its pool's integrated jump tail, operators at separated trigger scales identify the tail's index, and the geometry census decides pool by pool which reading is available.

The proposition is conditional on the per-pool geometry count, whose measured distribution is reported as data in Section 22. The single-operator tail shape (Corollary 5) and the cross-operator rate ratios (Proposition 15(iii)) estimate the same index, which supplies an overidentification check wherever a pool hosts both a fast operator and a spread of geometries.

Proposition 16 (Diffusive Corollary). *At $\nu = 0$ the instrument degenerates to a volatility sensor; the intensity representation of Proposition 1 recovers σ^2 per cell up to the operator's filter thinning, so population firing rates estimate diffusive variance after per-sensor calibration, and aggregation across sensors requires precision weighting rather than averaging.*

In words, with no jumps the population of detectors is a volatility sensor, and each sensor is calibrated through its own filter before any aggregation. The proposition folds the volatility-oracle reading into the paper as the zeroth-order inversion; its content at $\nu \neq 0$ is that the same calibration discipline governs the jump reading, which is why the estimators below never pool across pools.

21. Estimators and the bias budget

21.1. Definitions

All estimators operate per pool-era; their full definitions, admission thresholds, and interval mechanics are collected in Appendix H. The local scale is a trailing bipower estimate over H blocks ending strictly before the crossing block, so the candidate displacement is excluded; the realised-variance variant is reported alongside, and its inflation is a bias-budget row. The per-pool readings of Section 22 are captured with the realised-variance variant of the trailing scale; row 3 of the budget signs the difference, so the quoted excesses are conservative (Section 22.6). Per admitted event the overshoot is scored in trailing local scales, and the per-pool exceedance fraction beyond the cut $\kappa = 3$ carries a Wilson interval,⁴² with the jump share $\hat{\pi}_J$ read off by subtracting the folded-normal baseline and the tail functional taken as the empirical survival of exceedance sizes above $o_{\min}$. Within a pool, strata combine by event count, the precision weights under the common- p null; across pools nothing is pooled. Each pool is its own estimand, its target functional occupation-weighted through that pool's ψ_i , so aggregation would mix estimands where it appears to reduce variance, and the per-pool functionals differ by measurement as well.

21.2. The agreement design

Policy heterogeneity is handled by replication rather than subtraction. At small delay the policy term is a multiplicative thinning of the diffusive branch (Lemma 11), not an additive offset, so every stratum’s small-delay exceedances estimate the same pool-level tail. Partition a pool’s operators into strata (fast and slow by median actuation delay, in the empirical work), estimate the exceedance functional separately per stratum, and test equality by Fisher’s exact test with a familywise correction at the tested-pool count. Agreement supports compatibility with a common pool-level tail; it is a non-rejection, not a demonstration that the tail is a property of the price law. A rejection localises a violated hypothesis, era drift, kernel mis-specification, or an operator artefact, to the offending cell. The design needs no model of the policy offsets, and every per-pool reading is internally replicated.

21.3. The bias budget

Table 3 states the signed budget. Three rows carry synthetic evidence from the harness and one was forced by the data. Row 3 is measured at a median factor 1.198 on the census pools, so the realised-variance readings run in the conservative direction for the exceedance statistic, the cell-level effect being carried in the budget and not assumed uniform; in the harness’s planted world, where the jump variance share is extreme, the realised-variance pipeline fails the planted-recovery check outright while the bipower pipeline passes. Row 8 is the data-forced addition. Where $\hat{\sigma}\sqrt{\bar{a}}$ is small against the tick spacing, the cut fires on quantisation quanta rather than jumps, so a per-pool reading is admitted only where

$$\hat{\sigma}\sqrt{\bar{a}} \geq c_{\text{tick}} \vartheta; \quad (25)$$

the harness measures $c_{\text{tick}} = 16$ under a trade-quantum null with two-tick mean moves, the constant scaling with the per-trade move distribution.

Table 3. The signed bias budget of the exceedance estimator.

Term	Sign	Control	Certification
1. diffusive leakage	+, subtracted	folded baseline $2Q(\kappa)$	drift small per (F2)
2. resolution floor	–	shrink $\bar{a}$ against m	tail estimate at $o_{\min}$
3. $\hat{\sigma}$ jump inflation	–	bipower estimator	RV/BV ratio per pool
4. crossing staleness	–	dense admission	per-event bracket (F6)
5. Cox level ambiguity	levels only	relative-rate form	(F4) refutable implication
6. policy masquerade	+, bounded	inside row 1’s bound	agreement test
7. era drift	either	per-era statements	era-half replication (F1)
8. tick discreteness	+ in quiet pools	admit $\hat{\sigma}\sqrt{\bar{a}} \geq c_{\text{tick}} \vartheta$	harness sweep

With m admitted events the binomial rate is $(Nq_{\text{sd}})^{-1/2}$ in the cell’s firing count N and small-delay share q_{sd} , and the jump-share error is the Wilson half-width scaled by $(1 - 2Q(\kappa))^{-1}$ plus the deterministic bracket of Theorem 6. The rightmost column repeats the certification pairing of Table 2.

22. The population read

The population is the sender-keyed census of production concentrated liquidity operators on one chain, 322 cells across 50 pools with recovered trigger geometries spanning trigger fractions 0.10 to 0.42,³⁴ observed over one era of roughly 120 days. The census is keyed by its Aerodrome-side producer but its pool set is not single-venue; factory verification splits the pools with scored

small-delay events into 18 on Aerodrome Slipstream, 8 on PancakeSwap V3, 3 on Uniswap V3, and 4 on further V3-fork factories, so wherever this section says the census, the name denotes the operator layer rather than one venue’s pools, and the cross-venue comparison of Section 22.7 restricts its first side to the Aerodrome-factory members. The cell count is the full sender-keyed layer; the fingerprinting paper’s headline population, 81 operators over 102 clean cells, is its stricter recovery-grade subset, the quoted trigger-fraction span is measured there, and the same admitted-versus-clean cut replicates on the second deployment’s census (Section 22.7). Of the record, 139,382 events carry a validated firing state, 63,961 of them are trigger firings, and 56,167 of those carry a measured crossing. All numbers below are per era and reproduce from the paper’s own verification surface, deterministic and captured.

22.1. *The pooled inversion fails*

Per-cell-normalised overshoots across the full census match the diffusion-with-lag benchmark almost exactly, tail ratio $q99/q50$ of 3.80 against a folded-normal 3.82, and the events the diffusion-with-lag benchmark makes very unlikely, large overshoots within five blocks of the crossing, are 0.11 % of measured firings. The cause is the policy plateau of Lemma 10; the median squared overshoot per delay block falls by a factor 0.088 from short delays to long, so overshoot at firing is close to delay-invariant, which is what penetration and dwell rules produce and what a first-passage reading forbids. Fifty-four per cent of rebalances fire with the state strictly inside the trigger band, convention or schedule rather than triggers. The naive spectrometer does not exist; the instrument lives in the conditioned subsample.

22.2. *The conditioned excess*

Conditioning delivers the signal on two independent readings. Fast-operator strata fatten past the folded-normal benchmark, $q99/q50$ of 4.33 to 4.72 against 3.82, and the direct small-delay test reads the same way; with the local scale measured in a one-day trailing window ending before the crossing, candidate excluded, 18 to 22 % of overshoots within ten blocks of the crossing exceed three local scales against a folded normal baseline of 0.27 %, the span running over the two-, five-, and ten-block delay cuts and the one-day and seven-day windows, with ninety-ninth percentiles from 11.7 to 14 local scales, stable under the dense-crossing admission. The worst-case bracket of Theorem 6 at this cut is far looser, admitting diffusive leakage up to $2Q(\kappa/2) \approx 0.13$ at the largest drift the theorem tolerates; the folded-normal baseline is the operative one because the measured drift of the census pools sits deep inside the drift restriction, the certification of budget row 1, so the excess is read against the measured-drift baseline rather than the adversarial bracket. Even the median small-delay overshoot runs 1.1 to 1.3 local scales, across the census bulk and the within-pool strata, against a folded-normal 0.674, which is why the local scale is estimated from the tick stream and never from the firing bulk.

22.3. *Within-pool replication*

The design fixes the pool set. Of the 50 pools, 11 host both a fast and a slow stratum; within those pools the fast stratum’s small-delay exceedance is 0.144 (n 1,010), 53 times the folded baseline, and the slow stratum’s own rare small-delay events in the same pools carry 0.171 (n 940), the same order and slightly richer, exactly as Lemma 11 predicts for the heavier-filtered stratum. The agreement is consistent with a pool-level component rather than an operator-specific filter effect. Pool-level selection is real, fast-only pools run hotter (0.27 on n 117), so population aggregates keep the pool-census framing while per-pool readings are clean by measurement. Of six pools clearing the agreement test’s stratum threshold, two reject at the nominal level and none after the familywise correction.

22.4. Per-pool readings and the geometry census

Sixteen pools clear thirty admitted events. Their exceedance readings, plotted with Wilson intervals in Figure 5, span 0.058 to 0.522 with largely disjoint intervals, so the per-pool jump functional differs across pools far beyond sampling noise, which is evidence against a common cross-pool functional and is why the pool-specific estimands are kept.

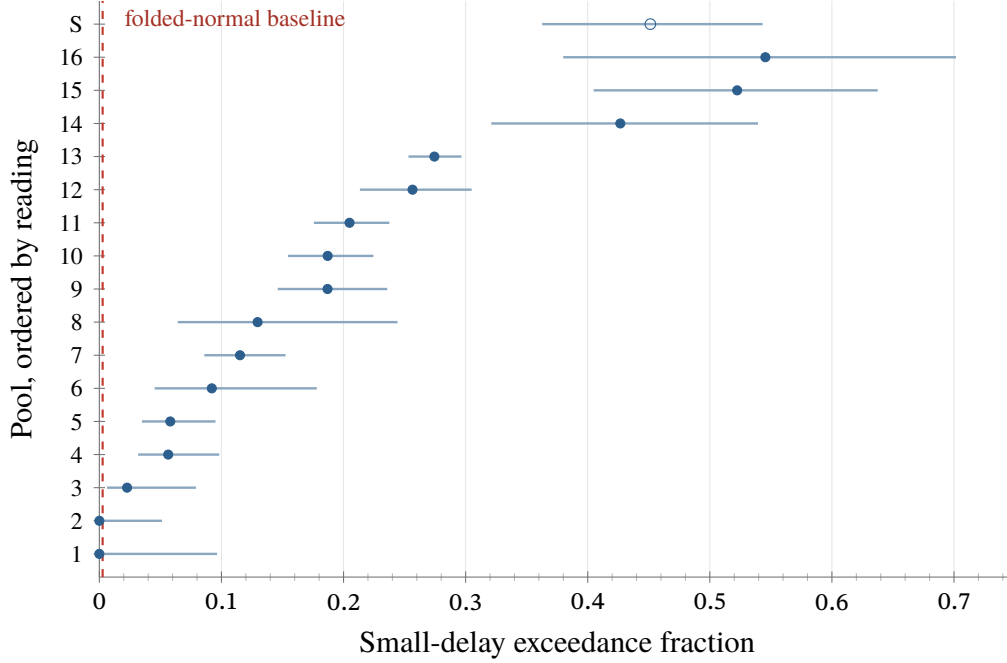


Figure 5. Per-pool small-delay exceedance readings with 95 % Wilson intervals, pools ordered by reading; the open marker S aggregates the seventeen pools below thirty admitted events. The dashed line is the folded-normal baseline 0.0027.

The geometry census settles the multiscale proposition’s conditionality; 24 of 50 pools host a single trigger scale, 26 host the two or more the hypothesis of Proposition 15(iii) names, and 19 host three or more. The conservative availability count is therefore the 19 pools, under two fifths, and the proposition is stated as conditional throughout.⁶

22.5. Tick-stream corroboration and its limit

Direct detection on the per-pool tick stream corroborates the magnitude; per-block displacements exceed three trailing local scales in every pool, median pool rate 0.33, so the price law at block scale is heavy-tailed against trailing-window diffusion unconditionally, which is directionally consistent with a price-process component in the firing-based excess rather than an operator artefact alone. The same exercise fails as a cross-pool ranking, Spearman -0.04 against the firing-based readings, because the per-block detector is discreteness-dominated in quiet pools, jump-size medians up to 32 scales beside low rates, the signature of one-tick quanta over a tiny local scale. The failure motivates budget row 8 and its admission constant.⁷

22.6. Certifications and cross-checks

Five certification probes close the loop between the hypotheses of Section 18.2 and the census. The era-half replication splits the era at its midpoint block; the population reading replicates,

⁶The two-scale margin is thin. The census counts scales at two significant figures and the ratio estimate degrades as adjacent scales approach, so the 26 is the count most exposed to that rounding and the 19 is the one to rely on.

⁷A crossing-conditioned validator, which would score the tick stream only where an operator’s line was actually crossed and so inherit the firing-based admission, is the natural redesign and is not attempted here.

exceedance 0.225 against 0.205, while of six pools with thirty admitted events in each half, three reject the one-functional null after familywise correction, so (F1) certifies at population level and binds as a per-pool admission criterion, with the failing pools carrying per-half statements only. The refutable implication of (F4) survives its first run; exceedance shape across cadence strata agrees to a Kolmogorov–Smirnov statistic of 0.17 ($p = 0.54$, 85 exceedances, matched medians and ninety-eth percentiles), a low-power pass that is now captured and cheap to rerun. The designated bipower estimator, rerun over the same events, raises every population cell, 25 to 30 % against the captured realised-variance 18 to 22 %, the direction and rough magnitude row 3’s measured inflation predicts, so the captured columns are confirmed conservative rather than assumed so. Two cross-checks anchor the inputs. Six census pools are independently indexed in the operator lake’s swap extension; on 555,020 shared era blocks the two tick streams agree within one tick on 91 to 99 % of blocks under the verified closing-tick convention, with one gross outlier flagged as a data-quality defect in one source, so the tick-stream provenance carries no systematic bias. And the discreteness admission constant, rebuilt per pool from real trade-size distributions where swap-level data exists, spans 8 for fine-trade pools to beyond 32 for coarse ones, bracketing the synthetic constant 16 of Section 24; the admission of budget row 8 is per-pool wherever the trade law is measurable, and coarse-trade pools admit no small-delay reading at production scales.

22.7. The second deployment and the cross-venue test

The instrument was ported unchanged to the population of a second deployment. An operator-keyed census of Uniswap V3 on Base over the same era admits 1,046 cells across 691 operators on 69 pools, of which 495 clear the clean-step line-recovery gate at thirty firings and 227, across 160 operators, clear the stricter bar of the fingerprinting convention, with recovered trigger fractions again spanning 0.10 to 0.42. The separating instrument is the clean-step gate, not the infrastructure filter, which removes almost nothing at the admission bar; the dense retail and bot population of the busiest pool passes admission in numbers and largely fails line recovery, which is what a census keyed on line-imprinting operators should show. The read replicates in structure. The clean cells supply 34,100 trigger firings, and the small-delay excess sits at 44 to 48 times the folded baseline under trailing local scales, higher again under the bipower rerun, on an independent population reading independent contracts, with the per-pool spread again far beyond binomial noise. It does not replicate in certification; every pool at the per-pool reading bar fails the discreteness admission of budget row 8, the scale ratios running 1.2 to 9.8 against admission constants of 16 to 32 with each observed reading at or below its pool’s own discreteness null, and the era-half criterion (F1) rejects on the only testable pool, so this venue-era admits no certified jump reading.

The cross-venue agreement test is read admission-first. One venue-level pair is testable at the bar, WETH/cbBTC, and it rejects equality (Fisher $p = 0.0022$); the rejection is fully localised to the pre-specified admission failure, the Uniswap pool’s scale ratio sitting at 1.2 with a discreteness null near 0.38 against the observed 0.41, and the within-venue heterogeneity across the pair’s three Aerodrome pools already exceeds the cross-venue gap. Where admission-comparable readings exist the venues agree. The one same-pool check powered on both pipelines, the census’s own Uniswap V3 WETH/USDC member read through two independent indexing, keying, and tick-cache chains, agrees within binomial noise ($p = 0.155$), the port-validity analogue of the provenance cross-check above; and the one pair powered across the census’s PancakeSwap and Aerodrome members, SOL/cbBTC, agrees across deployments. The certified verdict is that the agreement claim, that the signal is a property of the pool’s price law rather than of the operators reading it, survives where the instrument is admitted and is untested, not refuted, where it is not; a certified cross-venue equality test needs a denser-stream era or per-swap trade data to sharpen the admission constant.

23. The jump surcharge

The last result joins the three acts. Jumps do more than add a crossing channel to the intensity layer. A jump fires the trigger from beyond the line, so every crossing it causes is settled at a deeper displacement than the same policy would have paid diffusively, and the placement potentials of Section 11 price the difference exactly.

Proposition 17 (Jump Surcharge). *Under the era model with jumps and the hypotheses of Section 18.2, consider an isolated policy, of $\mathcal{A}_0$ or $\mathcal{A}_2$, with up-trigger displacement xh whose return convention keeps the upper branch binding at both the diffusive firing state $xh + O$ and the jump-shifted state $xh + O + O_J$. The kerf rate gains the additive term*

$$r_J = \lambda_J \mathbb{E} \left[\mathcal{C}^\uparrow(xh + O + O_J) - \mathcal{C}^\uparrow(xh + O) \right] \geq 0,$$

independent of the return point, where O is the monitoring overshoot of Proposition 4, O_J the straddle depth of Section 19.2, λ_J the occupation-averaged jump-crossing intensity of Lemma 8, and the expectation is taken under the jump-crossing law at that same weighting. In the narrow limit, on the event that the fired state remains inside the range,

$$r_J = \lambda_J \mathbb{E} \left[\ln \frac{(1-x)h - O}{(1-x)h - O - O_J} \right] \geq \frac{\lambda_J \mathbb{E}[O_J]}{h},$$

with the down branch symmetric. Straddles that carry the state beyond the range boundary pay the corner value of the exact potentials in the isolated class, and the swap-mediated cost at the landed share gap otherwise.

Proof sketch. The per-event cost difference between the jump-shifted and the diffusive firing state telescopes through the placement potentials of Section 11, cancelling the return point; positivity is strict monotonicity of $\mathcal{C}^\uparrow$, and the narrow-limit form is the logarithmic potential. Full proof in Appendix F. ■

In words, every jump crossing of an isolated policy whose binding branch does not switch pays at least its straddle depth over the half-width, in log-value, on top of everything the diffusive floor already charges, and no return convention avoids it.

The beyond-range clause carries a scope note. A landing beyond the range boundary leaves single-sided holdings from which no price-containing equal-width range can be minted, so its isolated-class log-cost is infinite; any jump law with support beyond $(1-x)h$ therefore gives every band-maintaining isolated-class policy an infinite kerf rate, the value-flux ledger staying finite, and isolated-class rates under such laws are quoted only under a stated per-event cap (Section 24).

Two consequences follow. The emission-based isolated floors of Sections 12 and 13 harden by the additive term r_J , since the surcharge is pathwise and per-event. The swap-mediated floor of Section 14 hardens too, but not by r_J ; a deeper landing enlarges the share displacement its impulse must correct, and that cost is the execution one of (21). The retention collapse of Section 15 is unaffected, because retention deletes the surrender, straddle-deep or not. The surcharge is also measurable. Its two inputs, the jump-crossing rate and the straddle law, are the functionals the instrument of the third act identifies per pool (Theorem 6 and Corollary 5), with the check-weighting caveat of Section 19.4 applying to the rate; the per-pool magnitudes belong to the population read (Section 22) and its disclosure gate.

The ledger can be worked on a live book. Take the operator's WETH/TOSHI position from the production anchor of Section 24, a 24 basis point pool at tick spacing 200 with median position value near \$535; Table 4 runs the ledger at the anchor window's measured parameters. The concavity leg, the price of holding concentrated inventory at that width and volatility, is recoverable only through

fees and emissions; the event floor at the pool’s own fee sits below it by the measured factor 0.24, which is $\eta/(2\rho)$ at this geometry, the ordering of Proposition 12 at production numbers; and the realised event-leg rate runs at twice its class floor, with the measured geometry, late trigger and small correction, in the corner Proposition 8 prices as cost-optimal in direction. The remaining gap is a control lever. In a subsequent trigger experiment on the same pool the operator moved the trigger fraction from one quarter of width to 0.12, the x of (23) from 0.5 to 0.76, and the churn share of gross revenue fell from 23.6 to 14.7 per cent with net revenue per day unchanged, the direction (23) predicts.

Table 4. The ledger worked on the WETH/TOSHI book.

Quantity	Form	Measured
narrow-range parameter	ρ	0.0051
sqrt-price volatility	$\sigma_s/\bar{s}$	0.25 per $\sqrt{\text{yr}}$
concavity leg	$\sigma_s^2/(2\bar{s}h)$	613 %/yr
event floor, pool fee	$\eta(1 - \rho_w)^2\sigma_s^2/w^2$	144 %/yr
leg ratio	$\eta/(2\rho)$	0.24
realised event-leg rate	(6) summed per year	290 %/yr
trigger fraction, correction	medians	0.69, 0.05

24. Numerical verification

Each act ships its own dependency-free reference implementation at a fixed seed, with captured output as the regression target. A fourth surface evaluates the first two acts on the operator’s production history, a fifth replays the exact layer against unmodified production bytecode on three deployments and two chains, a sixth replays the discrete-bin algebra of Appendix B against that venue class’s unmodified home deployment, a seventh calibrates the instrument and stress-tests the floors on synthetic ground truth at simulation scale, and an eighth carries the exact layer and the third-party identity onto Ethereum mainnet at a twelve-second block. Nothing below consumes on-chain data except the anchor, the fork suites, and the third act’s probes, which read the population layer of Section 22.

The eight surfaces are complementary rather than independent replications, and the distinction matters for how much each adds. The harnesses check implementations against formulas derived in this paper, so they test agreement and not the derivations; the anchor’s operator is the present author, which is why the same identity is evaluated on third parties below; and the seventh surface’s synthetic populations carry operator geometries drawn from the second deployment’s census, so its calibration inherits that population’s geometry rather than testing it. None of the eight replaces a proof, and none establishes validity outside the stated hypotheses.

The first act’s harness begins with the amplitude. The two-branch amplitude (8) is checked against the mint-minimum arithmetic over 20,000 random parameter draws with both branches evaluated, agreeing to 2.1×10^{-12} relative; the corner values of Corollary 3 agree to 1.1×10^{-14} , and the monotonicity threshold of Proposition 3 is located by bisection to 6.4×10^{-5} . The pathwise identity (2) is reproduced on a simulated path with 250 impulses at a relative residual of 1.551×10^{-14} ; the identity is exact path by path, which is also what Proposition 5 exploits. The remaining checks exercise the occupation-offset bound of Corollary 1 on a simulated tick path together with the monitoring dichotomy of Proposition 4(i): near-continuous monitoring concentrates the firing offset and coarse monitoring restores uniformity, and the scale-function direction split is reproduced within Monte Carlo and discrete-step overshoot error.

The second act ships two artefacts, both standard-library Python at fixed seeds. The scoping probe simulates the centred fixed-width family at four triggers and matches the exact coefficient

(17) within 3%; the minimum measured $c = 2.38$ at $x = 0.72$ and $\rho = 0.05$. Its extension simulates off-centre re-placements against the two-argument amplitude, yielding ratios 0.99 and 1.08, and grid-searches the full three-parameter renewal family analytically; the minimum $c = 2.397$ occurs at the symmetric point $(-0.72, 0.00, 0.72)$, a family whose single shared target cannot express the per-side reflection targets of Proposition 7.

The reference harness checks the algebraic layer first. The placement-family potential identity holds across twenty thousand random equal-width geometries with worst error 2.7×10^{-12} ; the share-potential identity holds across twenty thousand random width pairs with worst error 8.9×10^{-16} , at machine precision, confirming that Theorem 2 is an identity rather than an approximation; and the free locus holds across two thousand solved ratio-preserving widenings with worst cost 2.7×10^{-14} . The floor layer follows. The corridor constants quoted after Theorem 3 and its analytic corollary are reproduced; the centred Monte Carlo sweep and the renewal grid put every member above $c_{\min}$; and the widen-slide adversaries against the width-uniform floor at cap ratio two reach 5.51 at best against a floor of 5.12, the remaining margin consistent with a floor approached from inside the class, with a longer run of the full-recentre configuration at 644 paid events measuring 7.02 ± 0.28 . The clock and drift bounds of Lemma 5 hold over five thousand random geometries. In the swap class the constant A matches Table 1; edge-hugging converges to the fee-only floor at ratio 1.005; and Monte Carlo return-point policies match (23) to 2% at the deployed convention and to 14% at the near edge, where discrete-step overshoot lengthens cycles, the near-edge policy sitting 2% above the floor, consistent with the tightness of Proposition 8. No tested policy in any family beats its class floor.

The third act's synthetic harness seeds each check independently, so extensions never disturb captured draws. The delay-law mixture of Proposition 13 is reproduced to a relative error of 6×10^{-15} , and the joint flat-entry law is confirmed by exact Lévy first-passage sampling with no path discretisation, Kolmogorov–Smirnov statistics at the sampling floor for $n = 49,415$. The full pipeline threshold operator, trailing bipower scale with candidate exclusion, exceedance and jump-share estimators recovers a planted jump share inside Theorem 6's bracket and a planted Pareto tail index within a Hill error of 0.21 on 224 exceedances. The delay-invariance dichotomy separates a penetration operator (ratio 0.033) from a lag operator (2.3, the meander factor included), and jump enrichment measures 0.85 against 0.54. The agreement test holds empirical size 0.043 to 0.045 at nominal 0.05; the discreteness sweep delivers $c_{\text{tick}} = 16$ under a trade-quantum null; and Cox checking shifts the small-delay jump share while moving the recovered tail index by 0.07, inside noise, the level-shape split of (F4) verified end to end.

A fourth surface anchors the first two acts in production data; the operator is the present author. The position-manager history, 135,538 recorded rebalances with per-event burn, swap, mint, and residual amounts over the first 53 days of its era, is evaluated against the theory with no synthetic input. On the 78,908 events whose value chain closes without drawing on the standing residual balance, the share-potential identity holds with median absolute error 5.7×10^{-17} in k , at machine precision on production events, and residual conservation holds on all of them; the two fifths of events that do draw on the balance are the retention architecture of Section 15 operating live, priced in the rate evaluation but excluded from the identity subsample, whose hypothesis is isolated holdings. Per pool, no volatile-pair book runs below its class floor at a uniform production fee anchor; distances to floor are set by correction size and width as the benchmark reading of Section 14 predicts; and measured median trigger fractions, 0.22 to 1.14 across pools, confirm the late-firing direction. Figure 6 plots realised rates against class floors. The captured output supplies the worked example of Section 23.

The identity is therefore also evaluated on third-party operators: every same-transaction re-placement on the census pools of both deployments over the census era, reconstructed from public burn and mint events with the transaction sender attached, author and infrastructure addresses excluded. On the isolated interior re-placements that survive the anchor's own event discipline, 253 on Aerodrome Slipstream across ten operators and 345 on Uniswap V3 across eight, the identity holds with median absolute error 1.1×10^{-16} in k on each venue, the same machine precision as the anchor, with the recovered price self-consistent to 2.5×10^{-14} on Uniswap V3 and 1.6×10^{-11}

on Slipstream; the error tail (99th percentile near 4×10^{-4}) is in-transaction fee collection contaminating the isolated-holdings hypothesis at the margin. Twenty-two further isolated re-placements on Ethereum mainnet, read the same way over the mainnet era, hold at median 7.9×10^{-17} with maximum 2.0×10^{-14} , so the identity is evidenced on three venues across two chains. A further 2,385 and 33,008 single-sided limit placements satisfy the identity trivially, and the excluded classes, fee-compounding top-ups first among them, are tabulated in the captured output. The identity subsample is small because the behaviour is rare among third parties, not because it fails; no surviving event on either venue misses the prediction.

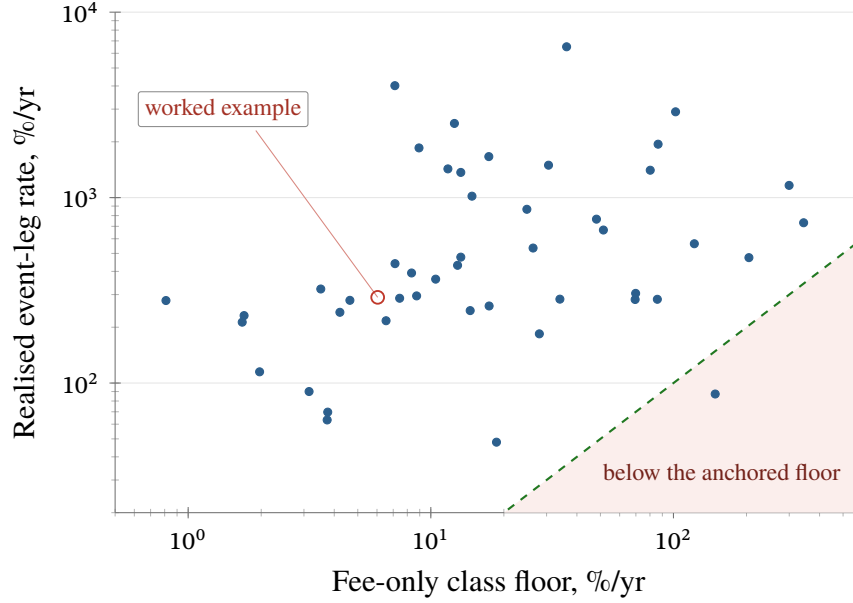


Figure 6. The production anchor’s per-pool realised event-leg rates against the fee-only class floor at the uniform $\eta = 10^{-4}$ anchor, the dashed identity, with the worked example’s book marked open. Points inside the tinted region are pegged pairs, where the uniform fee anchor and the realised-variance σ overstate the floor rather than the book undercutting it.

A fifth surface replays the exact layer against production bytecode. The fork suite of Section 25.1 replays four results against two unmodified position managers on Base and, in the mainnet arm, against the Uniswap V3 manager on Ethereum at a twelve-second block. On pseudo-random width pairs the re-mint liquidity equals the binding-side minimum and the mint amounts match the V3-exact prediction to a few wei on both deployments. The two-branch amplitude is realised as whole-tick translates of the old range with the pool price placed at the translate’s midpoint, so the only model gap is the translate’s width factor, and the closed form (8) agrees within that budget: twelve to forty-four parts per million at tick spacing ten and one to four and a half parts per thousand at spacing one hundred, with the binding side per branch as Lemma 3 states. The corner values are quantisation-free; the corner ratios of Corollary 3 agree to the wei, with the recentred corner mint reverting empty on both managers, and the identical re-placement of the free locus costs one raw token unit.

A sixth surface replays the discrete-bin algebra against its own deployment. The second fork suite replays Appendix B against an unmodified Liquidity Book pair on the class’s home chain. The bin price law is bit-exact against the pinned source libraries at ids spanning the active bin by five thousand, with the offset anchor exact and the step ratio inside the power library’s own truncation bound; the constant-sum depth, the atom amplitude, and the live fee quote and single-price execution are exact to the wei; and the per-traversal loss matches its price-ratio form with zero relative gap at the programme’s fixed-point tolerance.

The discrete-bin algebra is also evaluated on third-party operators. Over the three million blocks ending at the fork pin, every withdrawal and redeposit pair on the class's home chain falling within fifty blocks was reconstructed from public events, admitted only when the two transactions share a sender, neither contains a swap on the pair, and a closed-interval log query over the pair returns exactly those two transactions, so that no third party mutated the touched bins between them. 77 sequences survive, on nine pairs across twenty-one distinct senders. The burn law is exact in all 1,406 bin legs; the mint law holds with median residual 7.0×10^{-22} and maximum 1.0×10^{-20} ; and the composition fee, its protocol split, and the final-shares rule are exact on all forty-eight events that carry one. One caution follows from the exercise. The home deployment predates the source repository's correction to the composition-fee divisor, so the deployed final-shares rule divides by the original bin liquidity where the read commit divides by that liquidity plus the net fee; asserting the commit form against this deployment measures the version difference, at most 5.5×10^{-3} here, rather than the rule. The deployed form is the object, and the difference is reported beside it in the captured output.

A seventh surface calibrates the statistical layer on synthetic ground truth. Where the first six surfaces assert identities, this one measures the instrument; the worlds are diffusion with planted jump measures, no price model entering as an assumption; the populations carry operator geometries drawn from the second deployment's census; and the actual inversion pipeline of Appendix H runs unchanged on the synthetic firing records, each simulator validated event-exactly against the shipped harnesses before use. The recovered jump share sits inside the bracket of Theorem 6 in every cell of every sweep, 120 pool-inversions per planted world across population, era-length, and delay-cut sweeps. The bias is the resolution floor and behaves as the theorem prices it, flat in population and era and moving only with the delay cut, negative by the straddle mass below resolution and correctable through the recovered tail; precision is admitted-event-limited rather than operator-limited, a read to within two hundredths needing roughly two thousand admitted small-delay events per pool-era, which at production firing rates means on the order of tens to a hundred operators or pooled-era accumulation; and the tail read carries the finite-reference upward bias of its Hill kernel, with a stated breakdown regime, sparse heavy tails becoming unreadable below roughly four hundred thousand blocks of era at the thinner planted rate.

The same surface maps the statistical results beyond the shipped harnesses and ran the policy search behind Proposition 7. Across 141 planted jump, delay, and drift configurations, the flat-kernel form of the small-delay bracket holds on 4,669 of 5,046 admissible readings, every failure marginal and confined to high cut multipliers against small jump scales, exactly the regime in which Corollary 5 prescribes the estimated-kernel form in place of the flat leading order; the consistency clause of Theorem 6 manifests as convergence in the resolution ratio, block-quantised delays bounding the joint limit away; and the surcharge premium of Proposition 17 matches its potential form per event to a few parts per thousand with the lower bound holding throughout, beyond-range landings carried under a stated per-event cap of ten log-units. On the floor side, four independent instruments agree: the exact cycle algebra of the reflection family, a grid policy iteration with no structural restriction, a four-parameter renewal sweep, and a Monte Carlo zoo of 139 policies with exact mint arithmetic; the tested reflection and near-chattering members approach the floor across the simulated regimes, no member undercuts the era-banded floor for its visited band, and in planted jump regimes the infimum rises with tail weight while the optimal reflection point moves inward from the tangency, guarding against straddles.

An eighth surface carries the exact layer and the third-party identity onto Ethereum mainnet, at a twelve-second block and under a different sequencing regime. The fork track and the identity events are reported above. What the arm adds beyond them is a two-sided floor scatter. On the fifth to the third of mainnet firings that carry their corrective swap in the same pool, the swap's own constant-liquidity arithmetic marks the position before the swap, so the realised event cost is measured rather than bounded as it is on Base. Seven cells clear the firing bar, three operators over spans of eighteen to a hundred and fifteen days and 566 swap-carrying firings; all seven sit above the uniform fee-only

floor at the anchor's fee, at a median score of 57.6, and six of the seven sit above their own pool's fee tier. The sample is small and is not a census.

The same arm measures whether the event leg is somebody else's revenue. Against a within-block null, rebalance firings are neither backrun nor sandwiched on either chain, every adjacency ratio sitting at or below one, 0.97 and 0.81 on Base against 0.52 and 0.77 on mainnet; the raw shares alone would have read the other way, which is why the null is the statistic and the share is not. Mainnet firings sit late in their block, a frontrun ratio of 1.41 against a backrun ratio of 0.52, which reads as low-priority ordering rather than extraction and places the mint near the block's closing price, the convention the tick cache already assumes. The dissipation priced here is therefore not, on this evidence, an artefact of anyone taking the other side of it.

25. Machine verification

The verification package holds three kinds of artefact, and they are not interchangeable evidence. Two are executable against deployed bytecode, one is a kernel-checked proof, and the fixed-seed harnesses of Section 24 are finite-precision implementations checked against the stated formulas. The fork suites of the fifth and sixth surfaces run the exact layer against production bytecode, so the arithmetic they check against is the venue's rather than a model of it; this section says how they are built and what a reader has to do to rerun them. The formalisation is not a verification surface at all. It machine-checks the theorem layer over the reals, so a kernel checks those proofs, and it is reported here because the two artefacts answer complementary questions. Fixed-point pool arithmetic is outside the formalisation by construction, and the paper's stochastic steps are outside the fork suites by construction.

25.1. The fork suites

Twenty tests across four contracts, all against unmodified deployments at pinned blocks. Two tracks on Base at block 45,000,000 replay the exact layer against the Aerodrome Slipstream and Uniswap V3 position managers on the same tick-spacing-one-hundred WETH/USDC book, four tests each. A third track on Ethereum mainnet at block 25,200,000 replays the same four results against the Uniswap V3 manager on the tick-spacing-ten WETH/USDT book, where the corner ratio agrees to the wei and the free locus costs zero raw units against Base's one. One track on Avalanche C-Chain at block 92,430,000 replays Appendix B against the bin-step-ten WAVAX/USDC pair of the discrete-bin class's home deployment, eight tests. The mainnet pair is forced rather than chosen, the suite's tolerances being absolute raw units calibrated for a six-decimal token1, which mainnet USDC/WETH inverts. The Base pin sits inside both the census era and the production anchor's amounts window, so the fork tier, the census, and the anchor share an era.

Predictions come from an exact-tick-math reference contract synced to the live pool's state at the pinned block, never from a linearisation, so a disagreement is a disagreement with the deployed arithmetic and not with an approximation of it. Where quantisation makes exactness unavailable the tolerance is stated with its budget and the residual is logged, which is why the two-branch amplitude is reported at parts per million and per thousand by tick spacing while the corner ratios and the mint minimum are asserted to the wei. A clean checkout reproduces every fork result in the previous section; the captured run, with the test-to-result map, is the suite's regression target, and any archive-capable endpoint for each chain suffices, since the pins fix the state.

25.2. The formalisation

The theorem layer is machine-checked in Lean 4 against mathlib, over the reals. Every theorem stated inside the development is checked by the Lean kernel, subject to the foundational axioms the accompanying audit records, which closes on the three standard ones. The scope is the idealised model, and the mapping from file names to the paper's numbering ships with the development.

What is proved is the algebraic and variational spine. The central law is formalised in full, at any width pair and any admissible target range, together with the non-negativity of the kerf and the free locus as an exact characterisation; the abstract half, that for two share pairs summing to one the smaller cross ratio is at most one with equality exactly when the pairs agree, is isolated first and is the entire source of both. The amplitude layer follows, its two branches proved as the exact difference between withdrawn and re-minted value, not posited, with the corner values, the exact fractional slopes, and the down-branch monotonicity threshold in its iff form. The placement-family potentials carry their monotonicity and the impossibility of a free round trip. In the second act the corridor is checked, with the tangency constant coming out exactly two as an infimum and with its analytic corollary, and the impulse inequality is checked in the share coordinate as well, where it is exact at every width and carries no narrow-limit correction; that second coordinate is what shows the constant is not an artefact of the first. Both policy families are checked against the floor, as are the swap sandwich with its constant and its class-coverage argument, the retention collapse, the leg comparison, and the jump surcharge. In the third act the monitoring layer's two exact integrals are proved, and so is the analytic core of the exact-reproduction identity discussed in Section 19.1.

What the formalisation does not carry is the assembly. The floors of Sections 12–14 are stated under continuous monitoring, and the step from the corridor and the impulse inequality to a rate bound runs Itô between impulses, telescopes, and closes with a martingale argument, none of which mathlib supports. The corridor, the impulse inequalities, and the tangency constant are machine-checked; the floors those ingredients assemble into are not.

What is machine-checked in full is the counterpart, Proposition 5. For a quadratic verification function the second-order expansion is an exact identity with no remainder, so under discrete monitoring the scheme runs pathwise on finite sums, and the bound follows at each of the three classes with no stochastic analysis at all. That proposition entered the paper because the formalisation established it, and it is the one floor statement here whose proof a kernel has checked end to end.

The remaining gaps are stochastic, and the development records them. Theorem 1 needs the local-time field of a continuous semimartingale and Itô–Tanaka for a difference of convex functions, neither of which mathlib carries. Proposition 1 needs first-passage theory for the same reason, as does the renewal-reward step behind the cycle constants, so the family-optimum constants of Section 12 are the harness's rather than the formalisation's. The instrument's Gaussian branch estimates are not formalised, only the deterministic assembly of their conclusions. Offset uniformity and exact reproduction are each proved up to one standard step that is carried as a hypothesis: the identification of a pushforward density with a lattice sum in the first case, and an improper change of variables in the second. Nothing downstream of Theorem 1's impulse term depends on the unformalised parts.

Table 5 sets the two sections' surfaces beside each other by the layer of claim each one reaches, with what it does not reach recorded in the same row. The rows are independent, and the paper's strongest statements are the ones several of them support at once.

Table 5. The evidence ledger by claim layer.

Layer	Evidence	Establishes	Does not establish
algebraic	the formalisation and the seeded harnesses	the identities and the impulse inequalities	the stochastic assembly above them
contract	the fork suites, four deployments on three chains	agreement with deployed arithmetic	anything about populations
synthetic	fixed-seed simulation and the calibration surface	estimator behaviour against planted truth	that a real price law is the planted one
production	the operator's own event history	agreement on the recorded events	a law for venues not read
population	the admitted census cells	conditional per-pool readings	recovery of the full jump measure

26. Discussion

26.1. *The ledger and the benchmark frame*

The dissipation ledger sets two cost layers side by side in one forward expansion. The concavity leg is the slot the loss-versus-rebalancing literature occupies,⁴ benchmark-relative and generated by holding a concave payoff against a continuously rebalanced reference; the event leg is generated at re-placement instead, and is attributed to the venue's arithmetic and the operator's own parameters. Questions of venue design and of operating policy are questions about the event leg, the term the policy's own parameters enter. The event leg is also an instrument; its firing statistics invert for the price law and for the policy, as OF demonstrates and as Sections 20–22 carry out; the source-attributed firing record supplies the observables that inversion needs, which a benchmark-relative cost alone does not. Fees and emissions enter the completed ledger as revenue; the concavity leg, the event leg, and the actuation layer of Proposition 4 leave as cost. Each term is measurable from public data.

26.2. *Limitations*

The specialisations are stated for V3-style geometry and make no universality claim beyond it. Within the class the exact layer is asserted against three production deployments on two chains (Section 24), and the venue presentation is instantiated on one discrete-bin class whose algebra is in turn asserted against that class's unmodified home deployment (Appendix B), with no population read there; other architectures enter through Section 8 and must be reconstructed from their own source.

The identity is stated for continuous semimartingales, with jumps handled by the extension of Section 9.1; the jump-crossing term is calibrated per pool by the third act's instrument, behind its disclosure gate.

The offset lemma is conditional on regularity of the firing-state density, and the degenerate continuous-monitoring case shows the condition cannot be dropped. The independence of the corrective displacement from the coarse state holds to the fee-order error carried from OQM.

The second deployment's census era carries no certified jump reading. Every pool at the per-pool bar fails the discreteness admission, scale ratios of 1.2 to 9.8 against admission constants of 16 to 32, and the era-half criterion rejects on the only testable pool. The cross-venue comparison of Section 22.7 is therefore read admission-first, and its equality claim survives where the instrument is admitted and is untested rather than refuted where it is not. A venue-scale replication of the population read is outstanding, and the single certified census is the binding constraint on the third act's generality.

The mainnet arm reads a venue where the behaviour itself is rare. Active liquidity management there fires about thirty times less often than on Base, roughly 11,700 firings over the era against 364,098, so twenty-nine cells clear the firing bar and no mainnet census is available; the third act's population tiers are blocked by that sparsity rather than by the method, which is now measured rather than assumed. Two mainnet-specific filters are measured and not applied here. Just-in-time positions are 12.9 per cent of mainnet position events against 0.2 per cent of Base's, so a mainnet census must classify and exclude them; and only 32 per cent of mainnet firings carry their corrective swap in the pool, the rest routing through aggregators, which leaves their value chain unclosable from the pool's own logs and splits the population into an identified part and an unidentified one. Of the fifty-five busiest pools at the reading bar, eight carry V3 event signatures without belonging to the V3 factory and are excluded rather than mixed in.

The delay cut is matched in wall-clock seconds and not in blocks, because Theorem 6 bounds how far a price moves while the operator is uncorrected, which is elapsed time against diffusion rather than the chain's accounting unit. Base's ten-block cut is twenty seconds, so the mainnet cut is 1.67 blocks, and a small-delay class under two blocks wide is not powered on this population; the cross-chain delay-cut equivalence test is therefore not attempted. The mainnet discreteness ad-

mission is not computed either, so no mainnet reading is quoted against the admission constants of Section 22.7.

Population-level claims are inherited with their published scope. Trigger geometry transfers across operators, while the first-passage slope law is pinned on a single control operator, and nothing here upgrades that.

Further limits bound the third act. Identification beyond integrated tail functionals is conditional on per-pool geometry counts that production supplies in a minority of pools at the conservative census count; the multiscale reconstruction is therefore a conditional result plus a measured census, not a blanket capability, and I claim no recovery of the full jump measure. The level of the jump rate is check-weighted under Cox monitoring; shapes are clean under the exclusion restriction, whose refutable implication survives its first run at limited power (Section 22.6). Per-pool readings below the discreteness admission are quantisation measurements, the quantisation channel of OQM, not jump measurements, and are excluded rather than reinterpreted; the admission now has a venue-scale demonstration, the second deployment's era placing every pool at its reading bar within one decade of the tick quantum, so budget row 8 excludes an entire venue-era at once (Section 22.7). Era discipline binds throughout; the population's own policy drift means per-era statements only, and the era-half replication certifies (F1) at population level while failing it in half the testable pools, so finer-than-era homogeneity is a per-pool admission criterion, not an assumption (Section 22.6).

Dollar significance is deliberately out of scope. The identity prices structure, not size; the event leg's share of aggregate venue flow is a census quantity for the population line, and no result here depends on its magnitude.

26.3. *Operating and design implications*

For an operator the results reduce to three rules, all of them directional and all of them under the execution model. The trigger should be late, the direction the dwell rules of production populations already run in, though nothing here makes a deployed dwell rule optimal once revenue, out-of-range exposure and churn are counted. The correction should be small and the width large within mandate, and the benchmark score of Section 14 prices those two choices against the floor. Nothing in the isolated-class analysis, including $x^* = 0.7153$, applies once swaps are on.

For a designer the results place the decisive choice in the architecture. The floor's existence is a consequence of emission, and the retention architecture removes it from the system-value ledger altogether, at the residual costs Section 15 itemises; below that choice, the quantisation menu prices the placement and sizing rules per event and the cadence term prices the monitoring layer. The design question is whether the architecture keeps the floor, not how closely an operator can approach it.

26.4. *Consumers*

Within the paper the loop closes at Section 23, the jump surcharge priced by the instrument's own functionals. Beyond it, the fingerprinting instrument gains a price-law-conditioned null, an operator's overshoot tail against its pool's measured integrated jump-tail functional, and the defence stated alongside that instrument in OF applies unchanged, a sharper null tightening what jitter has to conceal rather than removing the cover it gives; risk mandates gain a per-pool jump-tail estimate as a direct input to width and trigger choices; the volatility-oracle channel is absorbed as Proposition 16; and the emission-side economics of the ledger upgrades its decision metric from emissions over the concavity leg alone to emissions over the sum of the concavity leg and the floor, the elasticity of the ratio unchanged.³⁵

26.5. *Future work*

On the measurement side, the placement-wedge attenuation and cadence-degradation predictions of Section 4 are testable on existing data. On the instrument side, the open items are the re-designed crossing-conditioned tick-stream validator that the discreteness diagnosis demands, the

drift diagnosis for the pools the era-half replication flags, the multiscale reconstruction beyond integrated tail functionals where the geometry census permits it, and a certified cross-venue equality test, which the executed port of Section 22.7 shows to need a denser-stream era or per-swap trade data on the second deployment. That deployment’s census itself sits outside this paper’s disclosure gate; over the same era, an operator-keyed census of Uniswap V3 on Base admits 1,046 pool-operator cells across 691 operators on 69 pools, 227 clean line-recovery cells across 160 operators at the fingerprinting paper’s bar, an independent evidence surface built by this paper’s own pipeline. On the portability side, each new venue enters the same formula through its own discontinuity set, and the recipe of Section 8 is the working form of that programme.

27. Conclusion

The paper proved one law and developed it in three directions. The share-potential identity prices composition motion exactly; the floor it implies bounds what every admissible policy must pay, hardened by the jump surcharge and deleted only by the architectural act of retention; and the instrument inverts the same law, reading the diffusive scale and the integrated jump-tail functional pool by pool from the firing record of the population that watches it. Each published law of GS, ME, and OQM returns as a specialisation.

What the three share is a consequence worth stating on its own. Because the cost is set by the venue’s arithmetic and not by the operator’s skill, the quantities an operator controls are the ones the ledger prices; what the ledger leaves out, opportunity cost and inventory exposure among it, is excluded by construction rather than absent from the decision. That is why the same formula supports a design bound and a measuring instrument at once. The market process supplies the motion, and deployed code supplies the surfaces, the amplitudes, the costs, and the detectors.

Any venue whose policy is readable from source is a candidate instance of the same formula; the recipe of Section 8 states what its reconstruction requires. Its exact layer has been carried out against a second production deployment of the V3 class, and its slot-filling on a first discrete-bin venue is asserted against that venue’s unmodified home deployment (Appendix B); the population read of a discrete-bin venue is the next demonstration.

Code availability

Everything described in Sections 24 and 25 is in one public repository, <https://github.com/g4nc0r/local-time-against-deployed-code>, with an immutable archival snapshot on Zenodo at <https://doi.org/10.5281/zenodo.21871949>.

The three act harnesses run anywhere Python runs, each printing its checks and exiting nonzero on a failure. The production anchor and the third act’s population probes are deterministic but read an event lake and a tick cache that are not vendored, and their inputs are documented where they live. The fork suites need one archive-capable endpoint per chain and nothing else. The Lean development builds against the mathlib revision pinned in its manifest, and its axiom audit runs as a separate command.

Two things do not ship. The operator layer behind Section 22 carries third-party addresses, and its release is governed by the companion paper’s disclosure terms rather than by this one. Fixed-point venue sources vendored for the fork suites are redistributed under their own licences at the commits named in the suite’s documentation.

The derived artefacts that do ship carry operators under stable pseudonyms rather than addresses, following the companion paper’s naming policy. Every reported quantity is the one computed from the identified data, and only the key differs. Pool, token and position-manager addresses are not obscured, being public contracts that identify no one.

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Appendix A: Notation

Symbols follow the convention of one symbol, one meaning across the companion papers. Where a symbol is anchored by a sibling paper, this paper keeps the anchor. Deliberate local scopes are recorded at the foot of the table.

Symbol	Meaning
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Price process and time

$X_t, \langle X \rangle$	price coordinate (tick coordinate in the concentrated liquidity instance); quadratic variation
ℓ_t^a, ψ_T	local time field; mean occupation density $\mathbb{E}[\ell_T^a]$
μ, σ, θ	era drift and volatility in tick units; $\theta = 2\mu/\sigma^2$
$\tau_k, N_T, \lambda, \Lambda$	firing times; firing count; firing rate; firing compensator
$(t_j), \Delta t, O, \beta$	check times; check interval; overshoot at firing; barrier-shift constant

Venue presentation

$\mathcal{V}, \mathcal{R}, R$	venue presentation; policy-state space; policy state
$c, f, f_R''(da)$	accrual rate; holding value; its second-derivative measure
$F(R), \Gamma, J$	firing set; impulse map; emitted flux amplitude
$\mathcal{D}(R), d_0$	discontinuity set (accrual c , curvature f , trigger imp); interior-placement gap
Φ_T	impulse flux $\sum_{\tau_k \leq T} J_k$
Ξ_T	dissipation measure (Definition 3)

Offset and grid

$\vartheta, \zeta_\vartheta, o$	grid spacing; offset coordinate; placement offset
$Z, q, \bar{p}_\vartheta, TV$	sampled state and its density (Lemmas 1–2); offset density; total variation
$\mathcal{G}, Y, v$	coarse σ -algebra; test variable; conditional total-variation bound

Amplitude geometry (GS frame)

$s, s_a, s_b, \bar{s}$	sqrt price; range bounds; midpoint
h, w, δ, ρ	half-width; width $2h$; displacement from midpoint; $h/\bar{s}$
$L, \phi, V, \Delta R$	liquidity; per-liquidity value; position value; residual amplitude. Primes denote the re-minted position, so L' and V' follow a re-placement

Channel laws (OQM frame)

$\varphi, W, t_\pm$	trigger fraction; range tick width; range tick bounds
$w_u, w_d, \bar{w}$	trigger-line distances, in ticks, from placement; their mean
$\pi^\pm, q^\pm$	firing-direction probabilities; leftover-side mix probabilities
ζ, Δ, g	within-tick offset; corrective swap tick displacement; tick sensitivity of the target share

Monitoring and jumps (Appendix C)

σ_1, n, Q	one-check scale $\sigma\sqrt{\Delta t}$; standard normal density; survival function
ν_0, g_O, p_0, p_1	flat-entry density; overshoot density; window and per-check detection bounds
X^J, μ^X, ν^X, ν	pure-jump part of (24); its jump measure; the compensator; its spatial part per era

The floor (second act)

η, L_p, N_1	pool fee rate; active liquidity; corrective-swap input value
$\sigma_s, [s_-, s_+]$	era sqrt-price volatility (hats denote era estimates); era band
Θ	cadence thinning credit (Proposition 11)
ω_0, ω_1	holdings value shares (5); primes denote target shares
$\mathcal{A}_0 \dots \mathcal{A}_3$	admissible policy classes (Definition 5)
k_j, r	per-event mint kerf, the log-cost (6); kerf rate (11)
$g, g^\uparrow, g^\downarrow$	emitted value fraction $1 - e^{-k}$, equal to the amplitude fraction $\Delta R/V$; its two branches (17)
$\mathcal{C}^\uparrow, \mathcal{C}^\downarrow, P$	one-sided cost potentials (13); their shorthand $P = 2s + h$
U	verification function
$E^\uparrow, E^\downarrow$	era-uniformised envelope slopes, the corridor bounds of (14); their era-price selections are $s^\uparrow, s^\downarrow$ (Appendix F)
$c(x), c_{\text{refl}}(x), c_{\text{min}}, c^*, x^*$	centred-family and reflection-family coefficients; floor constant, whose narrow limit is the tangency constant 2; centred-family optimum
$W_{\text{max}}, h_{\text{max}}, \rho_w, \rho_w$	width cap; its half; cap over era floor W_{max}/s_- ; the fixed-width analogue w/s_-
K_0	absolute constant of (20)
σ_0	state-clock volatility bound of the verification scheme (Appendix F)
G_{gas}, γ	per-event gas cost; gas fraction G_{gas}/V
I	footprint ratio $V/(sL_p)$ (impact coefficient)
A, d, d^*	swap-floor constant (22); share distance; its minimiser
m, x	correction size and trigger of the return-point family, in units of h

V_{sys}, V_d	system value $V + V_d$, position plus ledger; ledger value ($\mathcal{A}_3$)
$(x_d, y_d), (x_{\text{sw}}, y_{\text{sw}})$	ledger balances; corrective-swap token legs
r_J	jump surcharge rate (Proposition 17)
<i>The instrument (third act)</i>	
t_d, a	actuation delay, firing time minus most recent crossing time; its running value
τ_c, Y, O_J	crossing time; pre-jump gap; straddle depth
$\lambda_J, \lambda_c, \pi_J, \pi_c$	jump-crossing and continuous-crossing rates; check-weighted crossing-type shares
χ_t	check intensity (Cox-type admitted)
$g_{\text{act}}, g_{O a}, g_J$	delay density; conditional overshoot kernel; straddle density
$p_{\text{pen}}, d_{\text{dw}}$	penetration threshold; dwell requirement
ψ_b, K	gap-occupation kernel at the line; monitoring-plus-policy kernel
Z_a, Π_c, Π_J	normalised post-crossing displacement; crossing-type delay laws
$\bar{\nu}, m_1, \mathcal{J}(o)$	jump tail $\nu((\cdot, \infty))$; mean positive jump mass; integrated tail $\int_0^\infty \bar{\nu}$
$\mathcal{E}, \hat{\mathcal{E}}$	exceedance functional and its estimate
$\bar{O}, \hat{\pi}_J$	per-event standardised overshoot $O/(\hat{\sigma}\sqrt{t_d})$; jump-share estimate
$\kappa, \bar{a}, o_{\min}$	cut multiplier; delay cut; resolution floor $\kappa\sigma\sqrt{\bar{a}}$
$\hat{\sigma}_{\text{BV}}, \hat{\sigma}_{\text{RV}}$	bipower and realised-variance trailing local scales
d_i, H, S	window displacements; trailing window length; window span
α	regular-variation tail index
$\epsilon_{\text{ker}}, \epsilon_{\text{cens}}$	kernel mis-specification error; crossing-staleness error
c_{tick}	discreteness admission constant, applied in (25) with ϑ at one tick

Local scopes, recorded by symbol family.

- q and Z are confined to the lemmas of Section 4 and are unrelated to OQM's $q^\pm$.
- R is the policy state and never GS's token ratio; ΔR is GS's residual, not an increment of R .
- The Φ family is reserved across the companion papers for directional dust flows and N is the firing count, so the normal survival function is the Gaussian Q .
- g is the emitted value fraction throughout the first two acts; Section 18.1 uses it locally for the tick sensitivity of the target share, and g_O is the overshoot density.
- m is the correction size of the return-point family; m_1 is the mean positive jump mass, distinct from it.
- In Proposition 5 and its proof the subscripted δ_i and primed δ'_i are the displacements entering and leaving one monitored step, and n counts those steps; elsewhere δ carries no time index and n is not a step count.
- The arrow superscript marks the up and down branch throughout, on the potentials $\mathcal{C}^\uparrow, \mathcal{C}^\downarrow$, their envelope slopes $E^\uparrow, E^\downarrow$ and era-price selections $s^\uparrow, s^\downarrow$, and the amplitudes $\Delta R_\uparrow, \Delta R_\downarrow$ with their fractions $g_\uparrow, g_\downarrow$; the E family is distinct from the expectation $\mathbb{E}$ and the exceedance functional $\mathcal{E}$.
- The delay subscript in t_d is the only role of d in the third act; the share distance d , the placement gap d_0 , the dwell requirement d_{dw} , the window displacements d_i , and the ledger balances (x_d, y_d) are elsewhere.
- A hat denotes an estimate throughout ($\hat{\sigma}, \hat{\pi}_J, \bar{O}, \hat{\mathcal{E}}$); the quadratic-variation sampling $\hat{X}$ of Corollary 1 is the one exception.
- N is the firing count, split as $N^c + N^J$ by crossing type, with N_1 the corrective-swap notional; the normal law $N(0, \cdot)$ of Appendix G and the proof-local numerator N of Appendix F are the two scoped exceptions.
- The footprint ratio is I so that F stays the firing set and no calligraphic letter competes with $\mathcal{J}$.
- κ is the exceedance-cut multiplier, unrelated to the value-channel constant of OQM, which this paper never uses; the integrated tail is calligraphic $\mathcal{J}$ because Λ is the firing compensator; π_J, π_c extend the $\pi^\pm$ family from firing directions to crossing types; and the trailing-window length is H because L is liquidity.
- Four letters carry section-local senses beside their anchored ones and are always disambiguated in place; Y (test variable of Lemma 2; pre-jump gap in the straddle law), a (price point; delay value; proof-local interval endpoint), g (test function; amplitude fraction $g(\delta)$); the density family $g_{\text{act}}, g_{O|a}, g_J$, and c (accrual rate; policy coefficients $c(x)$ and $c_{\text{refl}}(x)$); the constants $c_{\min}, c^*, c_{\text{tick}}$.
- S is the estimator window span; Appendix C's scale function $S(x)$ and its constant κ , and Appendix D's branch retention ratios $Q^\uparrow, Q^\downarrow$, are proof-local.
- Appendix B reads the Liquidity Book in its own contract constants, scoped to that appendix; b (bin step), $P(i)$ (bin price), L_i, x_i, y_i (bin depth and reserves), B, C_v, v (base factor, variable-fee control, volatility accumulator), and the unit point mass $\delta_{P(i)}$. There η keeps its anchored sense of pool fee rate and ϑ its sense of grid spacing.

Appendix B: A discrete-bin instantiation of the venue presentation

This appendix reads a venue outside the V3 class through Definition 1, filling the five slots from contract source and displaying the discontinuity set and its amplitudes. The venue is the Trader Joe Liquidity Book,⁴³ whose deployed pair contract quotes a discrete book of bins rather than a continuous curve. It is the atom case the definition names and no instance in the body exercises, the holding value being piecewise linear in the price coordinate, so that its second-derivative measure is carried entirely on atoms. The appendix is algebra and source reading, and the algebra is bytecode-asserted; the fork suite of Section 24 replays the price law, depth, atom amplitude, per-traversal loss, and fee shape below against an unmodified deployed pair of this venue on its home chain, exact to the bit or the wei wherever the quantity is integer-valued. Symbols in this appendix are the venue's own contract constants and are scoped to it (Appendix A).

A pair contract of bin step b , an immutable constant in basis points, assigns to the bin of index i the single price

$$P(i) = (1 + b 10^{-4})^{i-2^{23}}$$

(`PriceHelper.getPriceFromId`), and the reserves x_i of the base token and y_i of the quote token in bin i carry the constant-sum invariant

$$L_i = P(i) x_i + y_i$$

(`BinHelper.getLiquidity`), the bin's depth in quote units. The whole bin trades at exactly $P(i)$; a swap consumes the active bin at its price and steps the active index to the next non-empty bin. Bins above the active index hold only the base token and bins below only the quote token (`BinHelper.verifyAmounts`), so the book is a ladder of single-price positions and the active index is the venue's price coordinate. In the tick coordinate of Section 3.1 the bin grid has spacing $\vartheta = \ln(1 + b 10^{-4}) / \ln 1.0001 \approx b$ ticks for small b .

The five slots of Definition 1 then read as follows.

- (i) *Policy state space.* A policy state is a finitely supported depth vector over bin indices together with the strategy's parameters, $R = ((L_i)_{i \in I}, \pi)$; the venue tracks the position as fungible per-bin shares (LBToken), and the pool-side composition of the active bin is venue state, not policy state.
- (ii) *Accrual rate.* Swap fees are credited to the reserves of the bin that executes the swap, pro rata to shares (`LBPair.swap`), so the accrual is supported on the held-bin prices and is activated by swap traversal rather than by occupancy alone, proportional to traversal volume at the fee rate η below, and $\mathcal{D}_c(R)$ is the set of held-bin prices. This is the first instance in the paper to populate the accrual slot.
- (iii) *Holding value.* A bin of depth L_i is worth L_i in quote units when the price is above $P(i)$, its reserves all quote, and $P L_i / P(i)$ when the price P is below, its reserves all base and marked at P , so

$$f(P, R) = \sum_{i \in I} \frac{L_i}{P(i)} \min(P, P(i)),$$

concave and piecewise linear in the price coordinate native to the venue's arithmetic.

- (iv) *Firing set.* The pair contract encodes no trigger; the firing set lives in operator strategy code. Deployed strategies are written on bin indices, a ladder on $[i_-, i_+]$ firing when the active index

leaves a sub-band, $F(R) = (0, P(i'_-)] \cup [P(i'_+), \infty)$ for grid thresholds $i_- \leq i'_- < i'_+ \leq i_+$, so the trigger surfaces are grid prices.

- (v) *Impulse map.* A re-placement burns the ladder, receiving per-bin pro-rata reserves under floor division (`BinHelper.getAmountOutOfBin`), and re-mints a new ladder centred on the active index. The active-bin leg pays the composition fee below, the emitted flux is the fee total plus the rounding remainders, and conservative emission (1) holds up to these source-derivable terms, the additive carve-out Section 3.3 anticipates.

The discontinuity set at state R therefore sits on one grid,

$$\mathcal{D}(R) \subset \{P(i) : i \in \mathbb{Z}\}, \quad f_R''(dP) = - \sum_{i \in I} \frac{L_i}{P(i)} \delta_{P(i)}(dP),$$

with $\delta_{P(i)}$ the unit point mass at $P(i)$. The second-derivative measure is purely atomic; the atom at $P(i)$ has amplitude $L_i/P(i)$, the bin's full base-token depth. The accrual discontinuities and the strategy trigger surfaces land on subsets of the same grid, so the venue's entire discontinuity structure is the bin grid.

The per-bin amplitude has a traversal form. The local-time term of (2) evaluates against the atoms as a sum, $-\frac{1}{2} \sum_k \sum_i (L_i/P(i)) \Delta_k \ell_T^{P(i)}$ in the price coordinate, local time read at grid points rather than integrated against a density. A complete downward traversal of bin i converts the depth L_i of quote into $L_i/P(i)$ of base at $P(i)$, which marked at the next grid price $P(i)/(1 + b 10^{-4})$ is a decline of

$$L_i \left(1 - \frac{1}{1 + b 10^{-4}} \right) = L_i \frac{b}{10^4 + b},$$

exact in the two contract constants, with the upward traversal its mirror in base units. Since local time at a level is the scaled limit of downcrossing counts,⁵ the atom term reproduces this per-traversal loss at first order in the bin step; it is the discrete-bin analogue of the in-range concavity cost of the V3 instance, concentrated on the grid instead of spread over a density.

The fee terms are also read from source, exact up to the fixed-point rounding of the fee libraries. The fee rate is $\eta = \eta_0 + \eta_v$ with base $\eta_0 = B b 10^{-8}$ for the pool's base factor B , and variable $\eta_v = C_v (vb)^2 10^{-20}$ for the pool's variable-fee control C_v (`PairParameterHelper`), where the volatility accumulator v rises by 10^4 per bin the swap crosses relative to a reference index, is capped, decays by the pool's reduction factor once the filter period passes, and resets past the decay period. The venue meters its own crossing count, the observable the third act's instrument reads, and prices fees quadratically in it. A mint into the active bin at a composition different from the bin's executes an implicit swap and pays a composition fee of $\eta(1 + \eta)$ per unit of the leg the swap moves (`FeeHelper.getCompositionFee`), the gross-quoted form of the venue's fee charge; a protocol share is removed from every fee and the remainder joins the executing bin's reserves. These are the impulse map's fee terms.

Assembled, the terms of (2) and (4) are the slots above, with the local-time term the atom sum. Equation (3) then prices a Liquidity Book ladder as it prices a V3 range, with occupation of a density replaced by local time at a grid. The reading carries the standing hypothesis of Section 3.1, that the price coordinate is a continuous semimartingale; what is atomic here is the venue's curvature and not the price. A ledger written on the pool's own quoted price, which moves only from grid point to grid point, is a different object and is not what this appendix instantiates. The amplitude specialisations of Act I were not used, the venue supplying piecewise-linear amplitudes directly. The empirical instantiation, an event-reconstructed census of Liquidity Book re-placements, is outside this paper's scope.

Appendix C: Proofs for the equidistribution, intensity, and monitoring layers

C.1. Lemma 1 (Offset Uniformity)

The pushforward density is $\bar{p}_\vartheta(z) = \vartheta \sum_{k \in \mathbb{Z}} q(\vartheta(k+z))$, which integrates to one over $[0, 1)$ by Tonelli. For $z < z'$ the points $\vartheta(k+z), \vartheta(k+z')$ interleave as a strictly increasing sequence, so

$$|\bar{p}_\vartheta(z) - \bar{p}_\vartheta(z')| \leq \vartheta \sum_k |q(\vartheta(k+z')) - q(\vartheta(k+z))| \leq \vartheta \text{TV}(q),$$

the increments along any increasing sequence being dominated by the total variation. Either $\bar{p}_\vartheta$ is identically one, when the sup bound is immediate, or, integrating to one on $[0, 1)$, it takes values on both sides of one and the oscillation bound gives the sup bound; the total-variation statement follows by integrating. ■

Throughout this appendix n and Q denote the standard normal density and survival function, and $\sigma_1 = \sigma\sqrt{\Delta t}$ the one-check diffusion scale.

C.2. Regularity of the mean occupation density

Lemma 12 (Drift-Uniform Regularity). *For the era model $X_t = x_0 + \mu t + \sigma B_t$, the mean occupation density satisfies $\text{TV}(\psi_T) \leq 2\sigma\sqrt{2T/\pi}$ for every drift μ , and $\mathbb{E}\langle X \rangle_T = \sigma^2 T$; the bound of Corollary 1 therefore holds for the drifted era model with the Brownian constant.*

Proof. The occupation-times formula and Fubini give $\psi_T(a) = \sigma^2 \int_0^T p_t(a) dt$ with p_t the density of X_t , a Gaussian of variance $\sigma^2 t$ whatever its mean. Total variation is subadditive under mixtures and $\text{TV}(p_t) = 2 \sup p_t = 2/(\sigma\sqrt{2\pi t})$, so

$$\text{TV}(\psi_T) \leq \sigma^2 \int_0^T \frac{2}{\sigma\sqrt{2\pi t}} dt = 2\sigma\sqrt{2T/\pi}.$$

■

C.3. Proposition 1 (cycles and the direction split)

For the integrability claim of (3), if f'_- is bounded on the interval visited by the stopped era path then the stochastic integral is a martingale by the standard localisation argument,⁵ and its expectation vanishes.

For finiteness of the mean cycle, let $w_u + w_d$ be the barrier separation and $t_0 = \Delta t + (w_u + w_d)^2/\sigma^2$. In any window of length t_0 there is a check time whose elapsed time from the window's start lies in $[t_0/2, t_0]$; from any interior state the traversal required to sit beyond a barrier at that check is at most $w_u + w_d$, so the probability that the cycle ends within the window is at least $p_0 = Q((w_u + w_d + |\mu|t_0)/(\sigma\sqrt{t_0/2})) > 0$, uniformly in the starting state. Hence $\mathbb{P}(\text{cycle} > kt_0) \leq (1-p_0)^k$ and the mean cycle is finite. Firings are regeneration times by the strong Markov property and era-constant policy, and where the policy returns to a common post-firing geometry the cycles are independent and identically distributed, so $N_T/T \rightarrow 1/\mathbb{E}[\text{cycle}]$ by the strong law for renewal processes. A policy whose post-firing state varies across firings, the grid placement offset of Proposition 10 being the case this paper meets, carries that state into the cycle and the statement is the corresponding Markov-renewal one, with the same limit under the era model. The continuous-monitoring direction split is the two-barrier exit computation through the scale function $S(x) = (1 - e^{-\theta x})/\theta$,^{6,44} the monitoring correction is bounded in the next subsection.

C.4. Proposition 4 (overshoot, thinning, mis-attribution)

The flat-entry hypothesis is that, conditional on no earlier detection, the pre-check state has a density ν_0 that is constant up to $o(1)$ across the $O(\sigma_1)$ window below the barrier b which contributes to the next check's law, and that $|\mu|\Delta t = o(\sigma_1)$. Conditional on the next check detecting, the overshoot $O = X_{\text{check}} - b$ has density proportional to

$$\frac{1}{\sigma_1} \int_{y < b} n\left(\frac{o + b - y}{\sigma_1}\right) \nu_0(y) dy,$$

and substituting $\xi = (b - y)/\sigma_1$ under the hypothesis gives $g_O(o) \propto \int_0^\infty n(o/\sigma_1 + \xi) d\xi = Q(o/\sigma_1)$. Normalising with $\int_0^\infty Q(o/\sigma_1) do = \sigma_1/\sqrt{2\pi}$ yields

$$g_O(o) = \frac{\sqrt{2\pi}}{\sigma_1} Q(o/\sigma_1), \quad o \geq 0,$$

a decreasing density with $g_O(0) = \sqrt{2\pi}/(2\sigma_1)$, total variation $\text{TV}(g_O) = 2g_O(0) = \sqrt{2\pi}/\sigma_1$, and mean

$$\mathbb{E}[O] = \sqrt{2\pi} \sigma_1 \int_0^\infty v Q(v) dv = \frac{\sqrt{2\pi}}{4} \sigma_1 \approx 0.63 \sigma_1.$$

This is the density Lemma 2 consumes, and it degenerates as claimed when $\Delta t \rightarrow 0$. Conditioning on failures at earlier checks tilts the entry law near the barrier; the exact sequential constant is the continuity correction of Broadie, Glasserman, and Kou,³⁹ under which discrete monitoring is equivalent at leading order to continuous monitoring with each barrier shifted outward by $\beta\sigma_1$, $\beta \approx 0.5826$. The offset application uses only $\text{TV}(g_O) = O(1/\sigma_1)$, which both constants satisfy.

For the thinning statement, the barrier-shift equivalence makes the realised driftless cycle mean $(w_u + \beta\sigma_1)(w_d + \beta\sigma_1)/\sigma^2$, so cadence alone thins intensity by $w_u w_d / ((w_u + \beta\sigma_1)(w_d + \beta\sigma_1))$, close to one for production geometries. Firing rates far below the raw first-passage prediction are therefore carried by dead-band and dwell rules, which enlarge the continuation region and enter the framework through $F(R)$; their scale is an input, not a derivation.

For mis-attribution, assume the per-check detection probability during an ongoing barrier excursion is bounded below by p_1 (flat entry gives $p_1 \approx 1/2$). On the event that a cycle whose first touch is the upper barrier is recorded at the lower, detection failed at $k - 1$ consecutive checks for some $k \geq 1$ while the path travelled the separation $w_u + w_d$ downward in time at most $k\Delta t$, so

$$\mathbb{P}(\text{mis-attribution}) \leq \sum_{k \geq 1} (1 - p_1)^{k-1} \cdot 2 Q\left(\frac{w_u + w_d}{\sigma\sqrt{k\Delta t}}\right).$$

With $Q(x) \leq e^{-x^2/2}$ and $\kappa = \log(1/(1 - p_1))$, the k -th term is at most $2 \exp(-(k - 1)\kappa - (w_u + w_d)^2/(2\sigma^2 k\Delta t))$; the exponent is minimised near $k^* = (w_u + w_d)/(\sigma\sqrt{2\kappa\Delta t})$ and the sum is $O(\exp(-\sqrt{2\kappa}(w_u + w_d)/(\sigma\sqrt{\Delta t})))$, exponentially small in $(w_u + w_d)/(\sigma\sqrt{\Delta t})$ as stated. ■

C.5. The jump-crossing compensator

For X a semimartingale with jump measure μ^X and predictable compensator ν^X , the Meyer–Itô formula³⁸ replaces Itô–Tanaka in Theorem 1, adding the jump sum of Section 9.1 with ℓ the local

time of the continuous martingale part. The firing process splits as $N = N^c + N^J$ with

$$N^J(dt) = \int_{\mathbb{R}} \mathbf{1}\{X_{t-} \notin F(R_{t-}), X_{t-} + z \in F(R_{t-})\} \mu^X(dt, dz),$$

whose predictable compensator is the same integrand against ν^X . The impulse amplitude at a jump firing is evaluated at the landing state $X_{t-} + z$, whose overshoot past the surface is bounded below by the jump's straddle depth, which is the observability statement of Section 9.1.

Appendix D: Proofs for the cost structure

D.1. Lemma 4 (Exact Fractional Slopes)

The two-branch amplitude (8) has upper branch $\Delta R(\delta) = L\delta(2\bar{s} + h + \delta)/(\bar{s} + h)$ and position value $V(\bar{s}) = L(2\bar{s}h + h^2)/(\bar{s} + h) = Lh(2\bar{s} + h)/(\bar{s} + h)$, so $g'(0^+) = \Delta R'(0^+)/V(\bar{s}) = 1/h$ with the $(2\bar{s} + h)/(\bar{s} + h)$ factors cancelling exactly. The lower branch is the continued upper branch times s/s'_b , whose $\delta \rightarrow 0^-$ limit is $\bar{s}/s_b$. ■

D.2. Theorem 2 (Share-Potential Identity)

Withdrawn amounts (x, y) at price s , value $V = xs^2 + y$, holdings shares $\omega_0 = xs^2/V$, $\omega_1 = y/V$. The new range's per-liquidity amounts (x_u, y_u) have target value $v_t = x_us^2 + y_u$ and target shares $\omega'_0 = x_us^2/v_t$, $\omega'_1 = y_u/v_t$. If token0 binds, $L' = x/x_u$ and

$$\frac{V'}{V} = \frac{(x/x_u)v_t}{V} = \frac{xs^2}{V} \cdot \frac{v_t}{x_us^2} = \frac{\omega_0}{\omega'_0},$$

and symmetrically ω_1/ω'_1 if token1 binds. The mint minimum selects the smaller ratio, so the retained fraction is $\min(\omega_0/\omega'_0, \omega_1/\omega'_1)$ and (7) follows; non-negativity holds because the shares on each side sum to one, so the two ratios cannot both exceed one. Equality at matched shares is immediate. ■

D.3. Corollary 4 (Placement-Family Potentials)

For the equal-width family write the withdrawn amounts per unit liquidity as $x_w = (h - \delta)/(s s_b)$ and $y_w = h + \delta$ with $s_b = s + h - \delta$, the new-range units as $x_u = (h - \delta')/(s s'_b)$ and $y_u = h + \delta'$ with $s'_b = s + h - \delta'$, and the per-liquidity value at displacement z as $\phi(z) = (hP - z^2)/(s + h - z)$, an algebraic identity from $\phi = 2s - s^2/s_b - s_a$. The token0 branch retains

$$\frac{x_w}{x_u} \cdot \frac{\phi(\delta')}{\phi(\delta)} = \frac{(h - \delta)s'_b}{(h - \delta')s_b} \cdot \frac{(hP - \delta'^2)s_b}{(hP - \delta^2)s'_b} = \frac{Q^\uparrow(\delta)}{Q^\uparrow(\delta')}, \quad Q^\uparrow(z) = \frac{h - z}{hP - z^2},$$

and the token1 branch retains $Q^\downarrow(\delta)/Q^\downarrow(\delta')$ with $Q^\downarrow(z) = (h + z)(s + h - z)/(hP - z^2)$; taking $\mathcal{C}^\uparrow = -\ln Q^\uparrow$ and $\mathcal{C}^\downarrow = -\ln Q^\downarrow$ gives (12). Monotonicity under the standing hypothesis follows from

$$(\mathcal{C}^\uparrow)'(z) = \frac{1}{h - z} - \frac{2z}{hP - z^2}, \quad (\mathcal{C}^\downarrow)'(z) = -\frac{2z}{hP - z^2} - \frac{1}{h + z} + \frac{1}{s + h - z},$$

the second term of the first bounded by $1/s < 1/h$, and the outer terms of the second bounded by $2/s$ against $1/(h + z) \geq 1/(2h)$ and $2h < s/2$. Setting $\delta' = 0$ and clearing denominators reproduces

the two-branch amplitude on both branches; the reduction is checked independently in the reference harness. ■

Appendix E: Proofs for the amplitude layer

Notation as in Section 6; write h for the half-width, $s = \bar{s} + \delta$, withdrawn amounts $x_w = L(1/s - 1/s_b)$ and $y_w = L(s - s_a)$, and the mint minimum

$$L' = \min\left(\frac{x_w}{1/s - 1/(s+h)}, \frac{y_w}{h}\right).$$

E.1. Lemma 3 (Binding Side)

With $1/s - 1/s_b = (h - \delta)/(s s_b)$ and $1/s - 1/(s+h) = h/(s(s+h))$, the token0 term is $L(h - \delta)(s+h)/(h s_b)$ and the token1 term is $L(h + \delta)/h$. Their difference has the sign of

$$(h - \delta)(\bar{s} + \delta + h) - (h + \delta)(\bar{s} + h) = -\delta(2\bar{s} + h + \delta),$$

which is non-negative exactly when $\delta \leq 0$ (the bracket being positive for admissible parameters). Hence token1 binds for $\delta \leq 0$ and token0 for $\delta \geq 0$. ■

E.2. Proposition 2 (Two-Branch Amplitude)

The upper branch is GS's closed form, obtained by substituting the token0-limited liquidity into the value difference; its normal form $L\delta(2\bar{s} + h + \delta)/(\bar{s} + h)$ is machine-checked in GS's companion formalisation. For $\delta \in [-h, 0]$, token1 binds by Lemma 3, so $L' = L(h + \delta)/h$, and with the per-liquidity value of the new range

$$\phi(s, s-h, s+h) = s+h - \frac{s^2}{s+h} = \frac{h(2s+h)}{s+h},$$

the minted value is $V' = L(h + \delta)(2s+h)/(s+h)$. The withdrawn value is

$$V = L\left(2s - \frac{s^2}{s_b} - s_a\right) = L \frac{2\bar{s}h + h^2 + 2\delta h - \delta^2}{\bar{s} + h}.$$

Subtracting and clearing denominators, the numerator factorises,

$$(2\bar{s}h + h^2 + 2\delta h - \delta^2)(\bar{s} + h + \delta) - (h + \delta)(2\bar{s} + 2\delta + h)(\bar{s} + h) = -\delta(\bar{s} + \delta)(2\bar{s} + h + \delta),$$

which gives the lower branch of (8). The ratio to the continued upper branch is $(\bar{s} + \delta)/(\bar{s} + h + \delta) = s/s'_b$. After conversion of OQM's token0-denominated swap-free credit to token1 value at the current price, the lower branch agrees with that paper's published form, as it must. ■

E.3. Corollary 3 (Corner Values)

At $\delta = h$ the withdrawn position is single-sided in token1, $x_w = 0$, so the token0 term vanishes and $L' = 0$; the amplitude is the full withdrawn value $V(s_b) = L(s_b - s_a) = Lw$. At $\delta = -h$ the position is single-sided in token0, $y_w = 0$, the token1 term vanishes, $L' = 0$, and the amplitude is $V(s_a) = L(1/s_a - 1/s_b) s_a^2 = Lw s_a/s_b$. The interior limit follows by expanding (8) at $\delta \rightarrow 0^-$, where the correction factor s/s'_b tends to $\bar{s}/s_b$. ■

E.4. Proposition 3 (Down-Branch Monotonicity)

Write $m = -\delta \in (0, h]$, with $0 < h < \bar{s}$ under the standing hypothesis so that $\bar{s} - m > 0$ throughout, and $G(m) = m(\bar{s} - m)(2\bar{s} + h - m)/(\bar{s} + h - m)$, so $\Delta R_\downarrow = L G(m)/(\bar{s} + h)$. The logarithmic derivative

$$\frac{G'(m)}{G(m)} = \frac{1}{m} - \frac{1}{\bar{s} - m} - \frac{1}{2\bar{s} + h - m} + \frac{1}{\bar{s} + h - m}$$

is strictly decreasing in m , since its derivative is $-1/m^2 - 1/(\bar{s} - m)^2 - 1/(2\bar{s} + h - m)^2 + 1/(\bar{s} + h - m)^2 < 0$, the last term being dominated by the second. Its minimum over $(0, h]$ is therefore at $m = h$, where its sign is that of

$$\frac{1}{h} + \frac{1}{2\bar{s}} - \frac{1}{\bar{s} - h} \propto 2\bar{s}^2 - 3\bar{s}h - h^2.$$

Strict monotonicity on $(0, h]$ is thus equivalent to $\rho^2 + 3\rho - 2 \leq 0$ with $\rho = h/\bar{s}$, that is $\rho \leq (\sqrt{17} - 3)/2$; in range-ratio terms $s_b/s_a = (1 + \rho)/(1 - \rho) \leq (3 + \sqrt{17})/2$. Above the threshold G'/G is negative at $m = h$ while the $1/m$ term carries it to $+\infty$ as $m \downarrow 0$, so, being strictly decreasing, it changes sign exactly once and G has the interior maximum. ■

Appendix F: Proofs for the floors

F.1. The verification scheme

Let ξ_t be a state process on an interval I , continuous between paid impulses, with $d\langle \xi \rangle_t \geq \sigma_0^2 dt$ and drift bounded by b_0 , and let $U \in C^{0,1}(I)$ have U' of bounded variation with $U'' \geq \mu_0$ as a measure and $\sup |U'| = U_1$. Suppose every paid impulse satisfies $U(\xi^-) - U(\xi^+) \leq k$. Telescoping U along the controlled path, Itô–Tanaka between impulses and the impulse inequality at them,

$$\sum_j k_j \geq U(\xi_0) - U(\xi_T) + \int_0^T \left[\frac{1}{2} \mu_0 \sigma_0^2 - U_1 b_0 \right] dt + M_T,$$

with M a true martingale when U' is bounded and $\mathbb{E}\langle \xi \rangle_T < \infty$. Every instantiation below runs on a bounded state interval with U bounded on it, so $T^{-1} \mathbb{E}|U(\xi_T) - U(\xi_0)| \rightarrow 0$ and the boundary term leaves the rate; expectations and $T \rightarrow \infty$ then give $r \geq \frac{1}{2} \mu_0 \sigma_0^2 - U_1 b_0$. Chattering sequences need no exclusion, every impulse being priced individually with non-negative cost, so accumulation points telescope by monotone convergence.

F.2. Theorems 3 (Fixed-Width Floor)

State $\xi = \delta$ on $[-h, h]$, $\sigma_0 = \sigma_s$, $b_0 = 0$. The era-uniformised envelopes are, for $z \in (-h, h)$,

$$E^\uparrow(z) = \frac{1}{h - z} - \frac{2z}{h(2s^\uparrow(z) + h) - z^2}, \quad E^\downarrow(z) = -\frac{2z}{h(2s^\downarrow(z) + h) - z^2} - \frac{1}{h + z} + \frac{1}{s_- + h - z},$$

with the era-price selections $s^\uparrow(z) = s_-$ for $z \geq 0$ and s_+ otherwise, and $s^\downarrow(z)$ the opposite, so that $E^\uparrow \leq (\mathcal{C}^\uparrow)'$ and $E^\downarrow \geq (\mathcal{C}^\downarrow)'$ for every era price, the lower bound term-wise and conservative by $O(1/s_-)$. The impulse inequality is equivalent to the corridor $E^\downarrow \leq U' \leq E^\uparrow$, since $U - \mathcal{C}^\uparrow$

nonincreasing prices upward translations and $U - \mathcal{C}^\downarrow$ nondecreasing the downward ones. With $m_0 = 2c_{\min}/h^2$ set

$$U'(\delta) = m_0 \delta + \sup_{\delta_1 \leq \delta} [E^\downarrow(\delta_1) - m_0 \delta_1],$$

a nondecreasing-plus-linear form, so $U'' \geq m_0$ as a measure; $U' \geq E^\downarrow$ by the $\delta_1 = \delta$ term, $U' \leq E^\uparrow$ by the defining infimum (14) applied to each pair, and U' is bounded because $E^\downarrow(\delta_1) - m_0 \delta_1$ is bounded above, its blow-up being to $-\infty$ at the lower corner. The scheme applies. Boundary behaviour needs no lemma. A policy whose displacement reaches $\pm h$ must impulse there, the isolated mint surrenders the entire position from a corner (the binding-side minimum vanishes), so such impulses cost more than any U increment, and free corner slides move nothing.

For the analytic corollary, convexity of $1/(h-z)$ gives the tangent bound $1/(h-\delta_2) \geq 4\delta_2/h^2$ and likewise $1/(h+\delta_1) \geq -4\delta_1/h^2$, so the uncorrected corridor gap dominates $\max(4(\delta_2 - \delta_1)/h^2, 1/h)$, each term of the sum being at least $1/(2h)$; the era corrections total at most $3/s_-$ in magnitude, two terms bounded by $1/s_-$ each and one by $1/s_-$. Splitting at $\delta_2 - \delta_1 = h/4$, both cases give gap over distance at least $(4/h^2)(1 - 3h/s_-)$, hence $c_{\min} \geq 2(1 - 3h/s_-)$. ■

F.3. Proposition 5 (Pathwise Discrete Floor)

For a quadratic U the second-order expansion is an identity rather than an approximation,

$$U(y) - U(x) = U'(x)(y - x) + \frac{1}{2}U''(y - x)^2,$$

with no remainder at any step size, which is the whole reason no stochastic calculus is needed. Applying it on the monitored move of step t and summing,

$$\sum_{t < n} [U(\delta'_t) - U(\delta_t)] = \sum_{t < n} U'(\delta_t)(\delta'_t - \delta_t) + \frac{1}{2}U'' \sum_{t < n} (\delta'_t - \delta_t)^2.$$

The monitored moves and the impulses alternate, so

$$U(\delta_n) - U(\delta_0) = \sum_{t < n} [U(\delta'_t) - U(\delta_t)] + \sum_{t < n} [U(\delta_{t+1}) - U(\delta'_t)],$$

and the impulse inequality bounds each term of the second sum below by $-k_t$. Substituting the first display and the drift hypothesis gives (16). Chattering needs no exclusion, since every impulse is priced individually and kerfs are non-negative; a step carrying no impulse contributes nothing to either sum. The three instantiations are the corridor's two verification functions, $U = 2\delta^2/h^2$ on $[-h, h]$ and $U(\omega) = 8(\omega - \frac{1}{2})^2$ on $[0, 1]$, and the swap class's quadratic at the sandwich constant. ■

F.4. Proposition 6 (Achievability)

Fires from the midpoint at $\pm xh$ are equally likely for the driftless era model, the mean cycle is $x^2 h^2 / \sigma_s^2$, and the per-event log-costs are $-\ln(1 - g_\pm(x))$ with the exact event fractions; renewal reward gives (17). In the narrow limit both fractions tend to x and minimising $-\ln(1 - x)/x^2$ gives the stated stationarity condition; solving it numerically gives $x^* = 0.7153$, $c^* = 2.4554$. ■

F.5. Proposition 7 (Sharpness of the Floor)

Fix $x \in (0, 1)$ and $0 < \varepsilon < x$. After every firing the displacement is $\pm(x - \varepsilon)h$, and by the symmetry of the driftless era model the pair (cycle cost, cycle length) has the same law from either

side, so renewal reward applies to the firing cycle. From $(x - \varepsilon)h$ the price exits $(-xh, xh)$ at the near barrier with probability $1 - \varepsilon/(2x)$ and at the far barrier with probability $\varepsilon/(2x)$, and the mean exit time is the exact Brownian product form

$$\mathbb{E}[\tau] = \frac{(xh - (x - \varepsilon)h)(xh + (x - \varepsilon)h)}{\sigma_s^2} = \frac{\varepsilon(2x - \varepsilon)h^2}{\sigma_s^2}.$$

In the narrow limit the potentials of Corollary 4 are $-\ln(h \mp \delta)$ up to constants, so the near-side impulse, from xh to $(x - \varepsilon)h$ on the same side, costs

$$\ln \frac{h - (x - \varepsilon)h}{h - xh} = \ln \left(1 + \frac{\varepsilon}{1 - x}\right) = \frac{\varepsilon}{1 - x} + O(\varepsilon^2),$$

and the far-side impulse costs the same by symmetry; the expected cycle cost is $\varepsilon/(1 - x) + O(\varepsilon^2)$. Renewal reward gives

$$r = \frac{\varepsilon/(1 - x) + O(\varepsilon^2)}{\varepsilon(2x - \varepsilon)h^2/\sigma_s^2} \longrightarrow \frac{\sigma_s^2}{2x(1 - x)h^2} \quad (\varepsilon \rightarrow 0),$$

which is (18); the minimum of $1/(2x(1 - x))$ over x is 2 at $x = 1/2$. Each member of the family is locally finite (its cycles have positive mean length) and admissible in $\mathcal{A}_0$, the price being interior to the re-placed range, so $\inf_{\mathcal{A}_0} r \leq 2\sigma_s^2/h^2$ in the narrow limit; Theorem 3 gives the matching lower bound. The infimum is not attained, the firing rate of the approximating policies diverging as $1/\varepsilon$, the standard boundary behaviour of singular control. ■

F.6. Lemma 5 and Theorem 4 (Width-Uniform Floor)

With $N = s - s^2/b$ and $\phi = 2s - s^2/b - a$, the quotient rule and the factorisation $N'\phi - N\phi' = -[(s-a)^2 + a(b-a)]/b$ give the closed form of ω'_0 ; ϕ is increasing on the range with $\phi(b) = w$, giving both $\phi \leq w$ and, at the other end, $\phi \geq \phi(a) = aw/b$. The clock bound follows from $(s - a)^2 \geq 0$ and $\phi \leq w$; the drift bound follows by differentiating ω'_0 , bounding every factor of the numerator by w or b (the bracketed sum is at most $6bw^2$) and the denominator below by $b(aw/b)^3$. For the theorem, run the scheme with $\xi = \omega_0$ on $(0, 1)$, $U = 8(\omega - \frac{1}{2})^2$, $\mu_0 = 16$, $U_1 = 8$. The impulse inequality is the corridor $-1/\omega \leq U' \leq 1/(1 - \omega)$, which holds with tangency by $(4\omega - 1)^2 \geq 0$ and $(4\omega - 3)^2 \geq 0$, and it is exact at every width by Theorem 2; free impulses do not move ω , so the path is continuous between paid impulses. The clock gives $\sigma_0^2 = (1 - \rho_W)^2 \sigma_s^2 / W_{\max}^2$ and the drift term is at most $6\rho_W/(1 - \rho_W)^2$ of the clock term, using $|U'| \leq 8$, the second-derivative bound, and $w \leq W_{\max}$; on $\rho_W \leq 1/5$ the assembled constant exceeds $8(1 - 10\rho_W)/W_{\max}^2 \cdot \sigma_s^2$. Boundary as in the fixed-width case. ■

F.7. Theorem 5 and Proposition 8 (Swap Class)

Coverage of the class first. An $\mathcal{A}_1$ impulse decomposes into a swap leg, priced by Assumption 1, and a mint leg, priced by Theorem 2. A mint leg that moves the holdings share by $d = |\omega' - \omega| > 0$ has, with ω_r and $\omega'_r = \omega_r + d$ the receiving token's shares before and after the move,

$$k = \ln \frac{\omega'_r}{\omega_r} = \ln \left(1 + \frac{d}{\omega_r}\right) \geq \frac{d}{\omega_r + d} \geq d,$$

at least one log-unit per unit of share against the swap leg's η . Coverage is conditional on $A \leq 1$, for which $4\gamma + 2\eta + I \leq 1$ suffices, since A is at most the value of the minimand of (22) at $d = \frac{1}{2}$;

the production anchors of Table 1 put that bound near 2×10^{-4} . Under it $A(1-d) \leq 1$ for every $d \in (0, 1)$ and the impulse inequality $U(\omega) - U(\omega') \leq A d(1-d) \leq d \leq k$ holds on mint legs, and a mixed impulse satisfies it leg by leg; every $\mathcal{A}_1$ impulse is therefore priced at or above the form of (21).

For the floor, $U = A(\omega - \frac{1}{2})^2$ has

$$U(\omega) - U(\omega') = A(\omega - \omega')(\omega + \omega' - 1) \leq A d(1-d)$$

at $|\omega - \omega'| = d$, the extreme pair having one point at an edge; the defining minimum of A makes this at most $\gamma + \eta d + Id^2$, which is (21), and the scheme runs with $\mu_0 = 2A$ on the fixed-width clock. For $\gamma = I = 0$ the minimand is $\eta/(1-d)$ with infimum η ; for $I = 0$ setting the derivative of $(\gamma + \eta d)/(d(1-d))$ to zero gives $\eta d^2 + 2\gamma d - \gamma = 0$ and the stated d^* . For achievability, the return-point cycle from $\pm(x-m)h$ to the trigger has mean $h^2 m(2x-m)/\sigma_s^2$, the share motion is $m/2$ at leading order with the exact shares carried by the harness, and (23) follows by renewal reward; monotonicity in x is inspection, and the $m \rightarrow 0$ limit at $x = 1$ is $\eta\sigma_s^2/(4h^2) = \eta\sigma_s^2/\omega^2$, an infimum over the family approached as $x \uparrow 1$ and $m \downarrow 0$ rather than a rate any member attains. Minimising over m at $\gamma > 0$ gives $m^* = 2\sqrt{\gamma/\eta}$ at leading order. ■

F.8. Proposition 12 (Leg Comparison)

The in-range second-derivative density is $-2L/s_b$, so the concavity leg's fractional rate is $\frac{1}{2}\sigma_s^2(2L/s_b)/V \approx \sigma_s^2/(2\bar{s}h)$ using $V \approx 2Lh$ and $s_b \approx \bar{s}$. Against $2\sigma_s^2/h^2$ the ratio is $4\bar{s}/h = 4/\rho$; against $\eta\sigma_s^2/(4h^2)$ it is $\eta\bar{s}/(2h) = \eta/(2\rho)$. ■

F.9. Proposition 17 (Jump Surcharge)

The per-event log-cost of an isolated re-placement from firing displacement δ to target δ' is the larger of the two potential differences of Section 11. When the upper branch binds at both the diffusive and the jump-shifted firing state, the difference of the two per-event costs telescopes to

$$\mathcal{C}^\dagger(xh + O + O_J) - \mathcal{C}^\dagger(xh + O),$$

in which the return point cancels; positivity is strict monotonicity of $\mathcal{C}^\dagger$. Multiplying by the jump-crossing rate and taking expectations over the straddle law gives the display. The narrow-limit form substitutes $\mathcal{C}^\dagger = -\ln(h-\delta) + \text{const}$, and the lower bound is $\ln(a/(a-v)) \geq v/a$, valid for $0 < v < a$, applied with $v = O_J$ and $a = (1-x)h - O \leq h$ on the event that the fired state stays inside the range. The beyond-range clause is the corner evaluation of the exact potentials, at which the isolated re-placement to any two-sided range surrenders the corner value, and the swap-mediated clause is the fee-and-impact cost of Section 14 evaluated at the share gap of the landing state. ■

Appendix G: Proofs for the instrument

G.1. Lemma 6 (Conditional Kernel)

By the strong Markov property at τ_c the post-crossing path is a fresh Brownian motion from b , driftless under the flat-entry condition $|\mu|\Delta t = o(\sigma_1)$ of Appendix C, which carries the μa shift as a correction rather than in the kernel, so $X_{\tau_c+a} - b \sim N(0, \sigma^2 a)$ unconditioned; the firing condition at the check restricts to $o > 0$ and renormalises by $\mathbb{P}(N(0, \sigma^2 a) > 0) = 1/2$. The decomposition is exact because every path with $X_{\text{check}} > b$ has a first crossing in the interval, and given it, the terminal displacement is free. ■

G.2. Lemma 7 (Flat-Entry Delay Law)

With entry states flat below b at density c , the crossing happens at time $s = \Delta t - a$ after entry with weight $\int_0^\infty c f_\tau(s | y) dy$, where $f_\tau(s | y) = y n(y/\sigma\sqrt{s})/(\sigma s^{3/2})$ is the first-passage density from distance y . The integral evaluates to $c\sigma/\sqrt{2\pi s}$, so the delay weight is proportional to $(\Delta t - a)^{-1/2}$; qualification contributes the constant $1/2$ of Lemma 6, and normalising gives the display. ■

G.3. Proposition 13 (Exact Reproduction)

Both sides marginalise the same joint law, which proves the identity structurally. The direct computation, with $\Delta t = \sigma = 1$, begins from the substitution $a = 1/(1 + s^2)$, under which $da/(\sqrt{a}\sqrt{1-a}) = 2 ds/(1 + s^2)$, so that

$$\int_0^1 \frac{n(o/\sqrt{a})}{\sqrt{a}\sqrt{1-a}} da = \int_0^\infty \frac{2 n(o\sqrt{1+s^2})}{1+s^2} ds = \frac{2}{\sqrt{2\pi}} \int_0^\infty \frac{e^{-o^2(1+s^2)/2}}{1+s^2} ds.$$

That integrand is Owen's T ,⁴¹ and the value the identity needs is its limit $T(o, \infty) = \frac{1}{2}Q(o)$. The limit has an elementary derivation that needs no special function. The antiderivative $\int_0^\infty t e^{-t^2 c/2} dt = e^{-o^2 c/2}/c$ at $c = 1 + s^2$ rewrites the integrand as an inner integral; the double integrand is non-negative, so Tonelli permits the interchange, and the inner Gaussian $\int_0^\infty e^{-t^2 s^2/2} ds = \sqrt{2\pi}/(2t)$ then cancels the factor t , leaving

$$\frac{2}{\sqrt{2\pi}} \int_0^\infty \int_0^\infty t e^{-t^2(1+s^2)/2} dt ds = \frac{2}{\sqrt{2\pi}} \int_0^\infty t e^{-t^2/2} \frac{\sqrt{2\pi}}{2t} dt = \int_0^\infty e^{-t^2/2} dt = \sqrt{2\pi} Q(o).$$

The second display is machine-checked in the formalisation, with no special function anywhere in its development; the substitution producing the first is improper at both endpoints and is not machine-checked. ■

G.4. Lemma 8 (Straddle Law)

The jump-crossing compensator of Appendix C integrates $\mathbf{1}\{X_{t-} = b - y, z > y\}$ against $\nu(dz)$ and the occupation of the gap; crossings with gap in dy arrive at rate $\psi_b(y) \bar{\nu}(y) dy$, and among them $O_J > o$ requires $z > y + o$, rate $\psi_b(y) \bar{\nu}(y + o) dy$. Dividing gives the ratio; the flat form substitutes $\psi_b \equiv \psi_b(0^+)$, cancels it, and integrates by parts. ■

G.5. Lemma 10 (Delay Invariance)

Under the quasi-stationarity hypothesis, conditional on the excursion surviving past its relaxation time to age a , the state at a qualifying check is distributed per the quasi-stationary law above the line, whatever the value of a ; the conditional law of O given $t_d = a$ is that law thinned by the filter's qualification event, an a -free object with finite second moment by hypothesis, so $\mathbb{E}[O^2 | t_d = a]/(\sigma^2 a) \rightarrow 0$. Convergence to quasi-stationarity for the era model's conditioned excursions is the standard renewal-and-regeneration argument on the cycle structure of Appendix C; the hypothesis packages it because the constants are not tracked here. For the pure-lag alternative the firing state at delay a is the unconditioned displacement at time a from the line, restricted to the crossing side, whose second moment is of order $\sigma^2 a$ exactly. ■

G.6. Lemma 11 (Jump Enrichment)

To qualify at delay a the continuous branch's diffusion must travel from b past $b + p_{\text{pen}}$ in time a , probability $Q(p_{\text{pen}}/\sigma\sqrt{a})$, bounded by the Gaussian tail; with $O_J > p_{\text{pen}}$ the jump branch's condition holds at the crossing instant and firing waits only for the next check. Dwell delays both branches identically. ■

G.7. Theorem 6 (Small-Delay Identification)

Split by crossing type under Proposition 14. Continuous branch; an event beyond the cut requires the post-crossing diffusion to reach $\kappa\sigma\sqrt{t_d}$ above the line by the firing check, probability at most $Q(\kappa - |\mu|\sqrt{a}/\sigma) \leq Q(\kappa/2)$ under the drift restriction, uniformly over policy filters, because a filter can only remove events or demand a larger excursion; qualification renormalisation costs at most the factor two of Lemma 6. Jump branch; an event fails the cut only if $O_J + \sigma\sqrt{a}Z_a + \mu a \leq \kappa\sigma\sqrt{a}$, which on $\{Z_a \geq -\kappa/2\}$, whose complement is absorbed into the first term, requires $O_J \leq 2\kappa\sigma\sqrt{a}$, the stated tail mass. Staleness mis-measures t_d upward and only shrinks measured ratios, giving the one-sided third term. ■

G.8. Corollary 5 (Tail Shape)

Numerator and denominator are jump-branch dominated by Theorem 6, continuous leakage at most $Q(\kappa/2)$ relative to the exceedance mass; the convolution with the $O(\sigma\sqrt{a})$ kernel perturbs the survival function by at most its Lipschitz modulus times the kernel scale. ■

G.9. Proposition 15 (Multiscale Identification)

(i) is Theorem 6 applied per operator with the relative-rate conversion. (ii) is linear algebra on the functional equations. (iii) substitutes the regularly varying form into the functionals; the kernels act as windows at their scales and Potter bounds⁴⁵ control the slowly varying part across the window. ■

Appendix H: Estimator definitions

The local scale is estimated in a trailing window of H blocks ending strictly before the crossing block, so the candidate displacement is excluded, by the bipower estimator

$$\hat{\sigma}_{\text{BV}} = \left(\frac{\pi}{2S} \sum_i |d_i| |d_{i-1}| \right)^{1/2},$$

with d_i the window's per-block displacements and S its span, admission at least 30 samples spanning at least $H/2$; the realised-variance variant is $\hat{\sigma}_{\text{RV}} = (\sum d_i^2/S)^{1/2}$. Events are admitted with $0 < t_d \leq \bar{a}$ and a dense-crossing bracket of at most ten blocks, and a pool-era is admitted only under the discreteness bar (25). Per event the standardised overshoot is $\tilde{O} = O/(\hat{\sigma}\sqrt{t_d})$; per pool-era with m admitted events,

$$\hat{\varepsilon} = \frac{\#\{\tilde{O} > \kappa\}}{m}, \quad \hat{\pi}_J = \max\left(0, \frac{\hat{\varepsilon} - 2Q(\kappa)}{1 - 2Q(\kappa)}\right),$$

with Wilson intervals on $\hat{\varepsilon}$ and the tail functional the empirical survival of exceedance sizes above o_{min} . The cut is $\kappa = 3$ throughout the empirical work.